Risk-certified class expansion through reference-anchored evidence alignment

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Abstract

Class expansion must integrate few-shot novel evidence while preserving base-class accuracy. We couple vocabulary-invariant reference alignment with finite-sample admission certification. Fixed references preserve existing rankings; query normalization induces an adaptive admission margin. Preserved base rankings make protection equivalent to the retention requirement, while a direct test credits replacement readouts' recoveries. On ImageNet-R, independently tuned query normalization adds 3.60 and 5.29 percentage points of novel capacity over uncentered fusion at 95% retention with L/14 and B/32. L/14 retained and dense capacities are comparable, yet directly certified policies differ by 4.85 points. Separately, original source-selected models reach 86.82% and 58.96% novel accuracy on ImageNet-Sketch with target calibration and empirical retention attainment in all ten partitions per encoder.

Keywords: Class expansion, Retention constraints, Incremental learning, Evidence fusion, Vision–language models, Long-tailed recognition

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1. Introduction

An expanding recognition system integrates unequal evidence: a trained classifier encodes accumulated base supervision, whereas arriving classes provide names and a few examples. Deployment requires novel admission without exceeding a specified loss of base recognition. Evidence must remain comparable as the vocabulary grows, and finite calibration data must support the resulting retention requirement.

Vocabulary growth can disturb evidence alignment even with frozen models. Standardizing a branch against active candidates changes its scale when a class arrives; after fusion, existing classes can exchange rank. Fixed base reference banks remove this dependence. Their query-specific statistics adapt learned, semantic and prototype evidence, while a scalar offset controls novel admission.

We develop *risk-certified class expansion through reference-anchored evidence alignment*. Preserving the original base ranking makes every lost correct base prediction go to a novel class and prevents recovery of base errors. This structure makes a sufficient protection criterion exactly match the retention requirement. A second, direct accuracy test credits replacement readouts' recoveries under the same requirement. The framework exposes how recognition capacity becomes useful certified admission.

The contributions are:

- A vocabulary-invariant alignment of asymmetric evidence, with an explicit query-adaptive admission margin and an exact retention envelope for fixed readouts.

- Finite-sample admission control for class-macro accuracy over prespecified arrivals, including a tail-coverage condition for the protection certificate and a direct test that credits recovered reference errors.

- A capacity-to-policy analysis connecting error structure, attainable recognition and selected admission. Controlled fusion comparisons establish the alignment mechanism; source-frozen ImageNet-Sketch evaluation demonstrates target calibration without model retraining.

On ImageNet-R L/14, retained and dense readouts have comparable capacity at the minimum retention target, while their directly certified policies differ by 4.85 points. The analysis therefore concerns both recognition and its use under an accuracy requirement.

2. Related work

Long-tailed recognition addresses unequal class supervision through balanced fitting and losses [1, 2, 3]. CLIP supplies semantic image–text evidence [4], while prompt adaptation [5, 6], prototypes [7, 8] and semantic caches [9] adapt pre-trained representations to labeled data. Our setting combines these evidence types according to their availability at class arrival.

Class-incremental and few-shot incremental learning study retention across successive tasks [10, 11]. SimpleCIL [12] and RanPAC [13] show the value of pre-trained prototypes and analytic readouts. KBS approaches develop multi-feature similarity supplementation [14], sharpness-aware prototype calibration [15], adaptive exploration [16], representation-enhanced analytic learning [17], and multimodal prompt fusion [18].

Recent work aligns task-specific expert inference in MoTE [19], separates expanding feature spaces through orthogonal constraints [20], and learns invariant representations through counterfactual distribution intervention [21]. We study accuracy-constrained admission with asymmetric base and novel evidence. CLIP-guided discovery [22] additionally uses unlabeled novel data, whereas our arriving classes provide names and labeled support. FeNeC [23] supplies a clustered-representative readout for controlled frozen-feature comparisons.

BiMC [24] and HyCal [25] calibrate multimodal or heterogeneous recognition metrics. Calibrated seen/unseen recognition [26] studies the related operating trade-off, and Learn then Test [27] supplies risk-controlled selection. Our design couples vocabulary-invariant evidence alignment with finite-sample admission control. Query moments shape the recognition frontier; independent calibration selects a policy. The statistical constructions build on risk-controlled testing and bounded-mean betting [28]. We derive their consequences for class-macro retention, unequal calibration coverage and retained versus replacement error structures. An author-code-checked HyCal transfer supplies a matched heterogeneous-metric comparison.

3. Reference alignment and certified admission

3.1. Vocabulary-invariant alignment of asymmetric evidence

Base classes $\mathcal{B}$ have labeled, potentially imbalanced fitting data. Arriving classes $\mathcal{N}$ provide a name and k labeled support images per class. Prediction selects from $\mathcal{B} \cup \mathcal{N}$ without a group oracle. A frozen encoder produces a normalized feature $v = f(x)$; normalized text anchors t_c average eight fixed

prompts. Learned base scores are $q_L(x, b) = v^\top w_b + b_b$, fitted by inverse-frequency-weighted ridge regression. Novel prototypes are normalized support means p_n, giving $q_P(x, n) = v^\top p_n$; semantic scores are $q_T(x, c) = v^\top t_c$.

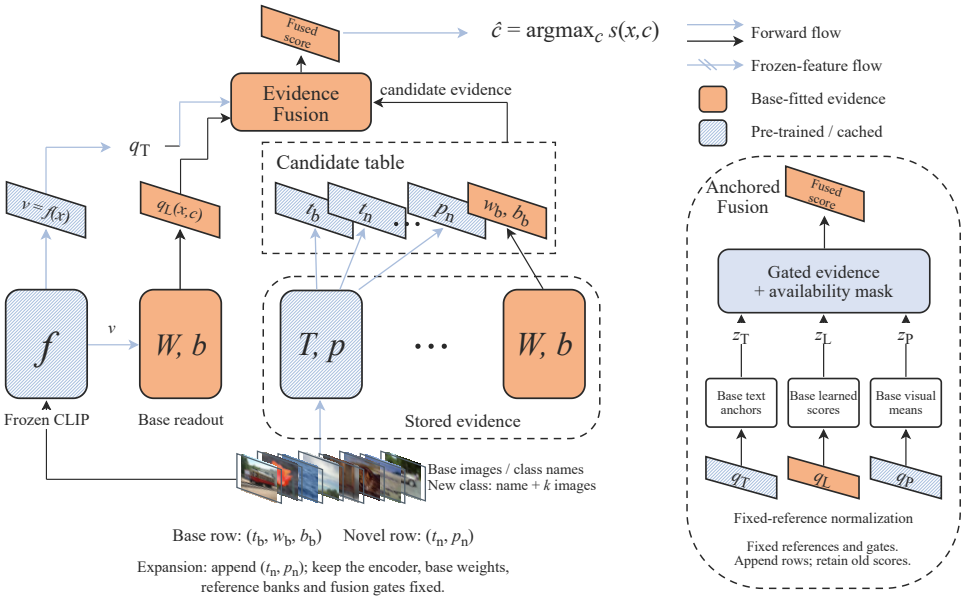


Figure 1: Reference-anchored asymmetric readout. Fixed base references set each branch's score frame; learned base columns remain stored as novel text anchors and support prototypes are appended. The separate novel offset controls admission. Orange marks base-fitted evidence; blue hatching marks pre-trained or cached evidence.

For each evidence source e, a fixed base reference bank $\mathcal{R}_e$ defines

$$\mu_e(x) = |\mathcal{R}_e|^{-1} \sum_{b \in \mathcal{R}_e} q_e(x, b), \tag{1}$$

$$\sigma_e^2(x) = |\mathcal{R}_e|^{-1} \sum_{b \in \mathcal{R}_e} (q_e(x, b) - \mu_e(x))^2. \tag{2}$$

We use base text anchors, learned columns and visual means as the text,

learned and prototype references. Normalized evidence is

$$\widehat{q}_e(x,c) = \frac{q_e(x,c) - \mu_e(x)}{\max(\sigma_e(x), 10^{-6})}. \tag{3}$$

The statistics adapt to each query while remaining independent of the active novel vocabulary. Fusion uses the evidence available to each row:

$$s(x,b) = \widehat{q}_T(x,b) + a\widehat{q}_L(x,b), \qquad b \in \mathcal{B}, \tag{4}$$

$$s(x,n) = \widehat{q}_T(x,n) + u\widehat{q}_P(x,n) + d, \qquad n \in \mathcal{N}. \tag{5}$$

The coefficients a, u combine evidence; d controls novel admission (Fig. 1). Removing the text-centering shift shared by every candidate exposes the effective offset

$$d_{\text{eff}}(x) = d + a\frac{\mu_L(x)}{\sigma_L(x)} - u\frac{\mu_P(x)}{\sigma_P(x)}, \tag{6}$$

using the floored scales in Eq. (3). Centering adjusts the group margin for each query; scaling also changes within-group fusion.

Why the reference must be fixed. Consider old text scores $(0, 2)$ and a learned contribution $(0.25, -0.25)$. Candidate-standardized text scores $(-1, 1)$ make the second old class win after fusion. Appending a text score of 100 changes the old standardized scores to approximately $(-0.728, -0.685)$, reversing their fused order without changing either old class's parameters. Fixed reference banks prevent this vocabulary-induced change.

Proposition 1 (Vocabulary-growth invariance)**.** *With fixed input, encoder, existing evidence, reference banks, coefficients and tie-breaking, appending classes preserves every existing score and its within-vocabulary ordering.*

Proof. Every term in Eqs. (2)–(5) for an existing class remains fixed. Hence each pairwise score difference remains fixed. □

This property concerns stored discrimination; a newly appended class can still win the expanded prediction. Raw and fixed-temperature readouts also preserve scores, providing controls that isolate query alignment.

Proposition 2 (Ridge identity and adaptive admission margin)**.** *Let the weighted ridge fit have one-hot targets, an unregularized intercept and a unique solution. Its learned reference mean satisfies* $\mu_L(x) = 1/B$*, where* $B = |\mathcal{B}|$*. Consequently,*

$$d_{\text{eff}}(x) = d + \frac{a}{B\sigma_L(x)} - u\frac{\mu_P(x)}{\sigma_P(x)}. \tag{7}$$

Proof. Let X include the constant feature, Y contain one-hot targets, Ω contain sample weights, and L regularize slopes. For the intercept vector e_0, $Xe_0 = \mathbf{1}$, $Le_0 = 0$ and $Y\mathbf{1} = \mathbf{1}$. The normal equations give $(X^\top\Omega X + L)W\mathbf{1} = (X^\top\Omega X + L)e_0$. Uniqueness implies $W\mathbf{1} = e_0$; summing the B learned scores gives one. Substitution into Eq. (6) completes the result. □

Thus the learned part of the adaptive offset varies through score dispersion, while the prototype reference contributes both its query mean and dispersion. The combined margin determines admission. Each new class needs $O(D)$ storage and $O(kD)$ prototype construction after encoding; scoring K classes costs $O(KD)$.

3.2. The retention envelope

A base-only pseudo-expansion episode splits base classes into pseudo-base and pseudo-novel groups. A classifier fitted on pseudo-base classes is combined with names and reserved support for pseudo-novel classes. Held-out base queries select a, u, d by harmonic macro accuracy,

$$H = 2A_B A_N/(A_B + A_N), \tag{8}$$

with zero assigned when both accuracies vanish. The coefficients then remain fixed across actual arrivals. Actual novel query labels and test labels are excluded from this selection; grids and split seeds are in Supplementary Section K.

Let $A_{B,\mathrm{ref}}^{\mathrm{closed}}$ denote the retained readout's closed-base macro accuracy. At retention fraction $0 < r < 1$, every readout receives the same absolute base target: a 90% reference gives an 85.5% target at 95% retention. Empirical expansion capacity is

$$N_r = \max_d A_N(d) \quad \text{subject to} \quad A_B(d) \geq r A_{B,\mathrm{ref}}^{\mathrm{closed}}. \tag{9}$$

For a fixed readout, let h_B, h_N be the best labels within each group and $m(x)$ their score difference before adding d. Increasing d switches each query from h_B to h_N once, at $m(x)$, subject to the fixed tie rule. Thus $A_B(d)$ is nonincreasing and $A_N(d)$ is nondecreasing. The largest feasible event maximizes Eq. (9) for retained and replacement readouts. Test-label sweeps describe capacity; independent calibration selects deployable thresholds.

Corollary 1 (Preserved base error structure)**.** *Let R indicate correctness of the fixed retained readout restricted to base classes and C_d correctness after novel admission. Then $C_d \leq R$ on base queries. Consequently recovery $G = \Pr(R = 0, C_d = 1) = 0$, and all destruction $D = \Pr(R = 1, C_d = 0)$ goes to novel labels:*

$$A_B(d) = A_{\mathrm{ref}} - D. \tag{10}$$

The reference in this statement uses the same frozen fusion coefficients. Independently retuned readouts are compared experimentally against the common original reference, relative to which recovery is allowed.

3.3. Finite-sample retention certification

For every readout, let R denote correctness of the same original closed-base reference and C_d correctness after expansion. With destruction D and recovery G, $A_B = A_{\text{ref}} - D + G$. Define the protection loss

$$L_d = R(1 - C_d) + (1 - r)(1 - R), \qquad \alpha = 1 - r. \tag{11}$$

Its risk is $D + \alpha(1 - A_{\text{ref}})$. Thus $\mathbb{E}L_d \leq \alpha$ is sufficient for retention and, when $G = 0$, equivalent to $A_B \geq rA_{\text{ref}}$. This is the direct connection between preserved base rankings and the accuracy-control objective.

Theorem 1 (Macro protection and tail coverage)**.** *For independent class-stratified calibration losses in* $[0, 1]$*, set* $w_i = 1/(Bn_{c(i)})$*,* $b = \max_i w_i$*, and* $q = \sum_i w_i L_i$*. A valid p-value for the null* $\sum_i w_i \mathbb{E}L_i \geq \alpha$ *is*

$$p(q) = \begin{cases} \exp[-\operatorname{kl}(q, \alpha)/b], & q < \alpha, \\ 1, & q \geq \alpha. \end{cases} \tag{12}$$

At level δ/T*, passing this bound requires*

$$n_{\text{eff}} = 1/b = B \min_c n_c \geq \frac{\log(\delta/T)}{\log r}. \tag{13}$$

Proof. For $\mu = \sum_i w_i \mathbb{E}L_i$ and $x < \mu$, the bounded-loss Chernoff argument yields $\Pr(q \leq x) \leq \exp[-\operatorname{kl}(x, \mu)/b]$ (Supplementary Section J). Monotonicity in $\mu \geq \alpha$ gives the p-value. Its minimum $p(0) = r^{1/b}$ gives Eq. (13). □

The least-covered class controls this certificate's effective count. The condition is necessary for this bound; observed losses determine whether a covered candidate passes.

Theorem 2 (Recovery-aware direct accuracy test)**.** *For independent correctness pairs with the same macro weights, let* $Z_i = C_{d,i} - rR_i$*. For any fixed* $0 \le \lambda < 1/r$*,*

$$E_\lambda = \prod_i [1 + \lambda(w_i/b)Z_i] \tag{14}$$

is nonnegative and has expectation at most one under $\sum_i w_i \mathbb{E}Z_i \le 0$*. A fixed convex mixture* E *therefore gives a valid fixed-sample p-value* $\min(1, 1/E)$ *for the unsafe accuracy null.*

Proof. Each factor is positive because $Z_i \ge -r$ and $w_i/b \le 1$. Independence and $1 + x \le e^x$ give $\mathbb{E}E_\lambda \le \exp[(\lambda/b)\sum_i w_i \mathbb{E}Z_i] \le 1$. Linearity and Markov's inequality establish the mixture result. □

This bounded-mean betting construction [28] credits corrected reference errors. We use twelve fixed mixture components, specified in Supplementary Section N. Sample sizes and mixture weights are fixed independently of calibration outcomes.

Both tests operate on thresholds fixed by fitting/tuning data, ordered from conservative to permissive. Fixed-sequence testing [27] stops at the first failure at level δ/T and returns the last accepted threshold. If none passes, the policy uses the original reference with zero novel admission. A union bound controls unsafe acceptance across $T = 5$ prespecified arrivals with probability at most δ. The statement holds per fixed method/order/partition/policy under independent target class-conditional sampling; calibration data are excluded from fitting and tuning. The reference fallback and finite admission are reported separately.

Error structure and paired-score variability. For a class-macro base query, let $A_C = \mathbb{E}C$ and $A_R = \mathbb{E}R$. Direct expansion of $Z = C - rR$ gives

$$\begin{aligned}\mathrm{Var}(Z) = A_R(1-r)^2 + (A_R - A_C)(2r-1) \\ + 2rG - (A_C - rA_R)^2.\end{aligned} \tag{15}$$

At fixed A_R, A_C, recovery accompanied by additional destruction increases variance by $2rG$ relative to a feasible zero-recovery coupling. Retention removes that component while also making protection exact. For stratified calibration, the same identity holds within each class; the variance of the weighted mean is $\sum_c \mathrm{Var}(Z \mid c)/(B^2 n_c)$. These conditional comparisons explain an error-structure advantage; actual certification also depends on accuracy slack, class coverage and the threshold family. Supplementary Section P derives the identity and its scope.

4. Experiments

4.1. Protocol and comparisons

We evaluate CIFAR-100 [29], Oxford-IIIT Pets [30] and ImageNet-R [31], using frozen OpenAI CLIP ViT-L/14 and ViT-B/32 features, five-shot support, ten frequency-stratified class partitions and two five-session arrival orders. Base/novel counts are 40/40, 15/15 and 100/100; each session adds eight, three or 20 classes. Base pools follow long-tailed sampling. ImageNet-R uses task-specific splits after removing byte-identical duplicates. Novelty is relative to task fitting, with CLIP's pre-training exposure retained. Detailed pools, preprocessing and seeds are in Supplementary Sections C, I and K.

Every comparator shares its encoder, partition and support. We compare fixed-gate classifier removal, independently recalibrated removal, dense an-

chored text–prototype fusion, summed/class-mean Tip-Adapter caches, and raw/fixed-temperature retained fusion. The latter uses a single base-validation scale per branch. HyCal uses the author's heterogeneous cosine/Mahalanobis metrics, count weighting and Gaussian-CDF calibration, transferred to the same eight-prompt anchors and frozen encoders. Its covariance regularization and novel offset are selected by base-only pseudo-expansion. Supplementary Section M documents the pinned implementation, numerical fixtures and transfer settings. All mixtures use base-only selection; FeNeC and earlier three-partition auxiliary studies are retained in Supplementary Sections A–H.

Orders are averaged within a partition. Means and standard deviations describe ten class assignments on fixed images. Paired differences use identical cells; constrained comparisons use jointly feasible sets. Conditional paired 95% bootstrap intervals resample partition means 20,000 times. Average incremental accuracy is the mean macro accuracy over five arrivals. These intervals characterize class-assignment variation, conditional on the fixed data and encoders.

4.2. Novel recognition at a shared base requirement

At the descriptive 95% retention budget, the ImageNet-R L/14 table reaches 84.31% novel accuracy at 86.42% base accuracy. Figure 2 compares novel recognition under shared base requirements across encoders and datasets. A more discriminative base readout can tolerate greater novel competition while meeting the same target, turning retained discrimination into admission capacity.

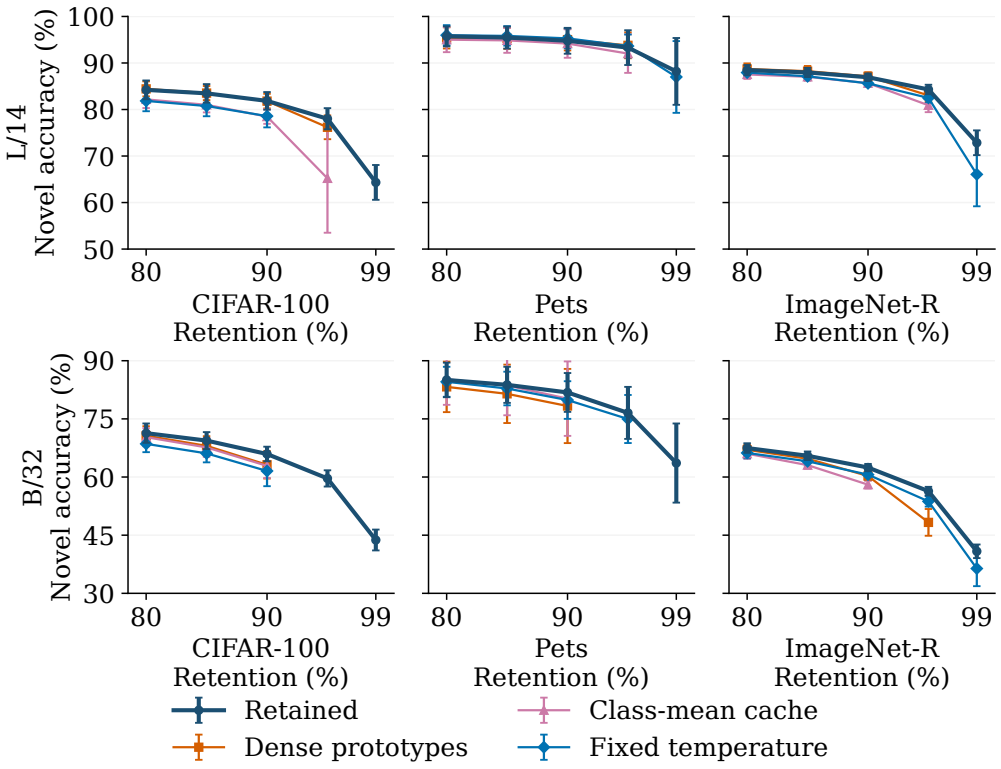


Figure 2: Novel recognition at shared base-retention requirements after the fifth arrival. Each readout must attain the stated fraction of the retained reference's closed-base accuracy within the same partition and encoder. Means and sample standard deviations cover ten partitions after averaging two arrival orders; points require all ten partitions to be feasible. Vertical ranges are 50–100% (L/14) and 30–90% (B/32); error bars stop at panel limits. Test-label threshold sweeps characterize descriptive capacity. Partial feasible sets and exact counts are reported in Supplementary Section I.

4.3. Query-adaptive alignment expands retention frontiers

Independent fusion tuning isolates query normalization under equal search budgets, with encoder, ridge fit, support and data roles fixed. At the common 95% base target, query normalization exceeds uncentered fusion by 3.60

[2.55,4.74] points on ImageNet-R L/14 and 5.29 [3.74,7.25] on B/32. Gains over fixed global normalization are 2.84 [2.42,3.28] and 4.65 [3.57,5.71] points. Each comparison includes ten partitions after averaging orders. Supplementary Section Q gives the factorial and search specification; Section M separately isolates centering and scaling with fusion held fixed.

The adaptive offset acts through the combined group margin in Eq. (7). On the original frozen models, adding its centering term raises base/novel ranking AUROC from .924 to .977 on ImageNet-R L/14 and from .856 to .904 on B/32. The term alone has novel-positive AUROC below .5 in all six conditions: its role is to align evidence within the decision margin. Section P supplies the conditional distributions and offset-reselected controls.

Independent tuning also gives a retained-minus-dense capacity gap of 9.69 [8.08,11.21] points on ImageNet-R B/32 and 0.25 [−0.46,0.92] on L/14. The next analyses examine how these recognition structures support repeated and certified admission.

4.4. The classifier's value across arrivals

Table 1 quantifies retained-minus-comparator average incremental accuracy with fixed base-only selected gates and offsets. On ImageNet-R, retaining the classifier adds 0.71 [0.46, 0.92] points over dense prototypes with L/14 and 0.90 [0.69, 1.11] with B/32. Corresponding gains over class-mean caches are 1.42 [1.15, 1.62] and 1.71 [1.44, 1.98] points. The result persists across both encoders and after calibration is restored to the classifier-free alternative.

CIFAR also shows gains over recalibrated removal with both encoders; its dense-prototype intervals span zero. Pets' average incremental intervals span zero for these comparators. The gain over prototype replacement is clearest

Table 1: The retained classifier's value across five arrivals. Cells show paired gains in average incremental accuracy (percentage points), with conditional paired 95% bootstrap intervals below. Positive values favor retention; bold estimates have intervals excluding zero. Two orders are averaged within each of ten partitions. Full final-session differences are in Supplementary Section I.

Comparator	Retained-minus-comparator gain (pp) ↑		
	CIFAR-100	Pets	ImageNet-R
CLIP ViT-L/14			
Recalibrated removal	**+3.00** [+2.23, +3.75]	-0.14 [-1.53, +0.86]	**+2.30** [+1.95, +2.59]
Dense prototypes	+0.15 [-0.46, +0.64]	-0.98 [-2.52, +0.22]	**+0.71** [+0.46, +0.92]
Class-mean cache	**+1.44** [+0.68, +2.10]	-0.06 [-1.76, +1.41]	**+1.42** [+1.15, +1.62]
CLIP ViT-B/32			
Recalibrated removal	**+2.38** [+1.44, +3.34]	-0.44 [-1.64, +0.81]	**+3.07** [+2.82, +3.34]
Dense prototypes	+0.86 [-0.15, +1.76]	-0.39 [-1.85, +1.02]	**+0.90** [+0.69, +1.11]
Class-mean cache	+0.96 [-0.04, +1.84]	-0.74 [-1.62, +0.12]	**+1.71** [+1.44, +1.98]

on ImageNet-R.

Supplementary Section M relates closed-base discrimination to class frequency and prototype quality and reports a controlled learned-branch intervention. It also evaluates the author-code-checked HyCal transfer: ImageNet-R average incremental accuracy is 76.36% with L/14 and 54.19% with B/32, with final 95% capacity feasible in two and zero partitions when both orders must qualify. These results describe the matched transfer under the shared

retention requirement.

4.5. Class coverage enables certified admission

Independent calibration separates fitting, base-only tuning and class-stratified certification. The readouts and threshold candidates are frozen before varying calibration coverage; all methods receive the same images. Nested per-class caps of 2, 5, 10, 20 and 40 evaluate the label budget, using $r = 0.95$, $\delta = 0.05$ and five arrivals. Exact data roles, subset aggregation and the development history of the source benchmarks are specified in Supplementary Sections J and L.

Equation (13) requires effective coverage of at least 89.78 for the protection bound. The original CIFAR and Pets allocations provide 80 and 45–60, respectively. Figure 3 shows how increasing class coverage enables finite admission. At cap 40, the required base-label counts are 1,600 for CIFAR, 600 for Pets and 3,636–3,778 for ImageNet-R. These calibration labels are additional to five-shot novel support and can come from a reserved labeled catalog audit pool.

At cap 40, retained readouts certify final-session expansion in all ten partitions of each dataset and encoder, with all test targets met (Table 2). L/14 novel accuracy is 72.76% on CIFAR, 88.51% on Pets and 80.74% on ImageNet-R. All 20 arrival paths per dataset/encoder meet every test target. Finite admission holds throughout all CIFAR and ImageNet-R paths and 17/20 L/14 and 16/20 B/32 Pets paths; the remaining Pets paths use an early-session fallback. Full comparisons distinguish policy gains, which include fallback, from accuracy differences on jointly admitted partitions.

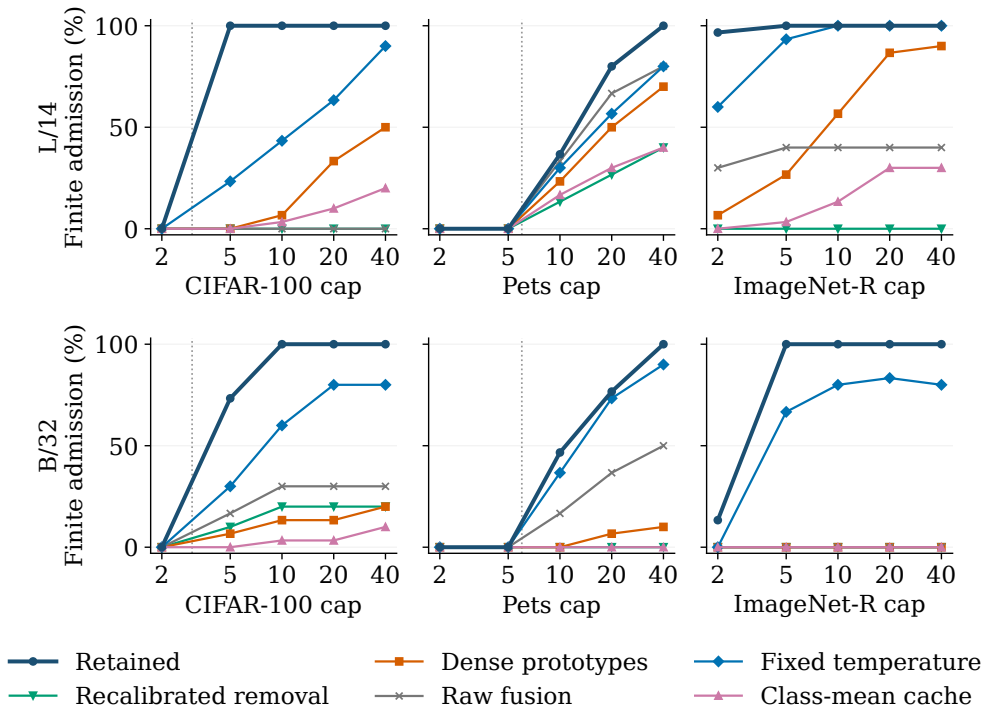


Figure 3: Finite certified admission versus calibration coverage, fifth arrival in the first order. All readouts share images and the protection rule. Grey lines show the necessary balanced cap from Eq. (13); ImageNet-R's floor lies below the plotted range. Below cap 40, three subset seeds are averaged within each of ten partitions; cap 40 uses the full pool once. Fallback counts as no finite admission. Novel-accuracy curves and exact counts appear in Supplementary Section L.

4.6. From recognition capacity to certified policies

Figure 4 holds calibration images, candidate thresholds and error allocation fixed while changing the statistical criterion. On ImageNet-R L/14, crediting recovered errors raises dense-prototype admission from 9/10 to 10/10 and novel accuracy from 48.58% to 76.76%, including fallback. Retained readouts attain 81.61%, a paired gain of 4.85 [2.78,6.93] points with all ten partitions

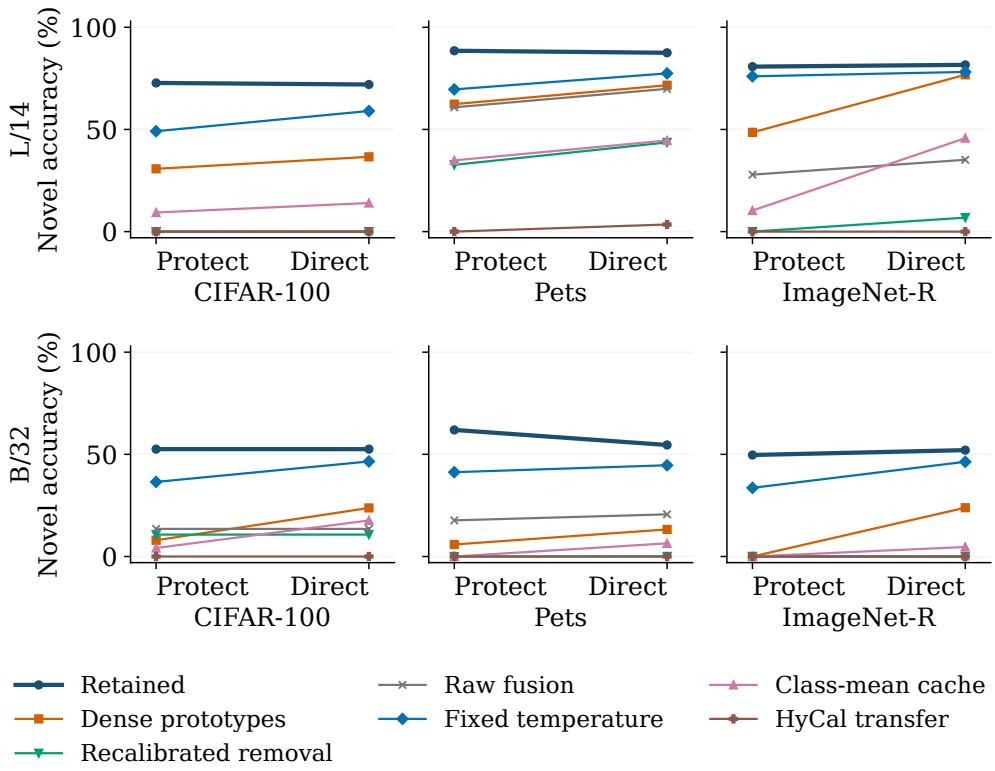


Figure 4: Calibration-rule comparison at identical cap-40 coverage, fifth arrival, first prespecified order. Lines connect the sufficient protection test and the direct paired accuracy test for each readout. Novel means cover all ten partitions, with zero for reference fallback. Each test uses 95% retention and 95% confidence over five prespecified arrivals within a fixed configuration. Admission counts, conditional accuracy and empirical target attainment are reported in Supplementary Section N.

admitted by both methods.

Let $F_m(t)$ be readout m's empirical novel capacity at base requirement t, N_m its selected-policy novel accuracy, and $H_m(t) = F_m(t) - N_m$ its remaining

Table 2: Protection-risk-calibrated admission after the fifth arrival in the first prespecified order, with up to 40 images per base class, 95% retention and 95% five-session confidence. Queries are actual base-calibration counts; Finite counts certified expansion and Test met counts attained test targets. Base/Novel are test percentages, mean ± sample standard deviation over ten partitions, including fallback. All six readouts are reported in Supplementary Section L.

Dataset	Calibration	Admission		Test accuracy (%) ↑	
	Queries	Finite	Test met	Base	Novel
CLIP ViT-L/14					
CIFAR-100	1600	10/10	10/10	85.86 ± 1.19	72.76 ± 4.76
Pets	600	10/10	10/10	94.10 ± 2.27	88.51 ± 4.85
ImageNet-R	3636–3778	10/10	10/10	89.02 ± 0.51	80.74 ± 1.83
CLIP ViT-B/32					
CIFAR-100	1600	10/10	10/10	76.74 ± 1.73	52.54 ± 4.34
Pets	600	10/10	10/10	88.10 ± 3.41	61.94 ± 14.07
ImageNet-R	3636–3778	10/10	10/10	72.98 ± 1.21	49.67 ± 2.38

capacity. For retained R and dense P readouts,

$$N_R - N_P = \underbrace{F_R(t) - F_P(t)}_{\text{capacity difference}} + \underbrace{H_P(t) - H_R(t)}_{\text{remaining-capacity difference}} . \tag{16}$$

On ImageNet-R L/14, capacities 83.93% and 83.64% differ by 0.29 points, while remaining capacities are 2.32 and 6.88 points. The latter difference accounts for 4.56 points of the 4.85-point policy gap. Comparable capacity can therefore coexist with different certified operating points. The remaining capacity combines threshold discretization, sampling and selection effects;

this decomposition does not assign all of it to test variance.

Table 3 extends the analysis across datasets and encoders. The 35.40-point CIFAR L/14 policy gap includes reference fallback, alongside a 3.68-point capacity gap. Pets L/14 has a −2.12-point retained-minus-dense capacity gap and a positive policy gap. ImageNet-R B/32 additionally supplies a 10.40-point capacity advantage. Figure 5 overlays selected policies on exact empirical trajectories, displaying both capacity and how it is used.

Table 3: Retained-minus-dense selected-policy and descriptive capacity gains (pp), fifth arrival, first order. Capacity uses the common 95% minimum base target. Within a row, both columns use the same jointly feasible partitions (count); policy means include reference fallback. Brackets give conditional paired 95% bootstrap intervals on fixed images.

Encoder	Data	Policy gap	Common-floor capacity gap	Matched count
L/14	CIFAR-100	+35.40 [+16.61,+54.61]	+3.68 [+1.08,+6.38]	10/10
L/14	Pets	+15.88 [-1.27,+40.52]	-2.12 [-4.07,-0.49]	10/10
L/14	ImageNet-R	+4.85 [+2.77,+6.93]	+0.29 [-0.55,+1.16]	10/10
B/32	CIFAR-100	+26.38 [+11.60,+41.41]	+3.50 [-0.47,+7.94]	9/10
B/32	Pets	+42.92 [+15.13,+66.79]	+1.17 [-5.91,+7.38]	8/10
B/32	ImageNet-R	+26.15 [+15.85,+37.54]	+10.40 [+7.31,+14.64]	9/10

Matching at the dense policy's actual, stricter base accuracy gives a 3.23 [1.48,5.02]-point retained-minus-selected-dense gain on ImageNet-R L/14, with mean base overshoot 0.027 points. Comparing both frontiers at that stricter target gives 3.10 [1.37,4.88] points. Together, the common-floor and realized-accuracy views identify where the advantage lies. Full comparisons report joint feasibility, admission-conditioned accuracy and fallback, including the single dense B/32 ImageNet-R test-target miss (Sections N and P).

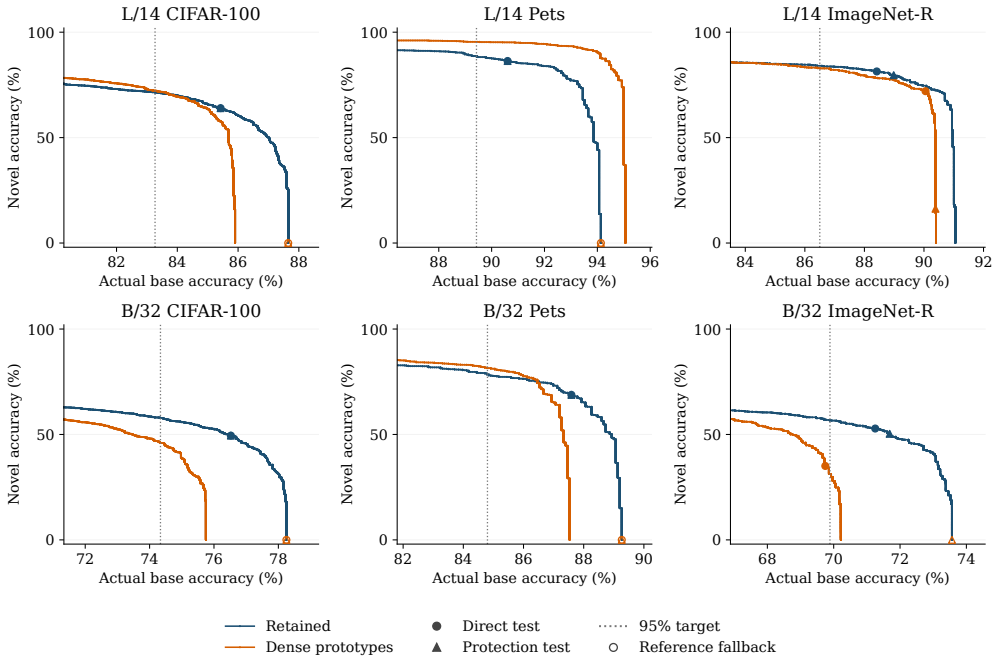


Figure 5: Capacity and certified operating points for prespecified partition 0, fifth arrival. The vertical line is the shared minimum base target. Triangles denote protection calibration, circles direct calibration, and hollow markers reference fallback. Curves use test labels to describe capacity; calibration chooses the points independently. Lines guide the eye; matching uses exact discrete events. Supplementary Section P provides all partitions and both orders.

4.7. Target calibration transfers frozen source readouts

The controlled CIFAR stress test uses clean fitting/support and noisy or blurred queries. Clean calibration meets 5/10 and 1/10 L/14 final targets; matching calibration restores 10/10 with finite admission for both encoders (Section O).

The separate transfer study freezes the original source-selected ImageNet-R readouts and evaluates matching ImageNet-Sketch classes [32]. Identity-

disjoint target calibration and test pools are fixed before scoring (Supplementary Section R). Source and Sketch calibration use equal per-class counts; model parameters and source support remain fixed. With direct Sketch calibration, retained readouts reach 86.82% and 58.96% novel accuracy with L/14 and B/32, respectively, with finite admission and target attainment in all ten partitions. Source calibration also meets these targets, at 82.99% and 54.87% novel accuracy. At the common 95% minimum target, L/14 retained and dense capacities both round to 88.84%; B/32 retention supplies a 5.78 [4.03,7.55]-point capacity gain on nine jointly feasible partitions. Section R reports the identity exclusions, absolute base accuracies, both calibration criteria and exact frontiers.

5. Conclusion

Reference-anchored evidence alignment supports class expansion with finite-sample retention control. Fixed base references preserve existing discrimination as the vocabulary grows, while query normalization induces an adaptive admission margin. Preserved base rankings make protection coincide with the accuracy requirement; a recovery-aware test extends certification to replacement readouts. Controlled comparisons establish the normalization mechanism, and capacity-to-policy analysis shows how comparable recognition frontiers can yield different certified admission. Target calibration transfers frozen source readouts to natural rendition shift without retraining. The framework connects stable evidence, explicit accuracy requirements and useful admission across successive class arrivals.

CRediT authorship contribution statement

Junyi Zhang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review and editing.

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Declaration of competing interest

The author declares that there are no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The experiments use public CIFAR-100, Oxford-IIIT Pets, ImageNet-R and ImageNet-Sketch data and the publicly released CLIP ViT-L/14 and ViT-B/32 checkpoints. Code, source-hashed experimental records, prediction archives, manuscript sources, and table/figure generation scripts are available at `https://github.com/Cheerssom/Reference-Anchored-Candidate-Tables`. Dataset images, pretrained weights, and feature caches are obtained separately using the repository instructions.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

OpenAI Codex was used to assist with literature discovery, manuscript drafting and revision, experiment and verification scripts, and LaTeX and table preparation. The author is responsible for the accuracy, interpretation and originality of the submitted work and for reviewing and approving the final manuscript and supporting materials.

References

[1] Z. Liu, Z. Miao, X. Zhan, J. Wang, B. Gong, S. X. Yu, Large-Scale Long-Tailed Recognition in an Open World, arXiv preprint arXiv:1904.05160 (2019).
URL https://arxiv.org/abs/1904.05160

[2] K. Cao, C. Wei, A. Gaidon, N. Arechiga, T. Ma, Learning Imbalanced Datasets with Label-Distribution-Aware Margin Loss, arXiv preprint arXiv:1906.07413 (2019).
URL https://arxiv.org/abs/1906.07413

[3] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, S. Belongie, Class-Balanced Loss Based on Effective Number of Samples, arXiv preprint arXiv:1901.05555 (2019).
URL https://arxiv.org/abs/1901.05555

[4] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever,

Learning Transferable Visual Models From Natural Language Supervision, in: International Conference on Machine Learning, 2021, pp. 8748–8763.
URL https://proceedings.mlr.press/v139/radford21a.html

[5] K. Zhou, J. Yang, C. C. Loy, Z. Liu, Learning to Prompt for Vision-Language Models, International Journal of Computer Vision 130 (9) (2022) 2337–2348. doi:10.1007/s11263-022-01653-1.
URL https://link.springer.com/article/10.1007/s11263-022-01653-1

[6] J.-X. Shi, T. Wei, Z. Zhou, J.-J. Shao, X.-Y. Han, Y.-F. Li, Long-Tail Learning with Foundation Model: Heavy Fine-Tuning Hurts, in: International Conference on Machine Learning, 2024, pp. 45014–45039.
URL https://proceedings.mlr.press/v235/shi24g.html

[7] J. Snell, K. Swersky, R. S. Zemel, Prototypical Networks for Few-shot Learning, arXiv preprint arXiv:1703.05175 (2017).
URL https://arxiv.org/abs/1703.05175

[8] Y. Wang, W.-L. Chao, K. Q. Weinberger, L. van der Maaten, SimpleShot: Revisiting Nearest-Neighbor Classification for Few-Shot Learning, arXiv preprint arXiv:1911.04623 (2019).
URL https://arxiv.org/abs/1911.04623

[9] R. Zhang, W. Zhang, R. Fang, P. Gao, K. Li, J. Dai, Y. Qiao, H. Li, Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification, arXiv preprint arXiv:2207.09519 (2022).
URL https://arxiv.org/abs/2207.09519

[10] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, C. H. Lampert, iCaRL: Incremental Classifier and Representation Learning, arXiv preprint arXiv:1611.07725 (2016).
URL https://arxiv.org/abs/1611.07725

[11] X. Tao, X. Hong, X. Chang, S. Dong, X. Wei, Y. Gong, Few-Shot Class-Incremental Learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 12183–12192.
URL https://openaccess.thecvf.com/content_CVPR_2020/html/Tao_Few-Shot_Class-Incremental_Learning_CVPR_2020_paper.html

[12] D.-W. Zhou, Z.-W. Cai, H.-J. Ye, D.-C. Zhan, Z. Liu, Revisiting Class-Incremental Learning with Pre-Trained Models: Generalizability and Adaptivity are All You Need, International Journal of Computer Vision (2024).
URL https://github.com/LAMDA-CL/RevisitingCIL

[13] M. D. McDonnell, D. Gong, A. Parvaneh, E. Abbasnejad, A. van den Hengel, RanPAC: Random Projections and Pre-trained Models for Continual Learning, in: Advances in Neural Information Processing Systems, Vol. 36, 2023, pp. 12022–12053. doi:10.52202/075280-0526.
URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/2793dc35e14003dd367684d93d236847-Abstract-Conference.html

[14] X. Xu, S. Niu, Z. Wang, W. Guo, L. Jing, H. Yang, Multi-feature space similarity supplement for few-shot class incremental learning, Knowledge-

Based Systems 265 (2023) 110394. doi:10.1016/j.knosys.2023.110394.

[15] R. Chen, X.-Y. Jing, F. Wu, H. Chen, Sharpness-aware gradient guidance for few-shot class-incremental learning, Knowledge-Based Systems 299 (2024) 112030. doi:10.1016/j.knosys.2024.112030.

[16] C. Han, Z. Gu, C. Xu, Z. Cui, Adaptive exploration for few-shot incremental learning, Knowledge-Based Systems 330 (2025) 114496. doi:10.1016/j.knosys.2025.114496.

[17] R. He, D. Fang, Y. Chen, K. Tong, C. Chen, Y. Wang, L.-p. Chau, H. Zhuang, REAL: Representation enhanced analytic learning for exemplar-free class-incremental learning, Knowledge-Based Systems 332 (2026) 114901. doi:10.1016/j.knosys.2025.114901.

[18] H. Fang, H. Cai, K. Mao, J. Zhao, X. Xiao, Few-shot class-incremental learning based on prompt guidance and multimodal fusion, Knowledge-Based Systems 342 (2026) 115862. doi:10.1016/j.knosys.2026.115862.

[19] L. Li, Z. Wu, Y. Ji, MoTE: Mixture of task-specific experts for pre-trained model-based Class-incremental learning, Knowledge-Based Systems 324 (2025) 113795. doi:10.1016/j.knosys.2025.113795.

[20] M. Dong, Z. Zhang, X. Tan, J. Qiu, Y. Xie, Dynamic expansion orthogonal network for class-incremental learning, Knowledge-Based Systems 341 (2026) 115868. doi:10.1016/j.knosys.2026.115868.

[21] J. Yuan, W. Li, L. Li, L. Zhang, E. Ning, X. Gao, X. Ning, Counterfactual distribution intervention for few-shot class-incremental learning, Knowledge-Based Systems 351 (2026) 116754. doi:10.1016/j.knosys.2026.116754.

[22] Q. Yan, Y. Yang, Y. Dai, X. Zhang, K. Wiltos, M. Woźniak, W. Dong, Y. Zhang, CLIP-guided continual novel class discovery, Knowledge-Based Systems 310 (2025) 112920. doi:10.1016/j.knosys.2024.112920.

[23] K. Książek, H. Jastrzębski, K. Pniaczek, B. Trojan, M. Karp, J. Tabor, FeNeC: Enhancing continual learning via feature clustering with neighbor- or logit-based classification, Knowledge-Based Systems 338 (2026) 115479. doi:10.1016/j.knosys.2026.115479.

[24] Y. Chen, T. Ding, L. Wang, J. Huo, Y. Gao, W. Li, Enhancing few-shot class-incremental learning via training-free bi-level modality calibration, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2025, pp. 9881–9890.
URL https://openaccess.thecvf.com/content/CVPR2025/html/Chen_Enhancing_Few-Shot_Class-Incremental_Learning_via_Training-Free_Bi-Level_Modality_Calibration_CVPR_2025_paper.html

[25] E. Lee, M. Kim, J. Kwon, Y. Lee, J. Kim, S. Jang, Y. Kim, Hycal: A training-free prototype calibration method for cross-discipline few-shot class-incremental learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026, pp. 29462–29471.
URL https://arxiv.org/abs/2604.15678

[26] W.-L. Chao, S. Changpinyo, B. Gong, F. Sha, An Empirical Study and Analysis of Generalized Zero-Shot Learning for Object Recognition in the Wild, arXiv preprint arXiv:1605.04253 (2016).
URL https://arxiv.org/abs/1605.04253

[27] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei, Learn then test: Calibrating predictive algorithms to achieve risk control, arXiv preprint arXiv:2110.01052 (2021). doi:10.48550/arXiv.2110.01052.

[28] I. Waudby-Smith, A. Ramdas, Estimating means of bounded random variables by betting, Journal of the Royal Statistical Society Series B: Statistical Methodology 86 (1) (2024) 1–27. doi:10.1093/jrsssb/qkad009.

[29] A. Krizhevsky, Learning multiple layers of features from tiny images, Tech. rep., University of Toronto (2009).
URL https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf

[30] O. M. Parkhi, A. Vedaldi, A. Zisserman, C. V. Jawahar, Cats and dogs, in: IEEE Conference on Computer Vision and Pattern Recognition, 2012, pp. 3498–3505. doi:10.1109/CVPR.2012.6248092.
URL https://www.robots.ox.ac.uk/~vgg/publications/2012/parkhi12a/

[31] D. Hendrycks, S. Basart, N. Mu, S. Kadavath, F. Wang, E. Dorundo, R. Desai, T. Zhu, S. Parajuli, M. Guo, D. Song, J. Steinhardt, J. Gilmer, The Many Faces of Robustness: A Critical Analysis of Out-of-Distribution

Generalization, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 8340–8349.
URL https://openaccess.thecvf.com/content/ICCV2021/html/Hendrycks_The_Many_Faces_of_Robustness_A_Critical_Analysis_of_Out-of-Distribution_ICCV_2021_paper.html

[32] H. Wang, S. Ge, Z. C. Lipton, E. P. Xing, Learning robust global representations by penalizing local predictive power, in: Advances in Neural Information Processing Systems, Vol. 32, 2019, pp. 10506–10518.

Supplementary material: Risk-certified class expansion through reference-anchored evidence alignment

Junyi Zhang

Evidence guide

The primary evidence separates source-domain mechanism tests from frozen-model transfer. The source study in Section Q independently selects fusion coefficients and offsets; the Sketch study in Section R fixes the original source-selected models and changes the calibration/query domain.

Question	Evidence and scope
Does query alignment help after tuning?	Section Q: equal-budget source comparisons. M and P: fixed-model controls and effective bias.
When does retaining the classifier add value?	I, Q and R: common-target capacities, paired gains and feasibility counts.
Can admission meet the original-class requirement?	J and N: statistical tests; L: calibration coverage and fallback.
Does frozen-model admission transfer?	R: identity-disjoint Sketch queries with fixed source models and matched calibration counts; O: controlled corruptions.
How are operating points compared?	P: exact discrete frontiers, selected policies and error destinations.

For the central case, read Q (alignment), N (admission) and R (transfer). Sections A–H provide auxiliary readout and validation studies; K collects implementation details. Intervals and confidence scopes are stated with their respective evaluations.

Appendix A. Auxiliary readout and mechanism analyses

For the auxiliary readout and support-budget comparisons, the gate grid is $a, u \in \{0, 0.25, 0.5, 1, 2\}$ and $d \in \{-2, -1.5, \ldots, 2\}$. Offset-calibrated controls use their separately stated grids. The main retention/value evaluation uses the common expanded grid in Section 3.2. All experiments share the same frozen representations, true class partitions and disjoint support pools.

Appendix A.1. Readouts and evaluation measures

All compared readouts use the same CLIP image embeddings, text anchors, class partitions, and novel support images. The text baseline uses cosine similarity to class names. The visual prototype baseline uses means of base fitting examples and means of novel support examples. Text–prototype fusion adds the two cosine branches with a mixing coefficient selected on the same pseudo-novel calibration episode; its grid is $\{0, 0.25, 0.5, 1, 2, 4\}$, including pure text.

Tip-Adapter uses a balanced k-image cache per candidate and scores $100q_T + \alpha q_{\text{cache}}$, with affinity $\exp[\beta(v^\top f(x_i) - 1)]$. The same base-only episode selects $\alpha \in \{0, 0.5, 1, 2, 4\}$ and $\beta \in \{1, 2, 5, 10\}$. Cache images come from the reserved pool for both groups; base-trained readouts use the full base fitting set.

We also compare raw evidence fusion and candidate-normalized fusion. Raw fusion uses unstandardized text, learned, and prototype scores with the same coefficient grid as the proposed table. Candidate-normalized fusion standardizes text over all active classes, learned scores over the base classes, and prototypes over active novel classes. It shares the base classifier and calibration protocol with reference-anchored fusion. This comparison changes

Table S1: Five-shot expansion from 40 base classes to 80 classes. All values are accuracy percentages, reported as mean ± sample standard deviation over three distinct class partitions. All readouts share frozen CLIP features; coefficients are selected using base-only calibration. Base and novel performance jointly specify each operating point.

Readout	Base	Novel	All	Harmonic
CLIP text	78.32 ± 0.25	76.86 ± 1.97	77.59 ± 0.86	77.57 ± 0.88
Visual prototypes	77.53 ± 2.03	60.92 ± 3.94	69.23 ± 2.09	68.17 ± 2.49
Text + prototypes	82.47 ± 0.87	77.53 ± 0.47	80.00 ± 0.56	79.92 ± 0.55
Tip-Adapter (tuned)	78.38 ± 0.24	77.42 ± 1.72	77.90 ± 0.83	77.89 ± 0.83
Raw evidence fusion	78.32 ± 0.25	76.86 ± 1.97	77.59 ± 0.86	77.57 ± 0.88
Candidate-normalized table	85.36 ± 0.66	70.83 ± 4.33	78.09 ± 2.02	77.36 ± 2.48
Reference-anchored table	86.03 ± 1.40	66.90 ± 4.23	76.46 ± 1.43	75.19 ± 2.14

the normalization population while holding the principal evidence sources fixed.

We report macro base accuracy A_B, macro novel accuracy A_N, all-class macro accuracy, and the harmonic mean in Eq. (8). Both datasets use equal numbers of base and novel classes, so all-class macro accuracy is $(A_B + A_N)/2$. On the balanced CIFAR test set this also equals sample accuracy; on Pets the macro measure gives each breed equal weight. Stability measures include maximum absolute change in an existing score, change of the maximizing class restricted to base classes, and prediction changes over the expanded vocabulary. Error flips are reported as fractions of all base test images, not conditional fractions of previously correct images.

Appendix A.2. Recognition after five-shot class expansion

Reference-anchored fusion reaches 86.03 ± 1.40 base accuracy and 66.90 ± 4.23 novel accuracy (Table S1). Its base accuracy is 7.71 percentage points above text-only classification and 3.56 points above text–prototype fusion, providing a retention-oriented operating point for class expansion.

Text–prototype fusion uses visual evidence for every class and reaches

Table S2: Five-shot Oxford-IIIT Pets expansion from 15 to 30 breeds under long-tailed base supervision. Values are macro accuracy percentages, mean ± sample standard deviation across three distinct breed partitions. Grids and the base-only calibration rule are transferred from the CIFAR experiment without dataset-specific search expansion.

Readout	Base	Novel	All	Harmonic
CLIP text	92.87 ± 3.37	94.05 ± 1.35	93.46 ± 1.59	93.43 ± 1.63
Visual prototypes	92.08 ± 2.61	68.30 ± 4.62	80.19 ± 2.33	78.35 ± 2.97
Text + prototypes	94.77 ± 2.91	93.91 ± 1.55	94.34 ± 1.43	94.31 ± 1.44
Tip-Adapter (tuned)	93.56 ± 2.45	94.07 ± 1.18	93.81 ± 1.27	93.80 ± 1.28
Raw evidence fusion	90.13 ± 8.01	91.18 ± 4.27	90.66 ± 6.13	90.63 ± 6.18
Candidate-normalized table	96.85 ± 1.50	86.49 ± 7.64	91.67 ± 4.56	91.29 ± 4.99
Reference-anchored table	97.18 ± 2.25	87.79 ± 6.18	92.49 ± 2.83	92.14 ± 3.23

80.00 ± 0.56 all-class accuracy and 77.53 ± 0.47 novel accuracy. Candidate-normalized fusion reaches 85.36 ± 0.66 base and 70.83 ± 4.33 novel accuracy. These points characterize the recognition trade-off accompanying the reference-anchored score system.

Raw fusion selects the text-only point in all five-shot CIFAR partitions. Both normalization rules activate learned evidence in their selected solutions, demonstrating the role of scale alignment when combining classifier logits with cosine similarities.

Appendix A.3. Fine-grained expansion on Oxford-IIIT Pets

On fine-grained Pets expansion, reference anchoring reaches 97.18 ± 2.25 base and 87.79 ± 6.18 novel accuracy (Table S2). Base accuracy is 2.41 points above text–prototype fusion. Mean base/novel accuracy also increases from 96.85%/86.49% with candidate normalization to 97.18%/87.79% with reference anchoring.

Text–prototype fusion reaches 94.34 ± 1.43 all-class macro accuracy; the reference-anchored table reaches 92.49 ± 2.83 with greater retention of base-breed discrimination. Both datasets use the same search grids and base-only

Table S3: Additional five-shot controls with the same frozen encoder, support images, and base-only calibration. Values are mean ± sample standard deviation across three partitions. The random-projection readout is a controlled frozen RanPAC-style adaptation described in the text.

CIFAR-100

Readout	Base	Novel	All
Text + group offset	78.93 ± 2.75	75.56 ± 4.40	77.24 ± 0.89
Anchored text + prototypes	82.40 ± 2.02	76.19 ± 3.18	79.30 ± 0.59
Frozen RP-ridge (2000)	70.18 ± 0.69	52.52 ± 4.27	61.35 ± 1.83

Oxford-IIIT Pets

Readout	Base	Novel	All
Text + group offset	95.29 ± 3.80	89.85 ± 4.86	92.57 ± 2.31
Anchored text + prototypes	94.58 ± 3.22	90.38 ± 5.41	92.48 ± 1.10
Frozen RP-ridge (2000)	93.49 ± 3.32	73.70 ± 5.13	83.60 ± 1.13

calibration rule.

Appendix A.4. Calibration controls and analytic incremental readouts

Offset-calibrated controls share the same pseudo-novel episode (Table S3). Text + group offset uses base-referenced text scores and a novel offset. Anchored text + prototypes uses fixed base references for both branches and visual evidence for every class. Harmonic selection searches offsets from -4 to 4 in steps of 0.25 and prototype weights $\{0, 0.25, 0.5, 1, 2, 4\}$. Fixed-reference text–prototype fusion also satisfies Proposition 1; the score invariant applies across availability masks.

Anchored text–prototype fusion attains 82.40% base and 76.19% novel accuracy on CIFAR, and 94.58%/90.38% on Pets. Offset-calibrated text attains 78.92%/75.56% and 95.29%/89.85%, respectively. Together with the heterogeneous table, these controls separate group calibration, evidence configuration, and the shared fixed-reference property.

The analytic baseline follows the random-projection phase of RanPAC [1]. A fixed Gaussian matrix maps CLIP features to 2000 ReLU coordinates, and $(G + \lambda I)W = Q$ accumulates Gram matrices and class cross-products from base fitting data and novel support. Base validation macro accuracy selects $\lambda \in \{10^{-8}, \ldots, 10^{8}\}$; the pseudo episode selects a novel offset. This frozen adaptation uses base validation in place of the original within-task

Table S4: Validation retention-targeted calibration at five shots. The pseudo-base target is 95% of the closed reference-table pseudo-base accuracy. Among feasible grid points, selection maximizes pseudo-novel accuracy; infeasible partitions use the point with highest pseudo-base accuracy, then pseudo-novel accuracy. Feasible counts refer to validation partitions. Test values are percentages, mean ± sample standard deviation; test retention is not constrained.

Data	Readout	Base	Novel	All	Feasible
CIFAR	Text + group offset	81.47 ± 1.13	71.49 ± 5.03	76.48 ± 1.99	1/3
CIFAR	Anchored text + prototypes	82.84 ± 1.50	77.11 ± 4.46	79.98 ± 1.61	3/3
CIFAR	Candidate-normalized table	84.69 ± 1.92	72.45 ± 4.14	78.57 ± 1.54	3/3
CIFAR	Reference-anchored table	84.56 ± 1.31	73.88 ± 5.01	79.22 ± 2.21	3/3
Pets	Text + group offset	95.29 ± 3.80	89.85 ± 4.86	92.57 ± 2.31	3/3
Pets	Anchored text + prototypes	94.03 ± 3.37	90.96 ± 5.91	92.50 ± 1.27	3/3
Pets	Candidate-normalized table	97.25 ± 1.78	85.26 ± 7.25	91.26 ± 4.40	3/3
Pets	Reference-anchored table	97.18 ± 2.25	87.79 ± 6.18	92.49 ± 2.83	3/3

MSE selection. It attains 70.18%/52.53% base/novel accuracy on CIFAR and 93.49%/73.70% on Pets. Novel statistics refit the analytic readout; candidate-table expansion appends rows to its fixed readout.

Under the common 95% validation retention target, reference anchoring reaches 84.56% base and 73.87% novel test accuracy on CIFAR, compared with 82.84%/77.11% for anchored text–prototype fusion (Table S4). Both are feasible on all three validation partitions. On Pets, the corresponding test points are 97.18%/87.79% and 94.03%/90.96%. Text alone reaches the CIFAR validation target in one partition. Targets constrain validation selection; the table reports the resulting test operating points. Complete 95% and 99% results accompany the experimental records.

Appendix A.5. Vocabulary growth and score preservation

The vocabulary test appends nested sets of 10, 20, and 40 novel classes to the 40 CIFAR base classes, using sorted novel IDs and fixed five-shot gates. At each stage, we compare existing scores with the original base decision.

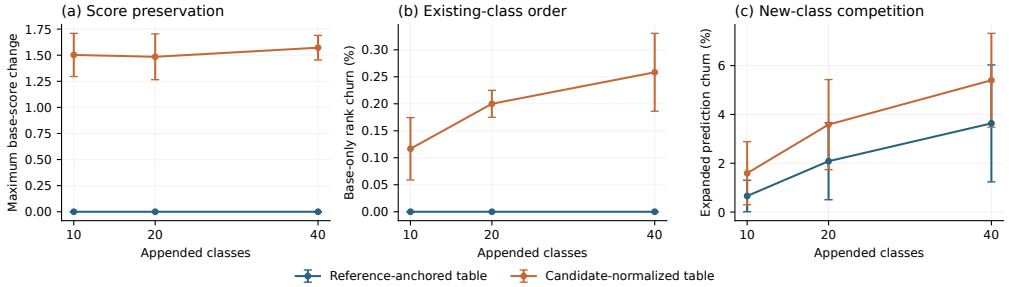


Figure S1: Vocabulary growth with five-shot novel prototypes. Each curve averages three true class partitions; error bars show sample standard deviation. Reference-anchored fusion preserves existing scores and base-only rankings. Expanded predictions can still change when a new candidate wins. Maximum score drift is measured across all test queries and existing base classes.

Reference anchoring gives zero measured base-score drift and zero base-only ranking churn across all nine CIFAR partition and vocabulary settings (Fig. S1(a,b)). Candidate normalization changes existing scores by up to approximately 1.29–1.74 normalized units and changes a fraction of base-only predictions. The Pets test appends 3, 7, and 15 breeds and likewise records zero anchored score drift and ranking churn in all nine conditions.

At 40 appended classes, expanded predictions change on $3.63 \pm 2.40\%$ of base images, including $1.84 \pm 1.54\%$ correct-to-wrong changes (Fig. S1(c)). Wrong-to-correct changes are zero, as specified by Corollary 1. The preserved base scores isolate these changes as competition from appended rows.

Five-session expansion adds eight CIFAR classes or three Pets breeds per session, with two random arrival orders per partition. Across 30 session records per dataset, comparisons of all previously admitted rows, including novel rows, give maximum floating-point score differences of 2.85×10^{-5} on CIFAR and 3.58×10^{-6} on Pets. Both arrival orders produce identical final predictions in every partition under class-ID tie breaking.

Table S5: Reference-anchored expansion across novel support budgets. Accuracy percentages are aggregated across partitions after averaging support subsets within each partition.

Shots	Base	Novel	All	Harmonic
1	84.57 ± 0.88	70.86 ± 5.41	77.72 ± 2.27	77.00 ± 2.85
2	85.65 ± 1.45	70.05 ± 1.13	77.85 ± 0.67	77.04 ± 0.67
5	86.03 ± 1.40	66.90 ± 4.23	76.46 ± 1.43	75.19 ± 2.14

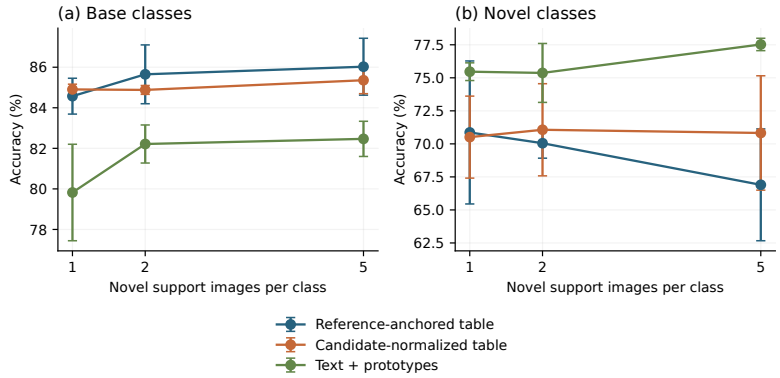


Figure S2: Base and novel accuracy at one, two, and five support images per novel class. The same fixed-reference scoring rule is used at every budget, with coefficients selected on the corresponding base-only pseudo episode. Error bars represent variation between class partitions.

Appendix A.6. Support budget and evidence availability

At one, two, and five shots, base accuracy remains between 84.57% and 86.03%, while novel accuracy ranges from 66.90% to 70.86% (Table S5; Fig. S2). Calibration includes zero branch coefficients, allowing the selected mixture to adapt to the support budget. Each budget defines a separately calibrated operating point: support determines the prototype evidence, and fusion determines its competition with learned base discrimination.

Table S6: Evidence removal at the selected five-shot reference-anchored operating point, with all remaining coefficients fixed. Accuracy percentages are mean ± sample standard deviation over three class partitions.

Readout	Base	Novel	All	Harmonic
Reference-anchored table	86.03 ± 1.40	66.90 ± 4.23	76.46 ± 1.43	75.19 ± 2.14
Without learned evidence	58.10 ± 32.21	80.72 ± 4.47	69.41 ± 13.87	62.85 ± 25.08
Without novel prototypes	86.42 ± 1.71	47.17 ± 30.85	66.79 ± 14.62	56.15 ± 30.05
Without offset	86.14 ± 1.69	65.58 ± 7.35	75.86 ± 2.83	74.25 ± 4.08

Appendix A.7. Mechanism ablation and operating trade-off

Removing learned evidence, novel prototypes, or the offset changes a different part of the selected decision rule (Table S6). With the remaining coefficients fixed, these interventions identify how each source contributes to the five-shot operating point. The independently calibrated text and text–prototype readouts provide the corresponding reduced-model comparisons.

Figure S3 sweeps d from -2 to 2 in steps of 0.5 with a and u fixed. Increasing d shifts the margin between the best base and novel candidates while preserving within-group rankings (Eq. (5)). The resulting curve identifies how the offset controls admission of new categories; a validation retention target selects the operating point.

Appendix A.8. Readout efficiency

Candidate scoring from cached 768-dimensional features takes approximately 1.76, 0.75, and 0.78 ms per batch for 50, 60, and 80 candidates. Measurements use batches of 256 on an NVIDIA RTX 4070 Laptop GPU, five warm-up iterations, and 50 timed iterations with GPU synchronization. They include reference statistics and fusion and exclude image/text encoding. At these vocabulary sizes, launch and reduction overheads dominate the measured latency.

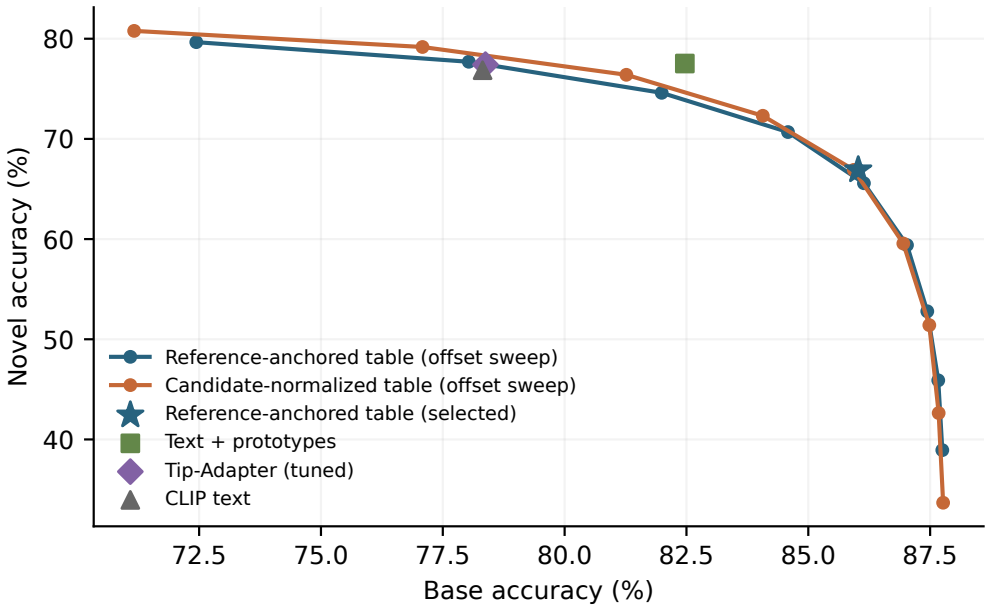


Figure S3: Base–novel operating curves for five-shot expansion. The offset is swept with the other selected coefficients fixed; these test curves are descriptive analyses and are not used to select the reported operating points. Symbols identify base-only calibrated points and matched baselines.

Appendix B. Validation-selected retention policy

Table S7 applies the common validation target after five arrivals. Raw records include all five target fractions and every arrival session. Infeasible targets are reported without substituting a zero novel accuracy. Comparisons with different feasible counts refer to different partition sets.

Table S7: Final-session test accuracies after the admission threshold is selected using base validation only. Budgets use the same closed-base validation reference for all methods within a partition. Values are percentages, mean ± sample standard deviation over feasible partitions; counts identify those partitions. Test accuracies are the outcomes of this operational policy and are not matched by test labels.

Data	Budget	Readout	Base	Novel	Feasible
CIFAR	90%	Retained classifier	78.33 ± 3.89	80.13 ± 2.25	3/3
CIFAR	90%	Classifier removed (fixed gates)	79.07 ± 3.94	70.79 ± 13.81	3/3
CIFAR	90%	No classifier (recalibrated)	78.73 ± 3.21	75.45 ± 5.43	3/3
CIFAR	90%	Anchored text + prototypes	80.18 ± 0.60	79.45 ± 0.76	3/3
CIFAR	95%	Retained classifier	82.99 ± 2.29	75.71 ± 3.14	3/3
CIFAR	95%	Classifier removed (fixed gates)	81.95 ± 0.04	71.54 ± 3.87	2/3
CIFAR	95%	No classifier (recalibrated)	81.85 ± 0.18	71.70 ± 4.10	2/3
CIFAR	95%	Anchored text + prototypes	81.95 ± 0.64	77.48 ± 1.52	2/3
CIFAR	99%	Retained classifier	86.48 ± 1.65	62.83 ± 10.11	3/3
CIFAR	99%	Classifier removed (fixed gates)	–	–	0/3
CIFAR	99%	No classifier (recalibrated)	–	–	0/3
CIFAR	99%	Anchored text + prototypes	86.85	53.85	1/3
Pets	90%	Retained classifier	84.87 ± 4.59	96.53 ± 0.78	3/3
Pets	90%	Classifier removed (fixed gates)	84.11 ± 2.76	96.44 ± 0.64	3/3
Pets	90%	No classifier (recalibrated)	82.00 ± 0.20	96.38 ± 0.52	3/3
Pets	90%	Anchored text + prototypes	85.69 ± 4.62	95.87 ± 1.57	3/3
Pets	95%	Retained classifier	90.29 ± 2.90	95.68 ± 0.83	3/3
Pets	95%	Classifier removed (fixed gates)	90.11 ± 1.17	95.71 ± 0.11	2/3
Pets	95%	No classifier (recalibrated)	89.88 ± 0.85	95.71 ± 0.11	2/3
Pets	95%	Anchored text + prototypes	89.37 ± 4.15	95.63 ± 1.55	3/3
Pets	99%	Retained classifier	96.43 ± 2.15	88.67 ± 7.89	3/3
Pets	99%	Classifier removed (fixed gates)	–	–	0/3
Pets	99%	No classifier (recalibrated)	–	–	0/3
Pets	99%	Anchored text + prototypes	97.70 ± 1.11	84.19 ± 12.46	2/3

Appendix C. Reproduction details

The expansion experiment is implemented in `src/experiments/fcs_expansion.py`. It can be run from any working directory. The manifest records the code hash, frozen-cache hash, image-pool hashes, class assignments, pseudo-class assignments, support indices, regularization choices, and selection grids. The frozen support features are computed with the same deterministic interpolation used for the frozen training cache. The cached image embeddings

are checked numerically before scoring.

Every novel support image comes from the five-image-per-class reserve created before long-tail sampling. This supplies a consistent support budget even for tail classes whose fitting pool contains fewer than five images. At five shots the three subset seeds select the same images in different orders, so the reported standard deviation is calculated across the three distinct class partitions. One- and two-shot results average the three subsets within each partition before computing the partition-level variation.

The main result JSON contains all compared readouts, including evidence-removal ablations. The vocabulary records contain the maximum score drift and both restricted and expanded prediction churn. The operating-curve records contain every swept offset, with no selection based on test outcomes. The efficiency records specify batch size and measured vocabulary size. The manuscript tables are generated directly from these files, and the summary CSV exposes both means and sample standard deviations.

The semantic prompts are "a photo of a {}.", "a bad photo of a {}.", "a cropped photo of a {}.", "a close-up photo of a {}.", "a bright photo of a {}.", "a good photo of a {}.", "a photo of the {}.", and a second occurrence of "a photo of a {}.". The repeated prompt follows the existing cached anchor construction and is retained consistently for all methods.

Frozen CLIP features are the sole image representation in the expansion experiments. Six fitting images are independently re-embedded and checked for cosine agreement above 0.999 with the cached features. Image hashes verify that fitting, validation, and reserved-support pools are disjoint. The fixed base text block is computed separately from novel text columns; sequential

prototype products can exhibit the reported floating-point residuals when matrix widths change.

Appendix D. Interpreting group competition

Let $M_B(x) = \max_{b \in \mathcal{B}}[\widehat{q}_T(x, b) + a\widehat{q}_L(x, b)]$ and $M_N(x) = \max_{n \in \mathcal{N}}[\widehat{q}_T(x, n) + u\widehat{q}_P(x, n)]$. The classifier selects a novel class precisely when $M_N(x) + d > M_B(x)$, subject to the fixed tie-breaking rule. Hence, the offset is a threshold on the margin $M_B - M_N$. Increasing d cannot change the winning base class or the winning novel class when all other quantities are fixed. It changes which group supplies the final winner. This explains the direction of the operating curves without assuming a calibrated probability interpretation for the fused score.

For nested candidate sets and fixed references, adding novel rows can increase M_N but cannot change M_B. An existing-class example may therefore cross the admission boundary as the vocabulary grows, while its base scores remain identical. This distinguishes the deployment behavior measured by expanded prediction churn from the normalization behavior measured by base-only ranking churn.

Appendix D.1. Exact empirical retention envelope

For a fixed readout, let b_i and n_i be the canonical winning base and novel class for query i, and let $\Delta_i = M_B(x_i) - M_N(x_i)$. Define $g_i(d) = \mathbf{1}[d > \Delta_i \text{ or } (d = \Delta_i \text{ and } n_i < b_i)]$. With the class-macro weights w_i, the two group

accuracies are

$$A_B(d) = \sum_{i:y_i \in \mathcal{B}} w_i \mathbf{1}[b_i = y_i](1 - g_i(d)), \tag{D.1}$$

$$A_N(d) = \sum_{i:y_i \in \mathcal{N}} w_i \mathbf{1}[n_i = y_i] g_i(d). \tag{D.2}$$

Thus A_B is nonincreasing and A_N is nondecreasing in d. For the stored finite-precision scores, an exact search need only consider each observed margin and its immediately preceding representable offset, together with the all-base and all-novel endpoints. Between adjacent margins every group decision is fixed. If τ exceeds $\sum_{i:y_i \in \mathcal{B}} w_i \mathbf{1}[b_i = y_i]$, the constraint is infeasible. Otherwise, sorting the margins of correctly classified base queries in descending order and accumulating their macro weights locates the boundary at which retained base mass first reaches τ. The tie rule selects that boundary or its predecessor. This is the largest feasible offset in the implementation's offset ordering. Monotonicity of A_N proves that it maximizes novel accuracy under the empirical constraint. The argument applies identically to base-validation selection; that policy's test outcome is evaluated on separate queries.

The envelope separates the within-base feasible ceiling, the admitted group threshold and the distribution of correctly classified novel-query margins. It therefore measures the joint operating value of a learned readout and its competition with arriving evidence, beyond the statement that stored scores are preserved.

Appendix E. Additional background and score-preservation proof

Appendix E.1. Long-tailed recognition and calibration

Long-tailed recognition addresses the bias induced by unequal class frequencies. Margin adjustment allocates different classification margins to

frequent and rare classes [2]; class-balanced weighting uses effective sample counts [3]; and prior-aware objectives modify the relationship between logits and class probabilities [4, 5]. Representation–classifier decoupling provides a complementary route: a representation is learned first and a balanced classifier is then fitted to that representation [6]. These developments motivate our use of a balanced base classifier as one source of evidence.

Foundation models change the representation-learning budget and the role of task-specific adaptation. LIFT studies lightweight adaptation of pre-trained models for long-tail learning [7]. Low-rank updates provide an efficient mechanism for adapting large parameter matrices [8], including transformer backbones [9]. Our candidate interface can accept weights obtained through these routes. The expansion experiments here fix the encoder throughout and isolate the effect of the readout.

Long-tailed calibration studies confidence under class imbalance [10], while robust long-tailed learning addresses its interaction with label noise [11]. We study the reference population used to standardize evidence as candidate classes change. Class-balanced validation and separate base/novel metrics expose how the resulting score system allocates discrimination.

Appendix E.2. Semantic and visual evidence for few-shot recognition

CLIP constructs classifiers from natural-language supervision and supports classification by changing the text candidates [12]. CoOp learns prompt context for target tasks, while CoCoOp introduces conditional prompting to improve transfer across categories [13, 14]. These methods concern the construction or adaptation of semantic evidence. Our table keeps semantic anchors available to every class and combines them with evidence that may

exist for only a subset of the vocabulary.

Prototype methods summarize class examples in an embedding space [15]. SimpleShot studies feature normalization for nearest-neighbor classification [16], and Tip-Adapter augments CLIP with a support-feature cache [17]. Few-shot transfer under distribution shift further motivates separating shared and task-specific knowledge [18]. Our comparisons use a shared encoder and novel support budget to examine evidence availability and its normalization reference.

Appendix E.3. Proof of the main score-preservation proposition

Proof. For an existing candidate c, each raw score $q_e(x, c)$ and mask m_{ec} is fixed. The means and variances in Eq. (2) are functions of x and $\mathcal{R}_e$, and neither depends on the appended rows. Therefore each normalized score in Eq. (3) is unchanged. The coefficients and offset of c are fixed, so their sum is unchanged. Subtracting the unchanged scores of any two existing rows gives an unchanged difference. A fixed tie-breaking rule preserves the maximizing existing row. □

The proposition covers both base rows and previously appended novel rows with fixed prototypes. The implementation computes the fixed base text block separately from novel columns to keep its numerical kernel independent of the appended width.

Corollary 1. *If every existing row has a preserved score, a changed final prediction after expansion must select an appended row. An example whose previous prediction was incorrect cannot become correct after adding classes when its true class belongs to the existing vocabulary.*

Proof. The best existing row is unchanged by Proposition 1. The expanded maximizer is either this row or a newly appended row. For an example labeled by an existing class, selecting a new row cannot correct a previous error, and keeping the existing maximizer leaves the previous error unchanged. □

Appendix F. Recent readout comparison and ImageNet-R extension

The extension protocol was recorded before ImageNet-R evaluation. Class partitions use seeds 0, 1 and 2, with exact frequency-stratified allocation of 100 base and 100 novel classes. Within each base group, seed $900 + p$ assigns 50 pseudo-base and 50 pseudo-novel classes. Base-only validation selects ridge regularization, branch mixtures and readout hyperparameters. No ImageNet-R test result selects these quantities. The retained table, removal controls and dense prototype replacement use the same grids as the CIFAR/Pets experiment.

FeNeC [19] is independently implemented from the paper's equations and checked against author-code commit `1575ea019fd583149c47c068f7f5ccc102ea4580`. Both variants use identity Tukey transformation for signed CLIP features and Euclidean k-means with a maximum of 100 iterations. The representative grid is $K \in \{1, 5, 25\}$ and the neighbor grid is $\{1, 3, 5\}$. FeNeC-M estimates one covariance per class, applies the author's ViT single-shrinkage and correlation normalization with $(\gamma_1, \gamma_2) \in \{(1, 1), (6.12, 8.10), (9.98, 0)\}$, then uses inverse squared-Mahalanobis-distance votes. FeNeC-E uses inverse unsquared-Euclidean-distance votes. Their native predictions are preserved when votes are converted to log votes, with a floor of 10^{-30}. The calibrated forms select a novel offset from $[-4, 4]$ in steps of 0.25 on the pseudo episode. Harmonic macro accuracy chooses the configuration.

The five-shot protocol caps a class's representative count at its sample count. A class with one fitting sample uses identity covariance, avoiding an undefined sample covariance. These are explicit sparse-data adaptations. A

per-class independent summary can be cached before its arrival; the voting neighborhood always contains only currently admitted classes. Native and calibrated forms share each current-vocabulary score matrix, so their complete matched-retention frontiers coincide, while fixed-offset trajectories can differ. The final grids are documented with the code. FeNeC frontiers use its sparse global-neighbor votes, with unvoted classes at the common floor; feasibility refers to this output interface.

All ImageNet-R images are content-hashed before pool assignment. Fifty-nine byte-identical duplicates are removed globally. No retained content hash appears in two fitting, validation, support or test pools. The source archive has SHA-256 `18c6bf493b39a0d975d48e587437f562caab9c52ae6327dcfa9dd8eb54aa1b52`. The class-name mapping, image paths, achieved fitting counts, source hashes, calibrated settings and prediction margins accompany the code and records. These are task-specific splits of ImageNet-R, whose source release is a rendition benchmark [20].

Table S8: ImageNet-R matched-retention recognition at three budgets after five arrivals. Values are percentages, mean ± partition-level sample standard deviation over feasible partitions. Counts specify available partition sets.

Budget	Readout	Base	Novel	Feasible
90%	Retained classifier	81.83 ± 0.29	86.80 ± 0.27	3/3
90%	Classifier removed (fixed gates)	81.81 ± 0.31	85.53 ± 0.49	3/3
90%	No classifier (recalibrated)	81.81 ± 0.31	85.34 ± 0.46	3/3
90%	Anchored text + prototypes	81.83 ± 0.32	86.59 ± 0.58	3/3
90%	Tip-Adapter sum + offset	81.82 ± 0.30	83.74 ± 0.25	3/3
90%	Tip-Adapter mean + offset	81.83 ± 0.30	85.30 ± 0.23	3/3
95%	Retained classifier	86.38 ± 0.32	84.48 ± 0.72	3/3
95%	Classifier removed (fixed gates)	86.37 ± 0.31	72.62 ± 12.51	3/3
95%	No classifier (recalibrated)	86.37 ± 0.31	72.69 ± 12.57	3/3
95%	Anchored text + prototypes	86.37 ± 0.32	83.32 ± 1.40	3/3
95%	Tip-Adapter sum + offset	86.41 ± 0.30	70.93 ± 10.32	3/3
95%	Tip-Adapter mean + offset	86.38 ± 0.33	81.58 ± 1.72	3/3
99%	Retained classifier	90.02 ± 0.34	73.69 ± 1.50	3/3
99%	Classifier removed (fixed gates)	–	–	0/3
99%	No classifier (recalibrated)	–	–	0/3
99%	Anchored text + prototypes	89.98 ± 0.47	47.62 ± 13.36	2/3
99%	Tip-Adapter sum + offset	–	–	0/3
99%	Tip-Adapter mean + offset	–	–	0/3

Table S9: ImageNet-R test outcomes with the 95% base-validation-only threshold policy. Validation feasibility is reported separately from achieved test accuracy. Tip-Adapter uses its base-only selected calibrated cache settings, followed by validation-only threshold selection.

Readout	Base test	Novel test	Validation feasible
Retained classifier	86.01 ± 0.59	84.73 ± 0.71	3/3
Classifier removed (fixed gates)	86.38 ± 0.18	57.35 ± 38.51	3/3
No classifier (recalibrated)	86.48 ± 0.20	57.10 ± 38.30	3/3
Anchored text + prototypes	86.36 ± 0.47	83.26 ± 1.49	3/3
Tip-Adapter sum + offset	86.10 ± 0.46	74.08 ± 5.48	3/3
Tip-Adapter mean + offset	86.68 ± 0.23	80.75 ± 2.09	3/3

Table S10: FeNeC matched-test-retention operating points at strict budgets. Values are percentages, mean ± sample standard deviation over feasible partitions; missing values have zero feasible partitions. Native and offset-calibrated forms share a score matrix and hence the same complete offset-sweep frontier.

Data	Budget	Readout	Base	Novel	Feasible
CIFAR-100	95%	FeNeC-M + offset	–	–	0/3
CIFAR-100	95%	FeNeC-E + offset	–	–	0/3
CIFAR-100	99%	FeNeC-M + offset	–	–	0/3
CIFAR-100	99%	FeNeC-E + offset	–	–	0/3
Pets	95%	FeNeC-M + offset	–	–	0/3
Pets	95%	FeNeC-E + offset	94.23	69.46	1/3
Pets	99%	FeNeC-M + offset	–	–	0/3
Pets	99%	FeNeC-E + offset	–	–	0/3
ImageNet-R	95%	FeNeC-M + offset	–	–	0/3
ImageNet-R	95%	FeNeC-E + offset	–	–	0/3
ImageNet-R	99%	FeNeC-M + offset	–	–	0/3
ImageNet-R	99%	FeNeC-E + offset	–	–	0/3

Appendix G. Dense caches, repeated calibration and measured readout cost

Appendix G.1. Dense semantic-cache protocol

The training-free Tip-Adapter score follows the ECCV paper and the author implementation [17]: cosine similarity to labeled cache keys is transformed by $\exp[-\beta(1 - \text{similarity})]$, summed into classes and added to 100 times text similarity with weight α. The author code is pinned at commit `d0e2d6f8c5feb8b6ce937b757810761f7155d4d5`. All permitted base fitting images and five reserved novel support images supply the keys. The sum variant retains the published aggregation. A separate class-mean variant divides each class's retrieval score by its key count, isolating the effect of unequal long-tailed cache sizes. Both reuse the same single-view frozen features and text anchors as the tables. This comparison transfers the published readout under controlled feature access; the original augmented-cache extraction and trainable Tip-Adapter-F are distinct settings.

On each original base-only pseudo episode, we search $\beta \in \{0.1, 1, 2, 5, 10, 20, 50\}$ and $\alpha \in \{0.1, 1, 5, 10, 20, 50, 100, 200\}$. The native form selects both parameters with zero offset. The calibrated form jointly selects these parameters and an offset from $[-40, 40]$ in steps of 2.5, using macro harmonic accuracy. This 33-point offset grid reflects the raw 100-scaled CLIP score system. No actual novel query or test label is used in selection. Both forms are reported because their selected cache settings can differ. All matched-retention targets are the original retained readout's partition-specific absolute targets.

Table S11: Complete five-shot fixed-gate readout comparison. Values are percentages, mean ± sample standard deviation over three class partitions after averaging the two arrival orders. The paired gain is retained minus comparator average incremental accuracy; a negative gain favors the comparator. Native and calibrated cache forms are selected separately.

Data	Readout	Incremental avg.	Final	Paired avg. gain
CIFAR-100	Retained classifier	82.02 ± 0.84	79.31 ± 1.14	0.00 ± 0.00
CIFAR-100	Classifier removed (fixed gates)	71.40 ± 14.78	68.03 ± 15.63	10.62 ± 15.22
CIFAR-100	No classifier (recalibrated)	79.84 ± 0.82	77.56 ± 1.02	2.18 ± 0.24
CIFAR-100	Anchored text + prototypes	81.56 ± 0.52	79.30 ± 0.59	0.46 ± 0.44
CIFAR-100	Tip-Adapter sum + offset	79.57 ± 0.41	77.21 ± 0.38	2.45 ± 0.63
CIFAR-100	Tip-Adapter mean + offset	81.74 ± 0.93	79.40 ± 1.11	0.28 ± 1.07
CIFAR-100	FeNeC-M + offset	66.60 ± 1.34	63.65 ± 1.55	15.42 ± 0.49
CIFAR-100	FeNeC-E + offset	72.63 ± 1.34	69.00 ± 1.97	9.39 ± 0.68
CIFAR-100	FeNeC-M (native)	59.11 ± 0.85	53.58 ± 1.27	22.91 ± 0.31
CIFAR-100	FeNeC-E (native)	72.63 ± 1.34	69.00 ± 1.97	9.39 ± 0.68
CIFAR-100	Tip-Adapter sum (native)	79.57 ± 0.41	77.21 ± 0.38	2.45 ± 0.63
CIFAR-100	Tip-Adapter mean (native)	81.74 ± 0.93	79.40 ± 1.11	0.28 ± 1.07
Pets	Retained classifier	94.09 ± 2.37	91.77 ± 3.16	0.00 ± 0.00
Pets	Classifier removed (fixed gates)	77.17 ± 26.01	76.85 ± 23.25	16.92 ± 28.33
Pets	No classifier (recalibrated)	94.20 ± 2.05	92.57 ± 2.31	-0.11 ± 0.56
Pets	Anchored text + prototypes	94.35 ± 0.60	92.48 ± 1.10	-0.26 ± 1.82
Pets	Tip-Adapter sum + offset	93.38 ± 2.32	91.50 ± 2.78	0.71 ± 1.21
Pets	Tip-Adapter mean + offset	94.47 ± 2.38	92.88 ± 2.51	-0.37 ± 1.61
Pets	FeNeC-M + offset	80.34 ± 1.29	75.66 ± 1.72	13.76 ± 1.61
Pets	FeNeC-E + offset	84.43 ± 2.37	79.82 ± 2.35	9.67 ± 1.50
Pets	FeNeC-M (native)	73.66 ± 4.73	66.74 ± 6.13	20.44 ± 2.69
Pets	FeNeC-E (native)	84.43 ± 2.37	79.82 ± 2.35	9.67 ± 1.50
Pets	Tip-Adapter sum (native)	94.35 ± 1.08	92.72 ± 1.65	-0.25 ± 1.32
Pets	Tip-Adapter mean (native)	95.15 ± 1.28	93.74 ± 1.29	-1.06 ± 1.33
ImageNet-R	Retained classifier	87.23 ± 0.21	85.40 ± 0.36	0.00 ± 0.00
ImageNet-R	Classifier removed (fixed gates)	77.13 ± 6.75	74.64 ± 7.19	10.09 ± 6.55
ImageNet-R	No classifier (recalibrated)	85.01 ± 0.57	83.65 ± 0.70	2.22 ± 0.40
ImageNet-R	Anchored text + prototypes	86.36 ± 0.24	84.67 ± 0.39	0.87 ± 0.29
ImageNet-R	Tip-Adapter sum + offset	84.62 ± 0.08	82.94 ± 0.07	2.61 ± 0.21
ImageNet-R	Tip-Adapter mean + offset	85.75 ± 0.30	84.19 ± 0.35	1.48 ± 0.35
ImageNet-R	FeNeC-M + offset	59.12 ± 0.54	55.79 ± 0.63	28.11 ± 0.46
ImageNet-R	FeNeC-E + offset	69.66 ± 0.97	66.09 ± 0.62	17.56 ± 0.80
ImageNet-R	FeNeC-M (native)	54.57 ± 0.47	49.71 ± 0.32	32.65 ± 0.37
ImageNet-R	FeNeC-E (native)	69.66 ± 0.97	66.09 ± 0.62	17.56 ± 0.80
ImageNet-R	Tip-Adapter sum (native)	84.62 ± 0.08	82.94 ± 0.07	2.61 ± 0.21
ImageNet-R	Tip-Adapter mean (native)	85.75 ± 0.30	84.19 ± 0.35	1.48 ± 0.35

Appendix G.2. Repeated calibration and support budgets

The 81 configurations span three datasets, three original class partitions, $k \in \{1, 2, 5\}$ and three joint calibration/support trials. Trial zero reuses the original pseudo split. The other trials select $\lfloor |\mathcal{B}|/2 \rfloor$ pseudo-base classes with seeds $1701 + p$ and $2701 + p$. Pseudo classifiers select ridge regularization on pseudo-base validation. Support-index seeds $7, 8, 9$ select one- and two-shot subsets from each five-image reserve. At five shots, the support set is identical across trials: variation comes from the pseudo split. These are correlated trials of the same underlying class partitions. We average the orders and trials within each partition and report the mean and sample standard deviation of the three partition means.

Each configuration independently calibrates the retained, recalibrated-removal, dense-prototype and dynamic-reference readouts on the same base-only episode. Fixed-gate removal preserves the retained mixture's remaining parameters. Dynamic normalization uses the current text candidates and current novel prototypes as references, with the same learned branch and calibration grid as the retained table. Its own mixture is selected independently and stays fixed across arrivals. All retention targets are shared within each configuration. Feasibility counts refer to nine partition/trial combinations per support budget. When only some trials are feasible, the group accuracies average those trials within each partition before summarizing partitions; paired constrained comparisons use the intersection of feasible trial IDs.

Fixed references have zero restricted pre-offset candidate-ranking changes in all 648 append checks (81 configurations, two orders, four appends). The largest floating-point score differences are 1.40×10^{-4} on CIFAR, 5.36×10^{-6} on

Table S12: Calibration and support robustness. Average incremental and final accuracies use fixed gates. Paired gains are retained minus each independently calibrated comparator, after averaging two arrival orders and three trials within each partition. Values are percentages or percentage points, mean ± sample standard deviation across three partitions.

Data	Shots	Retained avg.	Δ recalibrated	Δ dense	Δ dynamic	Retained final
CIFAR-100	1	79.77 ± 0.61	0.55 ± 1.15	0.08 ± 0.72	0.10 ± 0.74	76.32 ± 1.22
CIFAR-100	2	80.20 ± 0.32	0.84 ± 0.67	-0.76 ± 0.52	0.55 ± 0.84	77.00 ± 0.85
CIFAR-100	5	80.57 ± 0.96	1.22 ± 1.18	-0.79 ± 1.08	0.62 ± 0.59	77.25 ± 1.09
Pets	1	93.94 ± 2.37	0.05 ± 0.67	-0.24 ± 0.48	-0.54 ± 0.11	91.60 ± 3.20
Pets	2	92.64 ± 1.82	-1.45 ± 1.19	-1.62 ± 1.15	-1.43 ± 1.09	90.09 ± 1.87
Pets	5	94.56 ± 1.74	0.78 ± 1.03	0.43 ± 0.80	0.47 ± 1.43	92.75 ± 2.00
ImageNet-R	1	86.16 ± 0.22	1.81 ± 0.33	1.01 ± 0.29	-0.07 ± 0.29	83.83 ± 0.14
ImageNet-R	2	86.58 ± 0.37	2.10 ± 0.28	0.62 ± 0.34	-0.07 ± 0.28	84.43 ± 0.67
ImageNet-R	5	87.16 ± 0.12	2.14 ± 0.37	0.54 ± 0.23	0.15 ± 0.19	85.36 ± 0.25

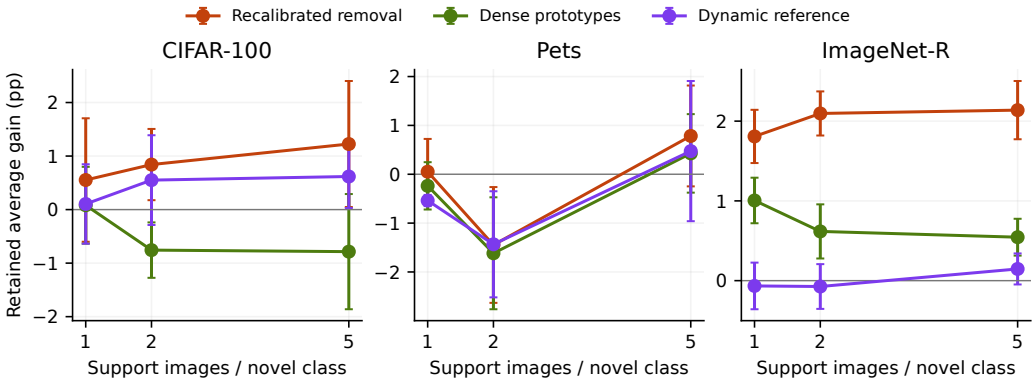


Figure S4: Paired average incremental gains under one-, two- and five-shot support. Error bars are partition-level sample standard deviations after averaging correlated calibration/support trials and arrival orders. The horizontal zero line identifies which readout has higher average accuracy in each setting.

Pets and 2.19×10^{-5} on ImageNet-R. Dynamic references produce maximum score changes of 2.40, 1.78 and 1.30, respectively, and up to 429, 206 and 118 restricted prediction changes per append over the full cached test query pool. These diagnostics compare the same previously active candidate IDs before adding the fixed group offset; they isolate changes induced by reference statistics. Recognition accuracy is reported separately in the fixed-gate and retention-policy tables.

Table S13: CIFAR-100: final-session outcomes in the repeated-calibration study at the 95% budget. Match uses base test labels descriptively; Val. uses base validation only and reports achieved test accuracy. Feasibility is counted over nine correlated partition/trial combinations.

Shots	Readout	Match base	Match novel	Feasible	Val. base	Val. novel	Feasible
1	Retained classifier	83.54 ± 0.81	73.03 ± 3.08	9/9	80.77 ± 2.34	76.30 ± 3.82	9/9
1	No classifier (recalibrated)	83.37 ± 0.57	59.17 ± 3.51	8/9	80.92 ± 1.45	72.28 ± 4.17	9/9
1	Anchored text + prototypes	83.37 ± 0.57	63.65 ± 3.03	8/9	80.38 ± 2.28	73.74 ± 5.16	9/9
1	Dynamic reference	83.54 ± 0.81	73.07 ± 3.81	9/9	81.10 ± 2.54	75.86 ± 4.56	9/9
2	Retained classifier	82.99 ± 0.46	73.84 ± 3.52	9/9	80.65 ± 1.93	76.40 ± 4.33	9/9
2	No classifier (recalibrated)	82.99 ± 0.46	61.20 ± 0.62	9/9	81.14 ± 1.01	71.44 ± 2.35	9/9
2	Anchored text + prototypes	82.99 ± 0.46	71.62 ± 6.74	9/9	80.81 ± 1.65	76.62 ± 4.51	9/9
2	Dynamic reference	82.99 ± 0.46	74.70 ± 3.56	9/9	80.37 ± 1.53	77.66 ± 2.64	9/9
5	Retained classifier	83.83 ± 0.76	75.71 ± 2.77	9/9	82.03 ± 1.38	77.69 ± 1.64	9/9
5	No classifier (recalibrated)	83.55 ± 0.29	62.19 ± 6.29	7/9	81.21 ± 1.18	73.83 ± 6.90	7/9
5	Anchored text + prototypes	83.83 ± 0.76	73.28 ± 3.14	9/9	81.95 ± 0.54	75.45 ± 1.14	8/9
5	Dynamic reference	83.83 ± 0.76	73.83 ± 0.74	9/9	82.44 ± 1.94	76.58 ± 2.18	9/9

Table S14: Oxford-IIIT Pets: final-session outcomes under the same repeated-calibration policies and aggregation as Table S13.

Shots	Readout	Match base	Match novel	Feasible	Val. base	Val. novel	Feasible
1	Retained classifier	92.58 ± 1.83	93.50 ± 0.89	9/9	88.07 ± 1.16	94.84 ± 0.89	9/9
1	No classifier (recalibrated)	92.58 ± 1.85	92.30 ± 2.36	9/9	88.56 ± 1.16	95.45 ± 0.70	6/9
1	Anchored text + prototypes	92.57 ± 1.84	93.03 ± 1.60	9/9	87.59 ± 1.34	94.59 ± 1.55	9/9
1	Dynamic reference	92.56 ± 1.82	93.74 ± 1.02	9/9	87.89 ± 1.50	95.47 ± 0.16	8/9
2	Retained classifier	92.28 ± 2.51	94.42 ± 1.14	9/9	88.05 ± 2.52	95.66 ± 0.49	9/9
2	No classifier (recalibrated)	92.27 ± 2.52	93.97 ± 0.96	9/9	87.04 ± 3.11	95.89 ± 0.27	7/9
2	Anchored text + prototypes	92.27 ± 2.50	94.33 ± 0.58	9/9	87.64 ± 2.55	95.51 ± 0.58	9/9
2	Dynamic reference	92.27 ± 2.50	94.62 ± 1.31	9/9	87.85 ± 2.40	95.77 ± 0.27	9/9
5	Retained classifier	92.40 ± 2.41	94.95 ± 1.39	9/9	88.31 ± 2.40	96.06 ± 0.47	9/9
5	No classifier (recalibrated)	92.40 ± 2.42	94.08 ± 0.92	9/9	87.39 ± 2.71	96.12 ± 0.41	7/9
5	Anchored text + prototypes	92.39 ± 2.39	94.79 ± 0.92	9/9	87.94 ± 2.68	95.99 ± 0.80	9/9
5	Dynamic reference	92.40 ± 2.40	94.84 ± 1.36	9/9	87.44 ± 2.34	96.12 ± 0.27	9/9

Table S15: ImageNet-R: final-session outcomes under the same repeated-calibration policies and aggregation as Table S13.

Shots	Readout	Match base	Match novel	Feasible	Val. base	Val. novel	Feasible
1	Retained classifier	86.24 ± 0.38	81.59 ± 0.50	9/9	85.73 ± 0.71	81.95 ± 0.67	9/9
1	No classifier (recalibrated)	86.25 ± 0.37	70.69 ± 10.24	9/9	86.13 ± 0.37	69.56 ± 13.12	9/9
1	Anchored text + prototypes	86.24 ± 0.36	78.57 ± 0.59	9/9	86.14 ± 0.54	78.97 ± 0.47	9/9
1	Dynamic reference	86.25 ± 0.36	81.78 ± 0.76	9/9	85.87 ± 0.61	82.11 ± 1.07	9/9
2	Retained classifier	86.38 ± 0.34	82.67 ± 1.21	9/9	86.10 ± 0.63	82.86 ± 1.30	9/9
2	No classifier (recalibrated)	86.38 ± 0.33	69.12 ± 14.06	9/9	86.27 ± 0.38	58.73 ± 33.47	9/9
2	Anchored text + prototypes	86.39 ± 0.34	81.53 ± 2.15	9/9	86.40 ± 0.40	81.50 ± 1.68	9/9
2	Dynamic reference	86.38 ± 0.33	82.80 ± 0.65	9/9	86.04 ± 0.67	83.02 ± 0.76	9/9
5	Retained classifier	86.35 ± 0.27	84.48 ± 0.35	9/9	86.08 ± 0.44	84.69 ± 0.46	9/9
5	No classifier (recalibrated)	86.34 ± 0.27	73.89 ± 9.72	9/9	86.36 ± 0.21	63.98 ± 26.37	9/9
5	Anchored text + prototypes	86.34 ± 0.27	83.45 ± 1.62	9/9	86.23 ± 0.52	83.65 ± 1.19	9/9
5	Dynamic reference	86.35 ± 0.29	84.42 ± 0.56	9/9	86.21 ± 0.47	84.50 ± 0.69	9/9

Appendix G.3. Readout cost and restart verification

Table S16 measures 512 cached queries on the RTX 4070 Laptop in float32. Each readout receives five warm-ups and 20 synchronized timing repetitions. Static cache values are constructed before timing, query chunks contain 256 images for retrieval, and unused score branches are skipped. An independently checked implementation verifies score agreement within 10^{-3} for every optimized readout. Persistent tensor bytes count text anchors, learned weights and references/prototypes for the tables, or text anchors, cache keys and the stored dense one-hot/class-mean value matrix for retrieval. These are implementation-level readout costs, excluding the common image encoder, query features and temporary workspaces. They do not represent complete model memory or end-to-end image inference time.

Every calibration and arrival-session result is written atomically. A live pause/resume check preserved a completed session's SHA-256 hash and modification time, then completed all 90 configuration units. The numerical verifier reconstructs group predictions, macro weights, selected thresholds and test metrics from archived sufficient statistics, including the base-validation

Table S16: Measured readout latency and persistent tensor storage for a final-session 512-query batch on the same laptop GPU. MiB denotes 2^{20} bytes. Cache keys number 4,937 on CIFAR, 553 on Pets and 3,830 on ImageNet-R in partition zero.

Data	Readout	Median (ms)	P90 (ms)	Stored tensors (MiB)
CIFAR-100	Retained classifier	0.715	0.763	0.586
CIFAR-100	Dense text + prototypes	0.515	0.542	0.469
CIFAR-100	Tip-Adapter sum	0.838	0.862	16.205
CIFAR-100	Tip-Adapter mean	0.841	0.857	16.205
Pets	Retained classifier	0.880	1.280	0.220
Pets	Dense text + prototypes	0.586	0.680	0.176
Pets	Tip-Adapter sum	0.564	0.673	1.771
Pets	Tip-Adapter mean	0.569	0.732	1.771
ImageNet-R	Retained classifier	0.864	0.918	1.465
ImageNet-R	Dense text + prototypes	0.631	0.790	1.172
ImageNet-R	Tip-Adapter sum	0.759	0.873	14.729
ImageNet-R	Tip-Adapter mean	0.732	0.898	14.729

selection arrays. The trial-zero five-shot table results agree exactly with the prior three-dataset records. Source, protocol and cache hashes, all outcomes and reproduction commands accompany the release.

Appendix H. Validation-query budget for admission thresholds

This experiment varies the old-class query labels used to select the admission threshold, conditional on the original five-shot classifier and fusion calibration. All fitted parameters, cache settings and branch mixtures stay fixed. The evaluation uses the original trial-zero score archives, class partitions and arrival order zero on each dataset. No image encoder or new classifier is fitted.

For each base class with n_c validation images, the nominal fraction f keeps $\max(1, \lfloor f n_c \rfloor)$ images. Fractions $f \in \{0.25, 0.5\}$ use three predeclared subset seeds, with nested class-wise permutations; the full budget uses all images once. Actual query counts are reported because rounding and the one-query

minimum change the realized fractions. Macro weights are recomputed from the selected class counts. The retained readout's closed-base accuracy on that subset defines a shared target $rA^{\text{closed}}_{B,\text{ref,subset}}$ for all six readouts, with $r \in \{0.95, 0.99\}$. The largest feasible offset uses only the selected base-validation margins and labels.

After selection, test queries measure both group accuracies. A test target is met when $A_{B,\text{test}} \geq rA^{\text{closed}}_{B,\text{ref,test}}$; this criterion is evaluated after the offset is fixed and never enters threshold selection. Table S17 separates validation feasibility from test target attainment. Subset repeats are averaged within each class partition before the mean and sample standard deviation over available partition means. The nine reduced-budget cases share three underlying partitions and are not independent repetitions. Small per-class counts quantize the selected threshold, so different target ratios can yield identical predictions and increasing the validation fraction need not change test target attainment monotonically. These empirical policies do not supply a finite-sample guarantee on test retention.

The full-budget policies reproduce the original validation-selected outcomes exactly. Saved predictions reconstruct every offset, macro accuracy and attainment flag across 315 session units and 3,780 metric records. The complete budgets, query fractions and test-retention ratios are in `validation_budget_summary.csv`. This experiment varies threshold-selection queries while holding the initial full-data fusion calibration fixed.

Table S17: Retained readout under different validation-query budgets after the final arrival. Accuracies are percentages, mean ± partition-level sample standard deviation after averaging subset repeats. Queries gives the actual old-class validation count range. Val. feasible counts partition/subset combinations meeting the validation constraint; Test met counts feasible combinations meeting the separate test criterion. The full budget uses one subset per partition.

Data	Nominal budget	Queries	Target	Base test	Novel test	Val. feasible	Test met
CIFAR-100	25%	40	95%	82.46 ± 2.25	75.14 ± 3.50	9/9	4/9
CIFAR-100	25%	40	99%	85.21 ± 2.64	66.99 ± 9.18	9/9	4/9
CIFAR-100	50%	68	95%	81.94 ± 3.23	76.92 ± 3.72	9/9	3/9
CIFAR-100	50%	68	99%	85.85 ± 2.61	64.30 ± 10.71	9/9	5/9
CIFAR-100	100%	161–163	95%	82.99 ± 2.29	75.71 ± 3.14	3/3	1/3
CIFAR-100	100%	161–163	99%	86.48 ± 1.65	62.83 ± 10.11	3/3	2/3
Pets	25%	15	95%	91.77 ± 2.39	92.42 ± 2.64	9/9	5/9
Pets	25%	15	99%	91.77 ± 2.39	92.42 ± 2.64	9/9	4/9
Pets	50%	25	95%	92.02 ± 0.95	94.75 ± 1.23	9/9	5/9
Pets	50%	25	99%	95.05 ± 0.48	90.49 ± 5.09	9/9	5/9
Pets	100%	60	95%	90.29 ± 2.90	95.68 ± 0.83	3/3	0/3
Pets	100%	60	99%	96.43 ± 2.15	88.67 ± 7.89	3/3	2/3
ImageNet-R	25%	696–731	95%	85.85 ± 0.38	84.98 ± 0.38	9/9	2/9
ImageNet-R	25%	696–731	99%	89.79 ± 0.16	74.81 ± 2.33	9/9	5/9
ImageNet-R	50%	1444–1519	95%	85.93 ± 0.65	84.87 ± 0.63	9/9	1/9
ImageNet-R	50%	1444–1519	99%	89.90 ± 0.33	74.24 ± 2.68	9/9	5/9
ImageNet-R	100%	2938–3087	95%	86.01 ± 0.59	84.73 ± 0.71	3/3	0/3
ImageNet-R	100%	2938–3087	99%	89.94 ± 0.25	74.48 ± 1.35	3/3	2/3

Table S18: All readouts at the nominal 25% validation-query budget and 95% target after the final arrival. All classifiers, mixtures and cache settings are held fixed from the original five-shot calibration. Accuracy aggregation and feasibility/attainment counts follow Table S17. Missing metrics correspond to zero feasible subsets.

Data	Readout	Base test	Novel test	Val. feasible	Test met
CIFAR-100	Retained classifier	82.46 ± 2.25	75.14 ± 3.50	9/9	4/9
CIFAR-100	Fixed-gate removal	79.39 ± 2.08	74.84 ± 0.15	5/9	1/5
CIFAR-100	Recalibrated removal	79.48 ± 2.19	73.22 ± 2.45	5/9	1/5
CIFAR-100	Dense text + prototypes	81.30 ± 2.08	77.12 ± 2.64	7/9	0/7
CIFAR-100	Tip-Adapter sum	79.32 ± 2.30	71.40 ± 7.86	6/9	1/6
CIFAR-100	Tip-Adapter mean	79.51 ± 3.58	77.12 ± 5.00	8/9	1/8
Pets	Retained classifier	91.77 ± 2.39	92.42 ± 2.64	9/9	5/9
Pets	Fixed-gate removal	88.87 ± 3.95	92.77 ± 5.91	6/9	1/6
Pets	Recalibrated removal	87.68 ± 3.95	89.64 ± 11.35	6/9	1/6
Pets	Dense text + prototypes	88.50 ± 6.27	93.41 ± 3.82	7/9	2/7
Pets	Tip-Adapter sum	89.86 ± 3.58	89.59 ± 5.68	7/9	2/7
Pets	Tip-Adapter mean	90.25 ± 1.80	91.90 ± 3.31	7/9	2/7
ImageNet-R	Retained classifier	85.85 ± 0.38	84.98 ± 0.38	9/9	2/9
ImageNet-R	Fixed-gate removal	85.66 ± 0.86	79.23 ± 1.90	7/9	3/7
ImageNet-R	Recalibrated removal	85.74 ± 0.90	79.08 ± 1.74	7/9	3/7
ImageNet-R	Dense text + prototypes	86.39 ± 0.50	83.05 ± 0.22	9/9	4/9
ImageNet-R	Tip-Adapter sum	85.47 ± 0.89	77.71 ± 1.44	8/9	4/8
ImageNet-R	Tip-Adapter mean	86.49 ± 0.54	81.22 ± 0.63	9/9	6/9

Appendix I. Ten-partition replication, encoder transfer and simple fusion

The expanded experiment fixes class-partition seeds $0, \ldots, 9$ on each of the three existing image pools. Assignments and pseudo splits $0, 1, 2$ are retained exactly, and the original L/14 results reconstruct with maximum metric difference zero. New seeds use the same frequency-stratified allocation, base/novel counts, five-shot reserve, pseudo-episode selection and two arrival-order rules. These are ten distinct class assignments on shared images, not ten independent datasets. Primary fixed-gate comparisons average both orders within a partition. The conditional paired percentile bootstrap uses 20,000 resamples of the ten paired partition values with seed 2026. Constrained differences use identical jointly feasible partition/order/session cells. JSON records retain the partition values and available sets, and CSV tables report every budget.

Appendix I.1. Second encoder and access controls

OpenAI CLIP ViT-B/32 uses cached revision `3d74acf9a28c67741b2f4f2ea7635f0aaf6f0268` with 512-dimensional projected features; L/14 uses 768 dimensions. Both consume the exact same fitting, validation, support and test images, dataset-specific preprocessing and eight text templates. Text anchors and ridge columns are recomputed for the selected encoder; no task-adapted features are used. The two encoders have separate closed-base references and hence separate absolute targets. Config, weights, tokenizer and pool hashes are recorded. Batch-16 feature shards support atomic restart with contiguous-index guards, and eight fresh test images are re-encoded per dataset to check each completed B/32 cache.

Appendix I.2. Raw and fixed-temperature controls

Raw fusion uses q_T, q_L and q_P directly in the same availability pattern as the retained table. Fixed-temperature fusion uses q_e/σ_e, where

$$\sigma_e = \max\left(10^{-6}, \frac{1}{|\mathcal{V}_B|}\sum_{x\in\mathcal{V}_B} \mathrm{Std}_{r\in\mathcal{R}_e}\, q_e(x,r)\right).$$

For the pseudo episode, $\mathcal{V}_B$ contains base-validation queries and $\mathcal{R}_e$ contains pseudo-base reference columns. Full-model scales use full base columns and base validation before arrivals. The three scalars remain fixed across sessions. Coefficient and offset grids are identical to the main retained model, with the same harmonic pseudo-episode objective. Neither control requires current-vocabulary statistics; both can preserve old scores. Their performance tests the empirical value of the per-query normalization, conditional on this specified calibration grid.

Appendix I.3. Full matched and validation-policy results

The main validation table includes retained, recalibrated-removal, dense-prototype, class-mean cache and simple-fusion alternatives. Summed cache and fixed-gate removal remain in the complete records. The following tables give both encoders' matched-test frontiers and the B/32 validation policy at the 95% budget. Every feasible set and test-attainment count refers to the final active vocabulary; order zero counts each partition once. Because means may cover different feasible sets, an unpaired difference of available means is not reported as a paired gain. Complete 80/85/90/95/99% outcomes are retained in `revision_summary.csv`; per-partition raw predictions supply both test and validation sufficient statistics.

Table S19: Complete eight-readout ViT-L/14 validation-selected 95% policy. Accuracies are achieved test percentages, mean ± sample standard deviation over feasible partitions. Feasible and Test met have the same meanings as the main validation-policy table.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	84.26 ± 1.91	77.10 ± 2.53	10/10	7/10
CIFAR-100	Fixed-gate removal	83.05 ± 1.64	65.32 ± 12.41	6/10	1/6
CIFAR-100	Recalibrated removal	82.91 ± 1.62	66.03 ± 13.04	6/10	1/6
CIFAR-100	Dense text + prototypes	84.00 ± 1.76	75.10 ± 7.15	8/10	4/8
CIFAR-100	Raw-score fusion	82.11 ± 2.51	64.45 ± 13.35	7/10	1/7
CIFAR-100	Fixed temperatures	83.72 ± 2.24	70.93 ± 10.56	10/10	5/10
CIFAR-100	Tip-Adapter sum	83.21 ± 1.71	65.36 ± 12.68	5/10	0/5
CIFAR-100	Tip-Adapter mean	83.19 ± 1.72	68.69 ± 10.39	9/10	2/9
Pets	Retained references	89.95 ± 2.67	93.79 ± 3.83	10/10	1/10
Pets	Fixed-gate removal	90.43 ± 3.16	93.81 ± 3.11	8/10	3/8
Pets	Recalibrated removal	90.48 ± 3.04	93.44 ± 3.56	8/10	3/8
Pets	Dense text + prototypes	89.46 ± 3.08	94.47 ± 2.60	10/10	2/10
Pets	Raw-score fusion	91.35 ± 2.91	92.72 ± 2.70	9/10	4/9
Pets	Fixed temperatures	90.48 ± 3.42	94.21 ± 3.19	9/10	3/9
Pets	Tip-Adapter sum	89.18 ± 1.73	91.61 ± 6.00	7/10	1/7
Pets	Tip-Adapter mean	90.73 ± 2.17	91.39 ± 4.32	10/10	4/10
ImageNet-R	Retained references	86.15 ± 0.94	84.63 ± 0.95	10/10	2/10
ImageNet-R	Fixed-gate removal	86.17 ± 0.65	66.89 ± 22.80	10/10	3/10
ImageNet-R	Recalibrated removal	86.16 ± 0.71	66.88 ± 22.79	10/10	3/10
ImageNet-R	Dense text + prototypes	86.01 ± 0.94	83.94 ± 1.10	10/10	3/10
ImageNet-R	Raw-score fusion	86.15 ± 0.68	65.55 ± 20.42	10/10	2/10
ImageNet-R	Fixed temperatures	85.99 ± 0.71	82.92 ± 1.03	10/10	1/10
ImageNet-R	Tip-Adapter sum	86.06 ± 0.78	74.52 ± 3.69	10/10	2/10
ImageNet-R	Tip-Adapter mean	86.08 ± 1.08	81.61 ± 1.51	10/10	4/10

Table S20: ViT-L/14 matched-test 95% retention frontier across ten class partitions. Values are percentages, mean ± sample standard deviation over feasible partitions after averaging orders. Test met equals feasibility by construction in this descriptive sweep.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	83.91 ± 1.42	78.02 ± 2.27	10/10	10/10
CIFAR-100	Fixed-gate removal	83.25 ± 1.17	61.68 ± 9.35	6/10	6/6
CIFAR-100	Recalibrated removal	83.25 ± 1.17	63.78 ± 6.52	6/10	6/6
CIFAR-100	Dense text + prototypes	83.91 ± 1.42	76.18 ± 2.56	10/10	10/10
CIFAR-100	Raw-score fusion	83.25 ± 1.17	62.37 ± 6.38	6/10	6/6
CIFAR-100	Fixed temperatures	83.64 ± 1.21	74.53 ± 2.70	9/10	9/9
CIFAR-100	Tip-Adapter sum	83.16 ± 1.28	61.83 ± 4.92	5/10	5/5
CIFAR-100	Tip-Adapter mean	83.91 ± 1.42	65.20 ± 11.69	10/10	10/10
Pets	Retained references	91.56 ± 1.87	93.30 ± 3.74	10/10	10/10
Pets	Fixed-gate removal	91.57 ± 1.88	93.78 ± 2.17	10/10	10/10
Pets	Recalibrated removal	91.56 ± 1.88	93.32 ± 2.10	10/10	10/10
Pets	Dense text + prototypes	91.56 ± 1.87	93.79 ± 2.78	10/10	10/10
Pets	Raw-score fusion	91.57 ± 1.88	92.39 ± 2.75	10/10	10/10
Pets	Fixed temperatures	91.56 ± 1.87	93.64 ± 2.48	10/10	10/10
Pets	Tip-Adapter sum	91.51 ± 1.98	87.78 ± 9.98	9/10	9/9
Pets	Tip-Adapter mean	91.57 ± 1.87	92.02 ± 4.15	10/10	10/10
ImageNet-R	Retained references	86.42 ± 0.53	84.31 ± 1.01	10/10	10/10
ImageNet-R	Fixed-gate removal	86.41 ± 0.56	73.82 ± 6.70	9/10	9/9
ImageNet-R	Recalibrated removal	86.42 ± 0.56	73.50 ± 6.79	9/10	9/9
ImageNet-R	Dense text + prototypes	86.42 ± 0.53	83.02 ± 1.40	10/10	10/10
ImageNet-R	Raw-score fusion	86.42 ± 0.56	70.85 ± 8.03	9/10	9/9
ImageNet-R	Fixed temperatures	86.43 ± 0.53	82.49 ± 1.11	10/10	10/10
ImageNet-R	Tip-Adapter sum	86.43 ± 0.56	73.16 ± 5.66	9/10	9/9
ImageNet-R	Tip-Adapter mean	86.43 ± 0.52	80.91 ± 1.48	10/10	10/10

Table S21: ViT-B/32 matched-test 95% frontier under the same access, aggregation and feasibility reporting as Table S20. Absolute targets are encoder-specific.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	76.16 ± 1.99	59.64 ± 2.10	10/10	10/10
CIFAR-100	Fixed-gate removal	74.84 ± 1.75	49.81 ± 10.48	4/10	4/4
CIFAR-100	Recalibrated removal	74.84 ± 1.75	49.81 ± 10.48	4/10	4/4
CIFAR-100	Dense text + prototypes	76.15 ± 2.22	51.78 ± 8.42	8/10	8/8
CIFAR-100	Raw-score fusion	75.12 ± 1.63	46.30 ± 9.93	5/10	5/5
CIFAR-100	Fixed temperatures	75.71 ± 1.46	54.14 ± 4.58	9/10	9/9
CIFAR-100	Tip-Adapter sum	74.08 ± 1.05	52.81 ± 5.29	3/10	3/3
CIFAR-100	Tip-Adapter mean	76.16 ± 2.11	46.50 ± 11.77	9/10	9/9
Pets	Retained references	86.92 ± 2.17	76.54 ± 6.71	10/10	10/10
Pets	Fixed-gate removal	86.45 ± 2.27	77.62 ± 6.10	5/10	5/5
Pets	Recalibrated removal	86.46 ± 2.29	78.61 ± 5.07	5/10	5/5
Pets	Dense text + prototypes	86.84 ± 2.33	71.48 ± 15.80	8/10	8/8
Pets	Raw-score fusion	87.12 ± 2.19	67.25 ± 11.41	9/10	9/9
Pets	Fixed temperatures	86.93 ± 2.16	74.97 ± 6.20	10/10	10/10
Pets	Tip-Adapter sum	86.83 ± 2.22	70.08 ± 9.72	6/10	6/6
Pets	Tip-Adapter mean	87.28 ± 2.29	68.64 ± 21.10	8/10	8/8
ImageNet-R	Retained references	70.38 ± 0.87	56.36 ± 1.11	10/10	10/10
ImageNet-R	Fixed-gate removal	69.72	31.36	1/10	1/1
ImageNet-R	Recalibrated removal	69.71	30.54	1/10	1/1
ImageNet-R	Dense text + prototypes	70.37 ± 0.88	48.30 ± 3.45	10/10	10/10
ImageNet-R	Raw-score fusion	69.71	28.06	1/10	1/1
ImageNet-R	Fixed temperatures	70.38 ± 0.87	53.72 ± 1.35	10/10	10/10
ImageNet-R	Tip-Adapter sum	69.67 ± 0.03	35.63 ± 8.58	2/10	2/2
ImageNet-R	Tip-Adapter mean	70.45 ± 0.92	41.88 ± 7.58	8/10	8/8

Table S22: ViT-B/32 validation-selected 95% policy. Values are achieved base/novel test percentages over validation-feasible partitions; Test met counts those meeting the independent test-retention criterion. Validation feasibility does not guarantee test target attainment.

Data	Readout	Base	Novel	Feasible	Test met
CIFAR-100	Retained references	75.98 ± 2.43	59.04 ± 3.66	10/10	5/10
CIFAR-100	Fixed-gate removal	73.46 ± 1.73	52.54 ± 10.71	7/10	1/7
CIFAR-100	Recalibrated removal	73.40 ± 1.79	54.20 ± 6.94	7/10	1/7
CIFAR-100	Dense text + prototypes	74.30 ± 2.85	55.73 ± 8.03	7/10	2/7
CIFAR-100	Raw-score fusion	73.82 ± 2.67	49.47 ± 12.37	8/10	2/8
CIFAR-100	Fixed temperatures	73.90 ± 2.39	56.16 ± 8.19	9/10	1/9
CIFAR-100	Tip-Adapter sum	73.55 ± 2.53	49.25 ± 12.78	7/10	1/7
CIFAR-100	Tip-Adapter mean	73.54 ± 2.88	57.47 ± 8.46	7/10	0/7
Pets	Retained references	88.16 ± 3.34	72.75 ± 8.65	10/10	8/10
Pets	Fixed-gate removal	86.17 ± 2.74	76.30 ± 6.46	5/10	3/5
Pets	Recalibrated removal	86.24 ± 2.14	77.70 ± 5.37	5/10	2/5
Pets	Dense text + prototypes	86.89 ± 3.09	70.27 ± 15.78	7/10	3/7
Pets	Raw-score fusion	86.93 ± 5.36	63.79 ± 15.01	9/10	5/9
Pets	Fixed temperatures	85.53 ± 5.27	74.57 ± 7.40	10/10	4/10
Pets	Tip-Adapter sum	86.87 ± 2.42	73.74 ± 4.16	4/10	2/4
Pets	Tip-Adapter mean	84.97 ± 3.13	81.94 ± 3.51	5/10	1/5
ImageNet-R	Retained references	70.18 ± 0.96	56.80 ± 1.33	10/10	3/10
ImageNet-R	Fixed-gate removal	68.23 ± 0.84	19.12 ± 6.82	2/10	0/2
ImageNet-R	Recalibrated removal	68.23 ± 0.84	19.12 ± 6.82	2/10	0/2
ImageNet-R	Dense text + prototypes	70.43 ± 1.50	46.95 ± 4.97	10/10	6/10
ImageNet-R	Raw-score fusion	68.16 ± 0.84	21.79 ± 6.87	2/10	0/2
ImageNet-R	Fixed temperatures	70.46 ± 1.26	53.38 ± 1.43	10/10	6/10
ImageNet-R	Tip-Adapter sum	68.89 ± 0.79	38.12 ± 2.27	3/10	0/3
ImageNet-R	Tip-Adapter mean	69.99 ± 1.61	46.39 ± 3.62	9/10	4/9

Table S23: Full ten-partition paired average and final gains. Entries are retained-minus-comparator percentage points [conditional paired 95% bootstrap CI], with orders averaged within each partition.

Encoder	Data	Comparator	Paired avg. gain [95% CI]	Paired final gain [95% CI]
L/14	CIFAR-100	Recalibrated removal	3.00 [2.23, 3.75]	2.62 [1.81, 3.43]
L/14	CIFAR-100	Dense text + prototypes	0.15 [-0.46, 0.64]	-0.21 [-0.85, 0.30]
L/14	CIFAR-100	Tip-Adapter mean	1.44 [0.68, 2.10]	1.24 [0.41, 2.01]
L/14	Pets	Recalibrated removal	-0.14 [-1.53, 0.86]	-0.78 [-2.84, 0.60]
L/14	Pets	Dense text + prototypes	-0.98 [-2.52, 0.22]	-1.54 [-4.08, 0.35]
L/14	Pets	Tip-Adapter mean	-0.06 [-1.76, 1.41]	-0.39 [-2.92, 1.71]
L/14	ImageNet-R	Recalibrated removal	2.30 [1.95, 2.59]	1.71 [1.28, 2.06]
L/14	ImageNet-R	Dense text + prototypes	0.71 [0.46, 0.92]	0.41 [0.03, 0.72]
L/14	ImageNet-R	Tip-Adapter mean	1.42 [1.15, 1.62]	1.13 [0.76, 1.43]
B/32	CIFAR-100	Recalibrated removal	2.38 [1.44, 3.34]	1.80 [1.01, 2.65]
B/32	CIFAR-100	Dense text + prototypes	0.86 [-0.15, 1.76]	0.43 [-0.47, 1.24]
B/32	CIFAR-100	Tip-Adapter mean	0.96 [-0.04, 1.84]	0.51 [-0.34, 1.33]
B/32	Pets	Recalibrated removal	-0.44 [-1.64, 0.81]	-2.00 [-3.80, -0.19]
B/32	Pets	Dense text + prototypes	-0.39 [-1.85, 1.02]	-1.17 [-3.22, 0.67]
B/32	Pets	Tip-Adapter mean	-0.74 [-1.62, 0.12]	-1.85 [-3.40, -0.59]
B/32	ImageNet-R	Recalibrated removal	3.07 [2.82, 3.34]	2.39 [2.19, 2.61]
B/32	ImageNet-R	Dense text + prototypes	0.90 [0.69, 1.11]	0.46 [0.26, 0.66]
B/32	ImageNet-R	Tip-Adapter mean	1.71 [1.44, 1.98]	1.34 [1.16, 1.52]

Table S24: Complete L/14 normalization controls. Fixed-gate averages cover five sessions; matched base/novel accuracy uses the descriptive 95% test-retention budget. Means and sample standard deviations use the reported feasible sets.

Data	Fusion	Fixed-gate avg.	Matched base	Matched novel	Feasible
CIFAR-100	Retained references	83.03 ± 1.19	83.91 ± 1.42	78.02 ± 2.27	10/10
CIFAR-100	Raw-score fusion	79.70 ± 2.64	83.25 ± 1.17	62.37 ± 6.38	6/10
CIFAR-100	Fixed temperatures	81.56 ± 0.98	83.64 ± 1.21	74.53 ± 2.70	9/10
Pets	Retained references	92.23 ± 2.87	91.56 ± 1.87	93.30 ± 3.74	10/10
Pets	Raw-score fusion	87.94 ± 14.58	91.57 ± 1.88	92.39 ± 2.75	10/10
Pets	Fixed temperatures	89.39 ± 10.15	91.56 ± 1.87	93.64 ± 2.48	10/10
ImageNet-R	Retained references	87.24 ± 0.59	86.42 ± 0.53	84.31 ± 1.01	10/10
ImageNet-R	Raw-score fusion	83.70 ± 2.58	86.42 ± 0.56	70.85 ± 8.03	9/10
ImageNet-R	Fixed temperatures	86.30 ± 0.38	86.43 ± 0.53	82.49 ± 1.11	10/10

Main Fig. 2 visualizes the shared-retention frontiers; the tables in this section additionally retain partially feasible sets and exact counts.

Appendix J. Independent protection-risk calibration

Appendix J.1. Disjoint access and refitting

For each dataset, original fitting and validation features are concatenated in that order. A generator with seed 662026 shuffles each class, reserving $\max(1, \lfloor 0.4N_c \rfloor)$ calibration and $\max(1, \lfloor 0.2N_c \rfloor)$ tuning observations; the remainder fits the readouts. Every class retains at least two fitting images. Identical indices are used by both encoders and every method. The global fitting/tuning/calibration sizes are 4,437/2,139/4,318 on CIFAR, 556/248/510 on Pets, and 5,096/2,390/4,863 on ImageNet-R. Each true partition uses only its base classes in these pools. Support and test observations remain outside all three. All ridge parameters, pseudo-episode mixtures, branch temperatures and semantic-cache hyperparameters are recomputed using fitting and tuning data exclusively.

Only frozen features are reused; previously tuned score models are not treated as independent of their old validation labels. The new calibration labels enter only threshold certification or the explicitly compared empirical threshold controls. The independent-pool experiment is a benchmark revision on previously studied test data, not a newly untouched final evaluation. The statistical analysis assumes that, conditional on each base class and the fitted models, calibration observations are independent draws from the class-conditional distribution to which the population statement applies. Dataset shift and adaptive support arrivals depending on calibration labels are outside that statement.

Appendix J.2. Weighted protection bound

Let B be the number of base classes and n_c their calibration counts. Assign each observation of class c weight $w_i = 1/(Bn_c)$, so $\sum_i w_i = 1$, and let $b = \max_i w_i$. For any fixed bounded losses $L_i \in [0, 1]$, write $q = \sum_i w_i L_i$ and $\mu = \sum_i w_i \mathbb{E}L_i$. Conditional on the score models and the independently selected threshold family, independence gives, for $t < 0$,

$$\log \mathbb{E}e^{tq} \le \sum_i \log(1 - \mu_i + \mu_i e^{tw_i}) \tag{J.1}$$

$$\le b^{-1} \sum_i w_i \log(1 - \mu_i + \mu_i e^{tb}) \tag{J.2}$$

$$\le b^{-1} \log(1 - \mu + \mu e^{tb}), \tag{J.3}$$

where $\mu_i = \mathbb{E}L_i$. The first inequality uses boundedness and the chord of the exponential; the second uses convexity of each log-moment-generating function in w_i and its zero value at zero; the third uses concavity in μ_i. Chernoff optimization yields

$$\Pr(q \le x) \le \exp[-\operatorname{kl}(x, \mu)/b], \qquad x < \mu, \tag{J.4}$$

where kl is Bernoulli relative entropy. Thus, for the null $\mu \ge \alpha$, the value

$$p(q) = \begin{cases} \exp[-\operatorname{kl}(q, \alpha)/b], & q < \alpha, \\ 1, & q \ge \alpha \end{cases} \tag{J.5}$$

is a valid lower-tail p-value. Unlike an ordinary pooled binomial calculation, this bound retains the unequal macro weights. Its dependence on the largest weight makes sparse class-conditional calibration sets conservative.

Apply the bound to the protection loss of main Eq. (11). Let $A_{\mathrm{ref}} = \Pr(R = 1)$, $D = \Pr(R = 1, C_d = 0)$ and $G = \Pr(R = 0, C_d = 1)$. Then

$$\mathbb{E}L_d = D + (1 - r)(1 - A_{\mathrm{ref}}), \qquad A_B(d) = A_{\mathrm{ref}} - D + G. \tag{J.6}$$

Consequently $\mathbb{E}L_d \leq 1-r$ implies $A_B(d) \geq rA_{\mathrm{ref}}$. The reference uncertainty is included in the paired loss rather than replaced by a point estimate. Retained base scores have $G = 0$; other readouts may have $G > 0$, so protection is sufficient and can be stricter than accuracy-ratio control for replacement models.

Appendix J.3. Threshold search and trajectory confidence

For each method and arrival, the independently fixed finite threshold family consists of unique quantiles at 0,1,. . . ,100% of base tuning margins, in increasing order of novel admission. Following fixed-sequence testing [21], test this family in order at level $\delta/5$, stop at the first failure, and return the last accepted threshold. A false certification requires rejecting the first true null in the fixed order, whose probability is at most $\delta/5$; independence between these tests is unnecessary. A union bound over the five prespecified arrivals gives error probability at most δ for each method/order/partition. The same calibration pool may be reused because the score models, threshold families and support arrivals do not adapt to its labels. No simultaneous confidence claim across methods, orders, partitions, datasets or separately reported δ values is made.

If the accepted finite set is empty, the policy returns the original reference classifier and admits no novel class. This fallback structurally preserves its original accuracy and has zero novel accuracy. It is reported separately from

useful certified expansion. The confidence levels 90,95,99% and empirical target fractions 95,97,99% were specified before the new experiment; the primary setting uses 95% retention and 95% confidence. Their complete results, worst shortfalls and five-session trajectories are saved in `independen t_risk_summary.json` and `independent_risk_summary.csv`.

Table S25: ViT-L/14 independent calibration under the original allocation in Section J. All operational means include every partition and base-only fallback; certification counts identify finite accepted thresholds. Section L expands calibration coverage with these score models frozen.

Dataset	Readout	Emp. met	Emp. novel	Cert. exp.	Test met	Cert. base	Cert. novel
CIFAR-100	Retained	7/10	76.28	0/10	10/10	88.29	0.00
CIFAR-100	Recalibrated removal	6/10	35.32	0/10	10/10	88.29	0.00
CIFAR-100	Dense prototypes	6/10	71.92	0/10	10/10	88.29	0.00
CIFAR-100	Raw fusion	5/10	34.46	0/10	10/10	88.29	0.00
CIFAR-100	Fixed temperature	8/10	71.36	0/10	10/10	88.29	0.00
CIFAR-100	Class-mean cache	4/10	65.73	0/10	10/10	88.29	0.00
Pets	Retained	1/10	93.07	0/10	10/10	96.05	0.00
Pets	Recalibrated removal	4/10	93.29	0/10	10/10	96.05	0.00
Pets	Dense prototypes	0/10	95.08	0/10	10/10	96.05	0.00
Pets	Raw fusion	4/10	86.90	0/10	10/10	96.05	0.00
Pets	Fixed temperature	3/10	91.87	0/10	10/10	96.05	0.00
Pets	Class-mean cache	1/10	92.50	0/10	10/10	96.05	0.00
ImageNet-R	Retained	7/10	83.45	10/10	10/10	89.73	77.64
ImageNet-R	Recalibrated removal	4/10	57.36	0/10	10/10	91.54	0.00
ImageNet-R	Dense prototypes	6/10	82.47	5/10	10/10	91.04	24.12
ImageNet-R	Raw fusion	3/10	64.80	4/10	10/10	91.08	26.42
ImageNet-R	Fixed temperature	5/10	81.28	10/10	10/10	89.76	65.69
ImageNet-R	Class-mean cache	4/10	79.37	2/10	10/10	91.18	6.20

Table S26: ViT-B/32 independent calibration under the original allocation. Infeasible certification triggers the shared base-only reference fallback and zero novel accuracy.

Dataset	Readout	Emp. met	Emp. novel	Cert. exp.	Test met	Cert. base	Cert. novel
CIFAR-100	Retained	1/10	60.64	0/10	10/10	79.08	0.00
CIFAR-100	Recalibrated removal	2/10	47.79	0/10	10/10	79.08	0.00
CIFAR-100	Dense prototypes	2/10	52.74	0/10	10/10	79.08	0.00
CIFAR-100	Raw fusion	2/10	50.33	0/10	10/10	79.08	0.00
CIFAR-100	Fixed temperature	5/10	55.06	0/10	10/10	79.08	0.00
CIFAR-100	Class-mean cache	2/10	57.79	0/10	10/10	79.08	0.00
Pets	Retained	1/10	75.10	0/10	10/10	90.24	0.00
Pets	Recalibrated removal	10/10	28.49	0/10	10/10	90.24	0.00
Pets	Dense prototypes	4/10	54.67	0/10	10/10	90.24	0.00
Pets	Raw fusion	6/10	53.57	0/10	10/10	90.24	0.00
Pets	Fixed temperature	5/10	64.90	0/10	10/10	90.24	0.00
Pets	Class-mean cache	7/10	42.00	0/10	10/10	90.24	0.00
ImageNet-R	Retained	2/10	56.51	10/10	10/10	73.66	45.30
ImageNet-R	Recalibrated removal	9/10	3.58	0/10	10/10	75.01	0.00
ImageNet-R	Dense prototypes	4/10	43.48	0/10	10/10	75.01	0.00
ImageNet-R	Raw fusion	9/10	3.75	0/10	10/10	75.01	0.00
ImageNet-R	Fixed temperature	2/10	52.45	7/10	10/10	74.24	27.28
ImageNet-R	Class-mean cache	5/10	33.38	0/10	10/10	75.01	0.00

Appendix K. Reproducibility details for the concise main text

Appendix K.1. Dataset construction and access

We evaluate expansion on CIFAR-100 [22] with long-tailed base supervision. The training class counts follow an exponential schedule with imbalance parameter $\rho = 100$ and nominal maximum 500 images. Five images per class are reserved before constructing the long-tailed pool, so the largest available pool contains 495 images. The remaining long-tailed pool contains 10,894 images, split into 10,487 fitting images and 407 held-out validation images. Validation reserves up to five examples per class, with smaller allocations for the rarest classes.

For seeds $\{0, \ldots, 9\}$, stratified sampling within head, medium, and tail frequency buckets yields distinct partitions of 40 base, 40 novel, and 20 unused classes. Proportional quotas use largest-remainder rounding after shuffling class membership. Evaluation uses the 80 base and novel classes.

Oxford-IIIT Pets [23] provides a complementary fine-grained task with 37 cat and dog breeds. We use its official trainval/test split and reserve five training images per breed as support. From the remaining training images, we construct an exponential long-tailed pool with $\rho = 10$ and maximum class count 90, giving 1,168 fitting and 146 validation images. Each true partition contains 15 base, 15 novel, and seven unused breeds, sampled with the same allocation rule. The official test split contains 3,669 images. Unlike CIFAR, its images have varied spatial resolutions; RGB images are resized to 224×224 by PIL bicubic interpolation before the same CLIP normalization. These features are freshly extracted for this study.

All primary expansion readouts use frozen OpenAI CLIP ViT-L/14 or ViT-B/32 features from a single 224×224 bicubic-resized image with CLIP channel normalization. Fitting, validation, and support pools are disjoint. Supplementary Section C gives feature-cache and image-pool verification details.

Base ridge regularization is selected from $\{0.01, 0.1, 1, 10\}$ by base validation macro accuracy; the pseudo-base classifier uses the same rule. The primary comparison uses all five reserved support images, ten distinct class partitions and two arrival orders per partition. Earlier one/two/five-shot and repeated-calibration studies use three original partitions and are reported separately in Supplementary Section G.

Classifier fitting, regularization selection, and gate calibration use base classes exclusively. Five-shot CIFAR expansion consumes 200 support images for 40 novel classes. Its balanced test set supplies 4,000 base and 4,000 novel images for the expanded metrics.

ImageNet-R [20] extends the active vocabulary to 200 rendition classes: 100 base classes followed by five arrivals of 20 classes. After removing 59 byte-identical duplicates, per-class seeded splits reserve approximately 20% each for validation and test, then five support images. The fitting pool is downsampled by class rank with a nominal 100:1 schedule, yielding 2–136 images per class after availability caps. The resulting pools contain 6,438 fitting, 5,911 validation, 1,000 support and 5,911 test images. Ten frequency-stratified class partitions and two arrival orders use the same selection and reporting rules as the other datasets. RGB images are resized to 224×224 with bicubic interpolation and encoded by the same selected frozen checkpoint within each encoder comparison.

The ridge readout uses an augmented constant feature, inverse-frequency sample weights and an unregularized intercept; only the feature slopes receive ridge regularization. Class prototypes are normalized means of the permitted fitting or support features.

Appendix K.2. Pseudo-expansion selection

Fusion calibration must respect the distinction between base supervision and novel support. We construct a pseudo-expansion episode by splitting the base classes into approximately equal pseudo-base and pseudo-novel groups: 20/20 on CIFAR-100, 7/8 on Pets and 50/50 on ImageNet-R. A separate classifier is fitted using the pseudo-base classes alone. The pseudo-novel rows contain names and k support examples from those base classes, while held-out base validation images supply query labels. This episode simulates missing learned evidence without consuming any actual novel-class validation labels.

For each support budget, candidate coefficients are evaluated on this

episode. The selection objective is the harmonic mean of macro accuracy on the two groups, $H = 2A_B A_N/(A_B + A_N)$ We set $H = 0$ when both terms vanish. The calibration grids are $a \in \{0, 0.25, 0.5, 1, 2\}$, $u \in \{0, 0.25, 0.5, 1, 2, 4\}$, and offsets $d \in [-4, 4]$ in steps of 0.25. Ties follow a fixed grid order. The full base classifier is then combined with actual novel rows using the selected coefficients. Neither actual novel validation queries nor test labels enter this selection rule.

This procedure uses one pseudo-class split per true partition and the same support-index seed as the evaluated support condition. CIFAR and Pets pseudo splits use seed $100 + p$ for class-partition seed p; ImageNet-R selects half the base classes with seed $900 + p$. Coefficients are chosen once for a support budget and stay fixed when the vocabulary-size stress test appends successive subsets of novel classes.

Appendix K.3. A normalization example

A two-class example illustrates the effect. Suppose the old text scores are $(0, 2)$, giving standardized scores $(-1, 1)$, while the learned branch contributes $(0.25, -0.25)$. The second old class has the larger fused score. Appending a text score of 100 changes the two old standardized scores to approximately $(-0.728, -0.685)$. Their text separation is now about 0.043, and the unchanged learned contribution reverses the old-class ordering. No classifier weight has changed. The example isolates a normalization effect independently of whether the new class ultimately wins the expanded decision.

Appendix L. Independent calibration coverage and label budgets

Appendix L.1. Frozen models and disjoint calibration images

The locked protocol in `experiments/kbs_calibration_budget_protocol.md` fixes all score models, tuning parameters, support, test predictions and candidate thresholds from Section J before varying calibration coverage. Six readouts receive identical class-wise subsets: retained tables, recalibrated classifier removal, dense anchored prototypes, raw fusion, fixed-temperature fusion and class-mean Tip-Adapter. No score-model refitting or hyperparameter selection uses these calibration labels. Previously cached calibration scores are reused exactly after checking reconstruction against the frozen models.

The pool prioritizes Section J's independent calibration images, then adds previously unused training images up to 40 per class. Image identities and content hashes exclude fitting, tuning, support and test images. The audit uses raw pixel hashes on CIFAR, resized-image pixel hashes on Pets, and file-content hashes on ImageNet-R. Across all classes, the resulting pools contain 4,000 CIFAR images, 1,480 Pets images and 7,376 ImageNet-R images, of which 1,805, 970 and 3,020 are supplemental. Each experiment selects its base classes from these pools. Both encoders reuse their pinned local checkpoints; only supplemental images require new encoding. Witness re-encodings verify agreement with the cached feature space.

Per-class caps $m \in \{2, 5, 10, 20, 40\}$ use nested random subsets selected without scores or correctness outcomes, with subset seeds 0, 1 and 2. At cap 40 these seeds select the same full pool, which is counted once. CIFAR and Pets have 40 available images per class; ImageNet-R has a minimum of 14, with 44 classes below 40. At cap 40 the actual base-calibration query

totals are 1,600, 600 and 3,636–3,778 across partitions, respectively. The cap changes both total sample size and tail-class coverage; the curves do not isolate a causal effect of total label count alone.

Appendix L.2. A necessary coverage condition for certification

For the weighted bound in Section J, even $q = 0$ gives

$$p(0) = \exp[-\operatorname{kl}(0, 1-r)/b] = r^{1/b}. \tag{L.1}$$

Since $p(q)$ increases on $0 \le q < 1 - r$, passing at level $\delta/5$ requires

$$n_{\text{eff}} = 1/b = B \min_c n_c \ge \frac{\log(\delta/5)}{\log r}. \tag{L.2}$$

At $r = 0.95$ and $\delta = 0.05$, this floor is 89.78. It is a necessary condition for this certificate, not a sufficient condition for successful expansion. Under the original allocation, CIFAR has $n_{\text{eff}} = 80$, Pets has 45 or 60 and ImageNet-R has 600, 700 or 800. Thus the original CIFAR/Pets calibration allocation cannot pass this bound even at zero observed loss. Balanced per-class caps must be at least 3 for CIFAR, 6 for Pets and 1 for ImageNet-R to clear the zero-loss floor. Actual protection losses require further evidence. A balanced allocation can improve the bound even with fewer total labels than a larger, uneven pool.

Appendix L.3. Aggregation, admission and complete comparisons

Figures S5 and S6 show the fifth arrival in order 0, the first prespecified order. Novel-accuracy means average three subset seeds within each of ten partitions below cap 40, and use the full pool once per partition at 40. Retained-table shading shows sample standard deviation across ten partition means, clipped to [0,100]. Finite-admission fractions count 30 correlated

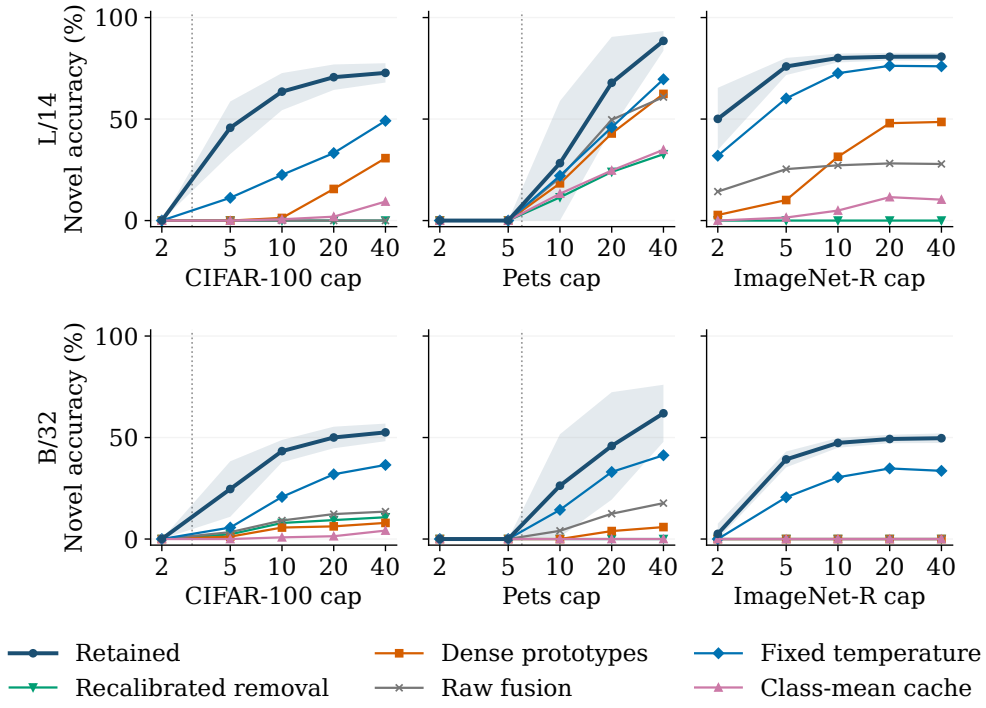


Figure S5: Protection-calibrated novel accuracy versus class-wise calibration cap, fifth arrival in order 0. Means include reference fallback, which contributes zero novel accuracy. Shading shows retained partition standard deviations. All six readouts share calibration images; three subset seeds are averaged below cap 40.

partition/subset cells below 40 and ten at 40, with equal partition weights; these cells are not independent Bernoulli trials. Confidence applies to five arrivals for each prespecified method/order/partition and budget separately, without a simultaneous statement across plotted budgets or methods.

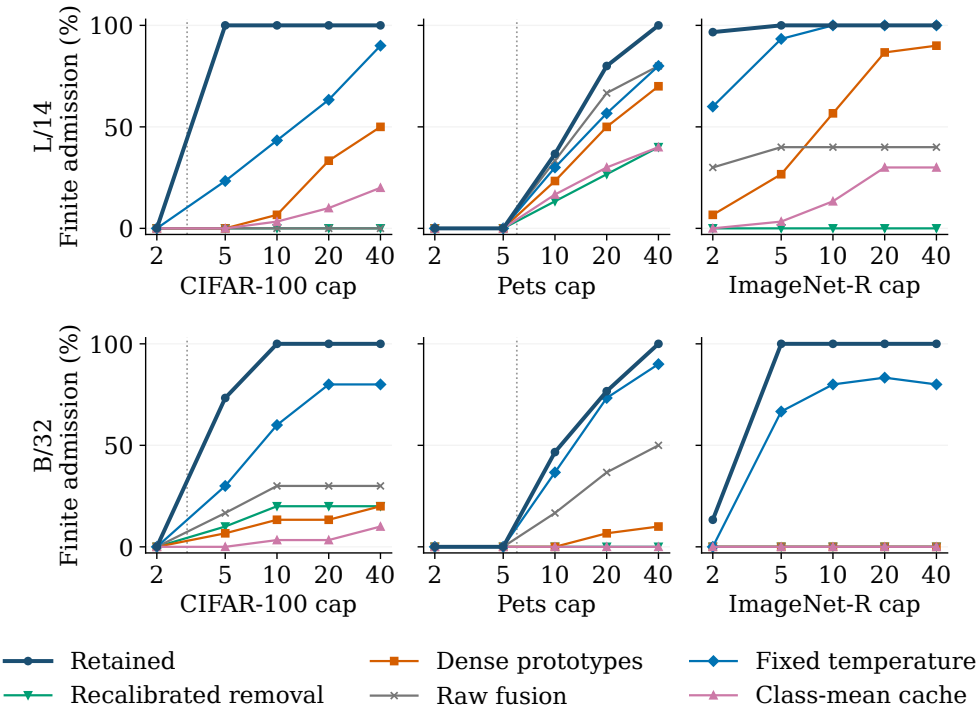


Figure S6: Finite certified admission after the fifth arrival in order 0, under 95% retention and 95% five-arrival confidence. Fractions use 30 partition/subset cells below cap 40 and ten full-pool cells at 40. Grey lines mark the necessary balanced cap for the zero-loss bound; ImageNet-R's floor of one lies below the plotted range. Every readout receives the same images and protection rule. Base-only fallback counts as no finite admission.

All six retained-table dataset/encoder conditions certify final expansion in 10/10 partitions and meet all ten final test targets. Across both orders, all 20 paths per condition meet the test target at every arrival. Finite admission throughout holds for all CIFAR and ImageNet-R paths, and for 17/20 Pets

Table S27: Complete L/14 cap-40 comparison after the fifth arrival in order 0. Empirical thresholds and independent risk certification share frozen score models and calibration pools. Accuracy means (%) include ten partitions and zero novel accuracy on fallback. Finite counts certified admissions; Test met counts attained test targets. Bold marks the largest certified-policy novel mean within each dataset; it does not denote a paired significance test.

Readout	Empirical policy		Certified policy			
	Test met	Novel ↑	Finite	Test met	Base ↑	Novel ↑
CIFAR-100						
Retained	8/10	76.90	10/10	10/10	85.86	**72.76**
Recalibrated removal	6/10	47.00	0/10	10/10	88.29	0.00
Dense prototypes	6/10	72.93	5/10	10/10	87.45	30.74
Raw fusion	6/10	43.13	0/10	10/10	88.29	0.00
Fixed temperature	6/10	71.75	9/10	10/10	86.66	49.14
Class-mean cache	5/10	66.31	2/10	10/10	87.94	9.35
Pets						
Retained	4/10	92.58	10/10	10/10	94.10	**88.51**
Recalibrated removal	7/10	92.62	4/10	10/10	95.48	32.68
Dense prototypes	3/10	94.54	7/10	10/10	94.93	62.39
Raw fusion	6/10	86.52	8/10	10/10	94.91	60.83
Fixed temperature	5/10	91.51	8/10	10/10	95.02	69.61
Class-mean cache	4/10	91.84	4/10	10/10	95.53	34.86
ImageNet-R						
Retained	8/10	83.53	10/10	10/10	89.02	**80.74**
Recalibrated removal	4/10	54.87	0/10	10/10	91.54	0.00
Dense prototypes	7/10	82.65	9/10	10/10	90.44	48.58
Raw fusion	4/10	64.01	4/10	10/10	90.84	27.88
Fixed temperature	5/10	81.38	10/10	10/10	89.19	76.00
Class-mean cache	5/10	79.08	3/10	10/10	90.99	10.31

Table S28: Complete B/32 cap-40 comparison, with access, accuracy units, fallback accounting and boldface convention as in the L/14 table.

Readout	Empirical policy		Certified policy			
	Test met	Novel ↑	Finite	Test met	Base ↑	Novel ↑
CIFAR-100						
Retained	1/10	59.19	10/10	10/10	76.74	**52.54**
Recalibrated removal	7/10	34.79	2/10	10/10	78.57	10.73
Dense prototypes	6/10	51.14	2/10	10/10	78.79	7.95
Raw fusion	5/10	37.98	3/10	10/10	78.38	13.51
Fixed temperature	4/10	55.06	8/10	10/10	77.87	36.50
Class-mean cache	3/10	54.52	1/10	10/10	79.21	4.16
Pets						
Retained	3/10	73.15	10/10	10/10	88.10	**61.94**
Recalibrated removal	10/10	35.20	0/10	10/10	90.24	0.00
Dense prototypes	5/10	54.36	1/10	10/10	90.17	5.89
Raw fusion	6/10	56.09	5/10	10/10	90.03	17.69
Fixed temperature	4/10	65.62	9/10	10/10	90.46	41.26
Class-mean cache	8/10	55.72	0/10	10/10	90.24	0.00
ImageNet-R						
Retained	3/10	56.13	10/10	10/10	72.98	**49.67**
Recalibrated removal	10/10	0.00	0/10	10/10	75.01	0.00
Dense prototypes	3/10	47.78	0/10	10/10	75.01	0.00
Raw fusion	10/10	0.00	0/10	10/10	75.01	0.00
Fixed temperature	4/10	52.37	8/10	10/10	73.68	33.60
Class-mean cache	4/10	32.26	0/10	10/10	75.01	0.00

paths with L/14 and 16/20 with B/32; other Pets paths use an early fallback. A path containing fallback therefore counts toward test protection but not uninterrupted useful admission.

Paired policy differences include all ten partitions, including fallback; `calibration_budget_paired.csv` additionally reports jointly finite partition counts and conditional accuracy differences. These distinguish improved ability to certify admission from predictive differences after both methods admit. For example, L/14 retained-minus-dense policy differences are 42.02 [23.83,60.07] points on CIFAR and 26.12 [−0.01,53.10] on Pets. On their jointly finite sets, the means are 12.84 points over five CIFAR partitions and −0.63 over seven Pets partitions. Conditional paired percentile intervals resample ten partition means 20,000 times with seed 2026; they describe class-assignment variation on fixed images. The sufficient protection criterion ignores recovery of reference errors by replacement models, as derived in Section J.

The 93,600 records cover 60 dataset/encoder/partition runs, two orders, five arrivals, 13 distinct cap/subset settings, six methods and two policies. The independent verifier reconstructs 88,216 threshold tests, selected thresholds and final predictions; it verifies zero protected-pool overlap and unchanged original models and test archives. A completed rerun preserves hashes and modification times of 1,996 checkpoint files. Complete cap-specific metrics, query counts and trajectories are supplied in `calibration_budget_summary.json` and its CSV export. The study uses previously examined benchmarks; the population confidence statement retains Section J's class-conditional sampling assumptions.

Appendix M. Controlled normalization, learned evidence and HyCal transfer

Appendix M.1. Frozen-gate factorial and learned-weight interventions

The locked protocol is `experiments/kbs_reviewer_revision_protocol.md`; its sampling clarification is `experiments/kbs_reviewer_revision_scope.md`. The four normalization cells independently replace query centering or query scaling with the corresponding branch moments averaged over base-tuning queries. Reference banks, models, support, gates and offsets are unchanged. Table S29 reports the resulting average incremental accuracy. This differs from the independently tuned fixed-temperature comparator in Section I. Halving or removing the learned coefficient additionally tests its contribution within the original score frame. Complete paired contrasts and fixed-gate means are in `reviewer_revision_summary.json`; bootstrap intervals average orders within each partition and condition on the fixed benchmark data.

Appendix M.2. Class-level quality diagnosis

For each observed base class, we average its test accuracy across partitions in which it occurs, then compare the retained and dense closed-base classifiers. Prototype quality is accuracy of the pure visual-mean readout; cohesion is mean within-class feature–prototype cosine similarity. Frequency is the permitted base-fitting count. Classes are sorted by frequency into three approximately equal groups. These are descriptive associations on overlapping benchmark tasks, not independent class-level significance tests or causal estimates. Table S30 reports all observed classes and correlations; Figure S7

Table S29: Controlled normalization factorial. Average incremental macro accuracy (%), mean ± sample standard deviation over ten partitions after averaging two orders. Only branch centering and scaling change; fusion gates and offsets stay fixed. Query/query is the retained table.

Center / scale	CIFAR-100	Pets	ImageNet-R
ViT-L/14			
Query / query	83.30 ± 1.11	91.32 ± 3.44	87.29 ± 0.56
Query / fixed	82.92 ± 1.25	91.20 ± 2.52	86.86 ± 0.45
Fixed / query	80.30 ± 2.31	90.33 ± 3.60	85.25 ± 1.16
Fixed / fixed	78.64 ± 3.00	89.94 ± 3.31	84.16 ± 1.64
ViT-B/32			
Query / query	71.73 ± 1.63	81.14 ± 3.37	67.93 ± 0.33
Query / fixed	71.33 ± 1.56	79.82 ± 3.35	67.57 ± 0.48
Fixed / query	70.35 ± 2.29	78.89 ± 4.56	67.05 ± 0.99
Fixed / fixed	68.99 ± 2.91	77.34 ± 4.65	66.23 ± 1.36

shows the class-level relationship. The distinct closed-base and incremental metrics should not be equated.

Table S30: Closed-base retained-minus-dense accuracy gap (pp) and Spearman correlations between class-level gain and fitting count, pure-prototype accuracy and cohesion. Classes counts distinct classes observed as base across ten partitions.

Encoder	Data	Closed gap	Count ρ	Prototype ρ	Cohesion ρ	Classes
l14	CIFAR-100	+2.20	0.04	-0.14	-0.17	99
l14	Pets	+0.43	0.55	0.03	-0.09	36
l14	ImageNet-R	+0.86	-0.12	-0.10	0.13	199
b32	CIFAR-100	+1.82	-0.03	-0.14	0.04	99
b32	Pets	+1.97	-0.04	-0.21	0.15	36
b32	ImageNet-R	+2.38	0.06	0.18	0.10	199

Appendix M.3. Recent HyCal baseline under matched access

We implement HyCal [24] from the author repository `https://github.com/EJLEE5826/HyCal-CIL`, pinned at `4558fb7fa6aa389310c9e9dc7db7c2`

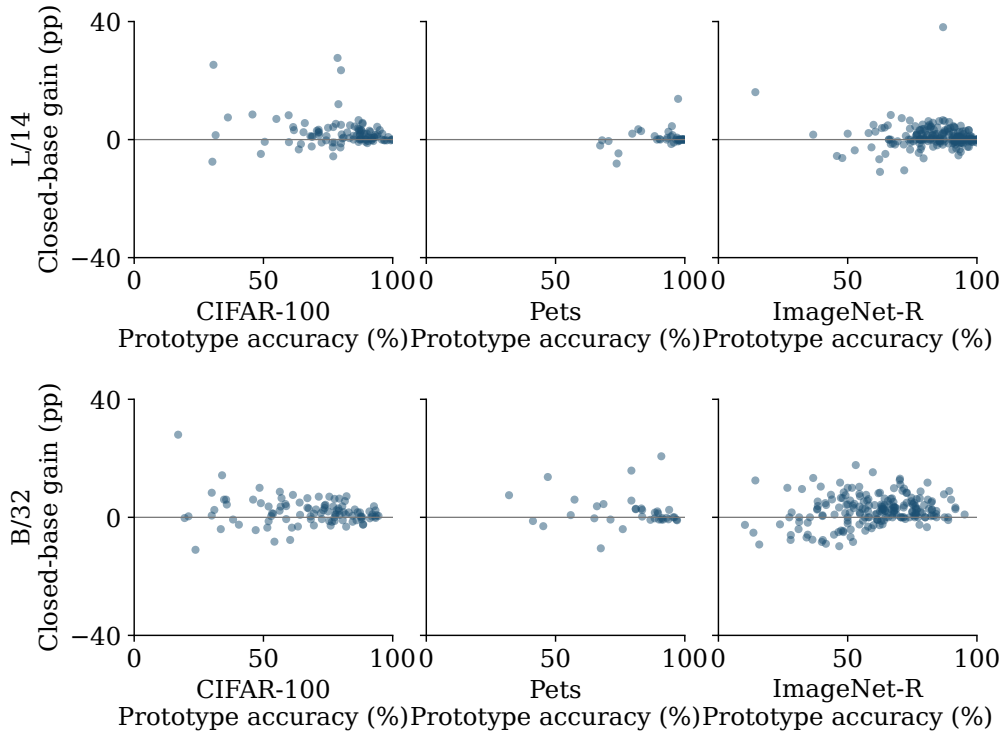


Figure S7: Pure-prototype class accuracy versus the retained-minus-dense closed-base accuracy gap. Each point is a distinct observed base class, averaged over its available partitions. Correlations and sample units are in Table S30.

`cbebc84a41`. The controlled transfer shares our encoders, eight-prompt text anchors, base-fitting pool, five novel supports, base-only tuning, calibration images, arrival orders and reference-accuracy target. The implementation checks establish numerical agreement of the transferred scoring components with the pinned author code. They do not reproduce the published benchmark setting or establish its published performance. The resulting comparison tests matched access in our long-tailed, retention-constrained task. It retains normalized visual/text sum prototypes, class-specific regularized covariance, count-dependent sigmoid weighting, unbiased cross-class score standardization and Gaussian-CDF calibration. The background Mahalanobis term is common to all classes of a query and cancels under this standardization. Precision computation uses the author's regularized covariance setting; sample-space inversion changes the computation rather than the score definition.

Covariance regularization is selected from $\{10^{-4}, 10^{-3}, 10^{-2}\}$ by base-only pseudo-expansion; the novel offset grid is $[-1, 1]$ in steps of .05. The author's count parameter is fixed at five and scale at one. No actual novel tuning queries are used. Actual novel fitting statistics are excluded; only their five support images construct their visual prototypes and covariance. Isolated author-code fixtures give maximum Mahalanobis-distance relative error 4.46×10^{-6} against the author's precision routine and zero Gaussian-CDF discrepancy. The archived field `precision_max_relative_error` names this distance comparison, rather than an elementwise precision-matrix comparison. Source provenance and numerical fixtures are in `hycal_author_fixture.json`. The author's published encoder/prompt protocol and balanced task differ from this matched transfer, so these are newly measured transfer results.

Table S31: Controlled HyCal transfer: unrestricted average incremental macro accuracy and final descriptive novel capacity at the common 95% base requirement (%). Capacity averages partitions feasible in both orders; HyCal feasible reports that count, and – denotes none. Unequal feasible sets are not paired accuracy comparisons.

Encoder	Data	Retained avg.	HyCal avg.	Retained $N_{.95}$	HyCal $N_{.95}$	HyCal feasible
l14	CIFAR-100	83.30	73.57	77.56	–	0/10
l14	Pets	91.32	83.66	91.92	75.29	6/10
l14	ImageNet-R	87.29	76.36	83.93	21.60	2/10
b32	CIFAR-100	71.73	62.67	58.38	–	0/10
b32	Pets	81.14	68.26	71.37	–	0/10
b32	ImageNet-R	67.93	54.19	55.44	–	0/10

Appendix N. Recovery-aware direct accuracy calibration and fair policy comparison

Appendix N.1. Direct paired accuracy calibration

Fix the models, arrivals and candidate thresholds independently of calibration. Let independent class-stratified calibration pairs have correctness indicators $R_i, C_i \in \{0, 1\}$ and deterministic macro weights $w_i = 1/(Bn_{c(i)})$, with $\sum_i w_i = 1$ and $b = \max_i w_i$. Define $A_X = \sum_i w_i \mathbb{E}X_i$ for $X \in \{C, R\}$. For a class-macro target, each class's observations must follow its target class-conditional law, or their averaged marginal law must equal that target. Disjoint, deduplicated benchmark pools establish separation of data access; independence and target-law correspondence remain explicit sampling assumptions. Set $Z_i = C_i - rR_i \in [-r, 1]$. Under the unsafe null $\sum_i w_i \mathbb{E}Z_i \leq 0$, for any fixed $0 \leq \lambda < 1/r$,

$$E_\lambda = \prod_i [1 + \lambda(w_i/b)Z_i] \geq 0, \tag{N.1}$$

$$\mathbb{E}E_\lambda = \prod_i [1 + \lambda(w_i/b)\mathbb{E}Z_i] \leq \exp\left(\frac{\lambda}{b}\sum_i w_i \mathbb{E}Z_i\right) \leq 1. \tag{N.2}$$

Independence gives the equality; $1 + x \leq e^x$ gives the first inequality. Every factor is positive since $w_i/b \leq 1$ and $\lambda r < 1$. For the prespecified grid $\Lambda = \{.01, .025, .05, .1, .2, .3, .4, .5, .65, .8, .95, 1\}$ at $r = .95$, $E = |\Lambda|^{-1} \sum_{\lambda \in \Lambda} E_\lambda$ also has null expectation at most one. Thus $p = \min(1, 1/E)$ is a valid fixed-sample p-value by Markov's inequality. Unlike the sufficient protection loss, Z_i credits recovery ($R_i = 0, C_i = 1$) while charging destruction ($R_i = 1, C_i = 0$).

The test directly addresses $A_C \geq rA_R$ for the class-macro population. It uses fixed sample sizes and mixture weights, with no optimization of λ or sample stopping based on the calibration observations. Thresholds are tested from conservative to permissive; stop at the first failure at level $\delta/5$. Rejecting any unsafe null entails rejecting the first unsafe null in this prespecified sequence, of probability at most $\delta/5$. A union bound covers five prespecified arrivals for each fixed method/order/partition/policy under the class-conditional sampling model. No independence between threshold tests or arrivals is required, and no simultaneous guarantee across experimental configurations is asserted.

Appendix N.2. Identical-access policy comparison

Both certificates use identical cap-40 calibration images, fixed models and independently fixed candidate sequences, $r = .95$ and $\delta = .05$. Each stops at the first failure and returns the retained reference if no finite threshold passes. The original sufficient criterion is preserved without modification. Tables below report final order-0 outcomes: Finite is the number of admitted partitions, Met counts finite-test target attainment including reference fallback, Novel and Base average all ten partitions, and Given admission averages novel accuracy over admitted partitions only. Thus high protection through

fallback is distinct from useful admission. Neither certificate dominates by construction; the paper reports both rather than choosing between them after inspecting test results. Confidence statements condition on a fixed method/order/partition/policy and five prespecified arrivals; they do not apply simultaneously across the full collection of reported configurations. Finite test attainment is an empirical diagnostic rather than the population guarantee itself.

Table S32: L14, CIFAR-100: sufficient protection and direct paired accuracy calibration. Accuracy columns are percentages; both admission and fallback are explicit.

Readout	Test	Finite	Met	Novel	Given admission	Base
Retained	Protect	10/10	10/10	72.76	72.76	85.86
	Paired	10/10	10/10	72.00	72.00	86.01
Dense prototypes	Protect	5/10	10/10	30.74	61.48	87.45
	Paired	6/10	10/10	36.60	61.00	86.85
Recalibrated removal	Protect	0/10	10/10	0.00	–	88.29
	Paired	0/10	10/10	0.00	–	88.29
Raw fusion	Protect	0/10	10/10	0.00	–	88.29
	Paired	0/10	10/10	0.00	–	88.29
Fixed temperature	Protect	9/10	10/10	49.14	54.60	86.66
	Paired	9/10	10/10	59.04	65.59	86.17
Class-mean cache	Protect	2/10	10/10	9.35	46.76	87.94
	Paired	3/10	10/10	14.00	46.65	87.57
HyCal transfer	Protect	0/10	10/10	0.00	–	88.29
	Paired	0/10	10/10	0.00	–	88.29

Table S33: L14, Pets: sufficient protection and direct paired accuracy calibration. Accuracy columns are percentages; both admission and fallback are explicit.

Readout	Test	Finite	Met	Novel	Given admission	Base
Retained	Protect	10/10	10/10	88.51	88.51	94.10
	Paired	10/10	10/10	87.52	87.52	94.40
Dense prototypes	Protect	7/10	10/10	62.39	89.13	94.93
	Paired	8/10	10/10	71.64	89.55	94.80
Recalibrated removal	Protect	4/10	10/10	32.68	81.71	95.48
	Paired	5/10	10/10	43.65	87.30	95.67
Raw fusion	Protect	8/10	10/10	60.83	76.03	94.91
	Paired	9/10	10/10	69.89	77.66	94.84
Fixed temperature	Protect	8/10	10/10	69.61	87.01	95.02
	Paired	9/10	10/10	77.43	86.04	95.41
Class-mean cache	Protect	4/10	10/10	34.86	87.15	95.53
	Paired	5/10	10/10	44.61	89.21	95.26
HyCal transfer	Protect	0/10	10/10	0.00	–	96.05
	Paired	1/10	10/10	3.49	34.87	95.82

Table S34: L14, ImageNet-R: sufficient protection and direct paired accuracy calibration. Accuracy columns are percentages; both admission and fallback are explicit.

Readout	Test	Finite	Met	Novel	Given admission	Base
Retained	Protect	10/10	10/10	80.74	80.74	89.02
	Paired	10/10	10/10	81.61	81.61	88.63
Dense prototypes	Protect	9/10	10/10	48.58	53.97	90.44
	Paired	10/10	10/10	76.76	76.76	89.21
Recalibrated removal	Protect	0/10	10/10	0.00	–	91.54
	Paired	2/10	10/10	6.84	34.18	90.83
Raw fusion	Protect	4/10	10/10	27.88	69.69	90.84
	Paired	5/10	10/10	35.15	70.29	90.19
Fixed temperature	Protect	10/10	10/10	76.00	76.00	89.19
	Paired	10/10	10/10	78.16	78.16	88.62
Class-mean cache	Protect	3/10	10/10	10.31	34.38	90.99
	Paired	9/10	10/10	45.75	50.83	89.23
HyCal transfer	Protect	0/10	10/10	0.00	–	91.54
	Paired	0/10	10/10	0.00	–	91.54

Table S35: B32, CIFAR-100: sufficient protection and direct paired accuracy calibration. Accuracy columns are percentages; both admission and fallback are explicit.

Readout	Test	Finite	Met	Novel	Given admission	Base
Retained	Protect	10/10	10/10	52.54	52.54	76.74
	Paired	10/10	10/10	52.54	52.54	76.74
Dense prototypes	Protect	2/10	10/10	7.95	39.76	78.79
	Paired	6/10	10/10	23.77	39.61	77.96
Recalibrated removal	Protect	2/10	10/10	10.73	53.63	78.57
	Paired	2/10	10/10	10.73	53.63	78.57
Raw fusion	Protect	3/10	10/10	13.51	45.02	78.38
	Paired	3/10	10/10	13.51	45.02	78.38
Fixed temperature	Protect	8/10	10/10	36.50	45.62	77.87
	Paired	10/10	10/10	46.52	46.52	77.14
Class-mean cache	Protect	1/10	10/10	4.16	41.55	79.21
	Paired	4/10	10/10	17.66	44.14	78.51
HyCal transfer	Protect	0/10	10/10	0.00	–	79.08
	Paired	0/10	10/10	0.00	–	79.08

Table S36: B32, Pets: sufficient protection and direct paired accuracy calibration. Accuracy columns are percentages; both admission and fallback are explicit.

Readout	Test	Finite	Met	Novel	Given admission	Base
Retained	Protect	10/10	10/10	61.94	61.94	88.10
	Paired	9/10	10/10	54.58	60.65	88.59
Dense prototypes	Protect	1/10	10/10	5.89	58.86	90.17
	Paired	2/10	10/10	13.25	66.26	90.28
Recalibrated removal	Protect	0/10	10/10	0.00	–	90.24
	Paired	0/10	10/10	0.00	–	90.24
Raw fusion	Protect	5/10	10/10	17.69	35.37	90.03
	Paired	5/10	10/10	20.63	41.25	89.85
Fixed temperature	Protect	9/10	10/10	41.26	45.85	90.46
	Paired	9/10	10/10	44.66	49.63	89.81
Class-mean cache	Protect	0/10	10/10	0.00	–	90.24
	Paired	1/10	10/10	6.44	64.36	90.56
HyCal transfer	Protect	0/10	10/10	0.00	–	90.24
	Paired	0/10	10/10	0.00	–	90.24

Table S37: B32, ImageNet-R: sufficient protection and direct paired accuracy calibration. Accuracy columns are percentages; both admission and fallback are explicit.

Readout	Test	Finite	Met	Novel	Given admission	Base
Retained	Protect	10/10	10/10	49.67	49.67	72.98
	Paired	10/10	10/10	52.06	52.06	72.42
Dense prototypes	Protect	0/10	10/10	0.00	–	75.01
	Paired	7/10	9/10	23.88	34.11	73.01
Recalibrated removal	Protect	0/10	10/10	0.00	–	75.01
	Paired	0/10	10/10	0.00	–	75.01
Raw fusion	Protect	0/10	10/10	0.00	–	75.01
	Paired	0/10	10/10	0.00	–	75.01
Fixed temperature	Protect	8/10	10/10	33.60	42.00	73.68
	Paired	10/10	10/10	46.35	46.35	72.41
Class-mean cache	Protect	0/10	10/10	0.00	–	75.01
	Paired	2/10	10/10	4.66	23.32	74.37
HyCal transfer	Protect	0/10	10/10	0.00	–	75.01
	Paired	0/10	10/10	0.00	–	75.01

Appendix O. Prespecified query-shift stress test

Two deterministic corruption mechanisms are applied to CIFAR RGB images before the original bicubic resize and CLIP normalization: additive Gaussian noise with standard deviation .08 in $[0, 1]$ intensity (clipped to the valid range), and Gaussian blur of radius 1.25 pixels at the original 32×32 resolution. The noise seed depends on image identity and role. The evaluation subset contains 20 original test images per class, sampled with seed 23102026 before outcomes, and the calibration pool contains 40 held-out images per class. Every method uses the same transformed images. Fitting, support and hyperparameter-tuning queries remain clean. These custom perturbations constitute a controlled covariate-shift stress test, rather than the official CIFAR-C suite or an untouched external dataset.

Clean calibration applies the original clean-query certificate to shifted queries; matched-shift calibration instead uses independently held-out images transformed by the same mechanism as test queries. Both preserve the clean-tuning candidate families. We evaluate ten partitions, the first prespecified order and all five arrivals, with both encoders and retained, dense-prototype and HyCal readouts. The direct-test retained policy meets every empirical session target on all ten matched-shift paths per corruption and encoder. Clean-to-shift target violations demonstrate why access separation alone cannot justify transfer of a population guarantee to a changed query law.

The tables give the fifth arrival, Finite admission and Met test-target counts out of ten, fallback-inclusive novel accuracy, and novel accuracy conditional on admission. All-session counts and base accuracies are provided in the machine-readable summary. The newly transformed 24,000 query embeddings are generated from existing local checkpoints; clean-image witness re-encodings verify feature-space consistency. No new models are trained or downloaded.

Table S38: L14, protection certificate under clean or matched-shift calibration. Novel columns are percentages.

Shift	Calibration	Readout	Finite	Met	Novel	Admitted novel
noise08	clean	Retained	10/10	4/10	38.45	38.45
noise08	clean	Dense prototypes	5/10	7/10	14.06	28.12
noise08	clean	HyCal transfer	0/10	10/10	0.00	–
noise08	matched shift	Retained	10/10	10/10	30.58	30.58
noise08	matched shift	Dense prototypes	0/10	10/10	0.00	–
noise08	matched shift	HyCal transfer	0/10	10/10	0.00	–
blur125	clean	Retained	10/10	1/10	28.59	28.59
blur125	clean	Dense prototypes	5/10	6/10	8.22	16.45
blur125	clean	HyCal transfer	0/10	10/10	0.00	–
blur125	matched shift	Retained	10/10	10/10	11.71	11.71
blur125	matched shift	Dense prototypes	0/10	10/10	0.00	–
blur125	matched shift	HyCal transfer	0/10	10/10	0.00	–

Table S39: L14, paired accuracy certificate under clean or matched-shift calibration. Novel columns are percentages.

Shift	Calibration	Readout	Finite	Met	Novel	Admitted novel
noise08	clean	Retained	10/10	5/10	37.76	37.76
noise08	clean	Dense prototypes	6/10	5/10	17.88	29.79
noise08	clean	HyCal transfer	0/10	10/10	0.00	–
noise08	matched shift	Retained	10/10	10/10	31.38	31.38
noise08	matched shift	Dense prototypes	0/10	10/10	0.00	–
noise08	matched shift	HyCal transfer	0/10	10/10	0.00	–
blur125	clean	Retained	10/10	1/10	27.91	27.91
blur125	clean	Dense prototypes	6/10	4/10	11.26	18.77
blur125	clean	HyCal transfer	0/10	10/10	0.00	–
blur125	matched shift	Retained	10/10	10/10	12.19	12.19
blur125	matched shift	Dense prototypes	0/10	10/10	0.00	–
blur125	matched shift	HyCal transfer	0/10	10/10	0.00	–

Table S40: B32, protection certificate under clean or matched-shift calibration. Novel columns are percentages.

Shift	Calibration	Readout	Finite	Met	Novel	Admitted novel
noise08	clean	Retained	10/10	4/10	12.97	12.97
noise08	clean	Dense prototypes	2/10	9/10	1.90	9.50
noise08	clean	HyCal transfer	0/10	10/10	0.00	–
noise08	matched shift	Retained	10/10	10/10	3.91	3.91
noise08	matched shift	Dense prototypes	1/10	10/10	0.81	8.12
noise08	matched shift	HyCal transfer	0/10	10/10	0.00	–
blur125	clean	Retained	10/10	6/10	15.79	15.79
blur125	clean	Dense prototypes	2/10	9/10	2.42	12.12
blur125	clean	HyCal transfer	0/10	10/10	0.00	–
blur125	matched shift	Retained	10/10	10/10	5.74	5.74
blur125	matched shift	Dense prototypes	1/10	10/10	0.30	3.00
blur125	matched shift	HyCal transfer	0/10	10/10	0.00	–

Table S41: B32, paired accuracy certificate under clean or matched-shift calibration. Novel columns are percentages.

Shift	Calibration	Readout	Finite	Met	Novel	Admitted novel
noise08	clean	Retained	10/10	4/10	12.97	12.97
noise08	clean	Dense prototypes	6/10	6/10	5.46	9.10
noise08	clean	HyCal transfer	0/10	10/10	0.00	–
noise08	matched shift	Retained	10/10	10/10	8.66	8.66
noise08	matched shift	Dense prototypes	1/10	10/10	1.00	10.00
noise08	matched shift	HyCal transfer	0/10	10/10	0.00	–
blur125	clean	Retained	10/10	6/10	15.79	15.79
blur125	clean	Dense prototypes	6/10	7/10	6.81	11.35
blur125	clean	HyCal transfer	0/10	10/10	0.00	–
blur125	matched shift	Retained	10/10	10/10	10.41	10.41
blur125	matched shift	Dense prototypes	2/10	10/10	1.57	7.88
blur125	matched shift	HyCal transfer	0/10	10/10	0.00	–

The five revision blocks contain 24,000 metric records. A separate reconstruction verifies 54,960 threshold tests, their first-failure selection, predictions, macro accuracies, unchanged original references and source hashes. Original experiments remain intact. These checks and the locked protocol are distributed with the code; raw image data, model weights and PT feature caches are obtained separately.

Appendix P. Frozen-model operating points and effective admission bias

Appendix P.1. Discrete matching and policy decomposition

This post-hoc diagnostic reuses the independently fitted models, coefficients, original predictions and cap-40 calibration records. It covers all six dataset/encoder conditions, ten partitions and both orders after the fifth arrival; order 0 is primary. The analysis plan was fixed before these new diagnostic outcomes, after the original benchmark results were known. It is not an external preregistration. All original data and scores remain unchanged.

For each fixed readout, enumerate every weighted-accuracy change as its offset crosses a sample margin, respecting canonical class-ID ties. Matching chooses the most permissive event with base accuracy at least the requested target, without interpolation. Figure S8 shows the prespecified partition 0; the complete trajectories for both orders are in the accompanying diagnostic records. Hollow markers indicate reference fallback, which uses a different readout and can lie outside the replacement trajectory.

Let $F_m(t)$ be readout m's descriptive novel capacity at base requirement t, and N_m its selected-policy novel accuracy. With the same reference and

target for both methods,

$$N_R - N_P = [F_R(t) - F_P(t)] + [F_P(t) - N_P] - [F_R(t) - N_R]. \qquad \text{(P.1)}$$

This is an accounting identity. Oracle headroom includes threshold-grid, sampling, conservatism and selection effects; it is not uniquely a certification penalty. Headroom can be negative when a selected policy misses the chosen empirical target. Table S42 evaluates the original common $t = .95A_{\text{ref}}$. On ImageNet-R L/14, capacities 83.93% and 83.64% give a gap of 0.29 [−0.55,1.16] points; retained and dense headrooms are 2.32 and 6.88 points. Their 4.56-point headroom difference accounts for most of the 4.85-point selected-policy gap at this target.

Matching instead to the dense policy's actual base accuracy asks a different question. On all ten L/14 ImageNet-R partitions the retained envelope attains at least that accuracy, with mean/max overshoots 0.027/0.067 points. Its novel accuracy is 79.99%, versus the dense selected 76.76%, a paired difference of 3.23 [1.48,5.02] points. Comparing both envelopes at that stricter target gives 3.10 [1.37,4.88] points. These target-specific comparisons do not establish uniform frontier dominance or isolate the learned weights from their full fusion readout.

The original .80/.85/.90/.95/.99 retention grid is also recomputed descriptively on these models, with joint feasibility counts. L/14 ImageNet-R gaps at .80–.95 have intervals spanning zero; its .99 gap is 6.06 [1.50,12.35] points on five jointly feasible partitions. This selected five-partition result does not establish an all-partition advantage or a newly certified 99% policy. Full tables and both arrival orders are in the diagnostic summary.

Table S42: Selected direct-policy gap and descriptive capacity gap at the common 95% minimum base target (pp). Each row uses the same jointly feasible partitions; conditional paired intervals resample partition means on fixed images. Counts expose infeasible comparisons. All-ten selected results remain separately available.

Encoder	Data	Policy gap	Common-floor capacity gap	Matched count
l14	CIFAR-100	+35.40 [+16.61,+54.61]	+3.68 [+1.08,+6.38]	10/10
l14	Pets	+15.88 [-1.27,+40.52]	-2.12 [-4.07,-0.49]	10/10
l14	ImageNet-R	+4.85 [+2.77,+6.93]	+0.29 [-0.55,+1.16]	10/10
b32	CIFAR-100	+26.38 [+11.60,+41.41]	+3.50 [-0.47,+7.94]	9/10
b32	Pets	+42.92 [+15.13,+66.79]	+1.17 [-5.91,+7.38]	8/10
b32	ImageNet-R	+26.15 [+15.85,+37.54]	+10.40 [+7.31,+14.64]	9/10

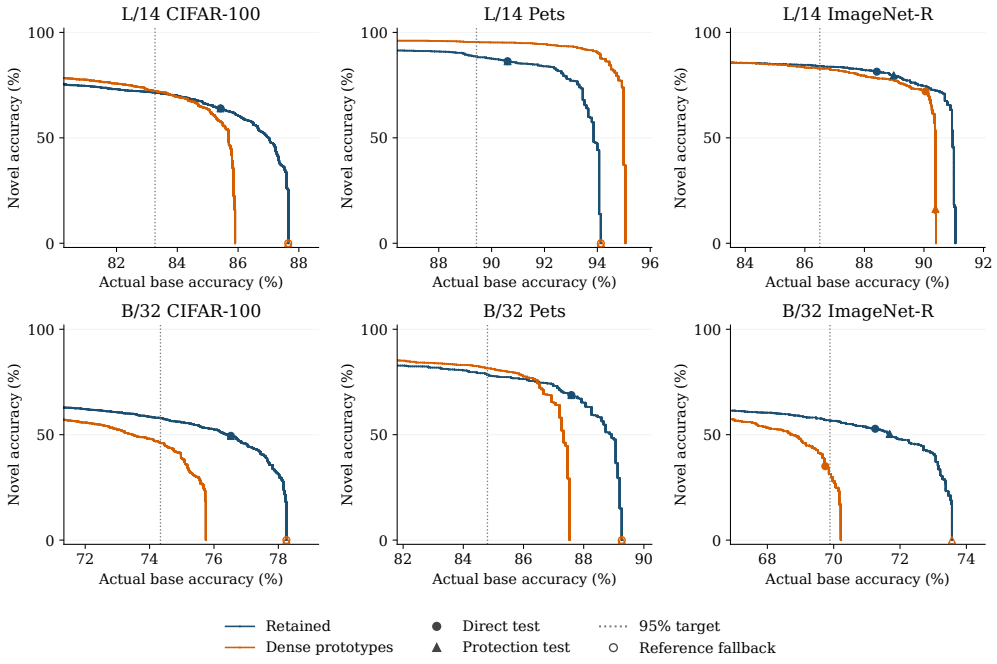


Figure S8: Prespecified partition-0 fixed-readout trajectories and independently selected cap-40 policy points. Panels zoom into the retention region; connecting lines guide the eye and are not interpolation used in matching. A triangle denotes protection calibration and a circle direct calibration. A hollow point is reference fallback. The vertical line is the common minimum requirement.

Appendix P.2. What centering changes algebraically

Suppress the text-centering shift shared by every candidate. Define uncentered, scaled scores

$$B_b(x) = q_T(x, b)/\sigma_T(x) + aq_L(x, b)/\sigma_L(x), \tag{P.2}$$

$$N_n(x) = q_T(x, n)/\sigma_T(x) + uq_P(x, n)/\sigma_P(x). \tag{P.3}$$

The centered decision admits novel if the group margin $\max_b B_b - \max_n N_n$ is below

$$d_{\text{eff}}(x) = d + \Delta(x), \quad \Delta(x) = a\mu_L(x)/\sigma_L(x) - u\mu_P(x)/\sigma_P(x), \tag{P.4}$$

with the original tie rule and floored standard deviations. Query scales still affect within-group fusion. Fixing a nonzero center while retaining query scale does not generally make this effective offset constant.

The fitted ridge model imposes a further constraint. Its augmented design X includes an unregularized intercept and one-hot targets Y. Since $Y\mathbf{1} = \mathbf{1}$ and the slope regularizer annihilates the intercept vector e_0, the normal equations imply

$$(X^\top \Omega X + L)W\mathbf{1} = X^\top \Omega \mathbf{1} = (X^\top \Omega X + L)e_0. \tag{P.5}$$

For the unique regularized solution, $W\mathbf{1} = e_0$ and $\mu_L(x) = 1/|\mathcal{B}|$ in exact arithmetic. The learned contribution to Δ varies through its scale, not a freely changing learned-score mean. The prototype reference mean can vary with the query. Numerical and imbalanced synthetic checks are supplied with the code; they are not recognition benchmarks.

Across ten order-0 partitions, the novel-positive Δ AUROCs are .353, .374 and .409 for L/14 on CIFAR, Pets and ImageNet-R, respectively; the B/32

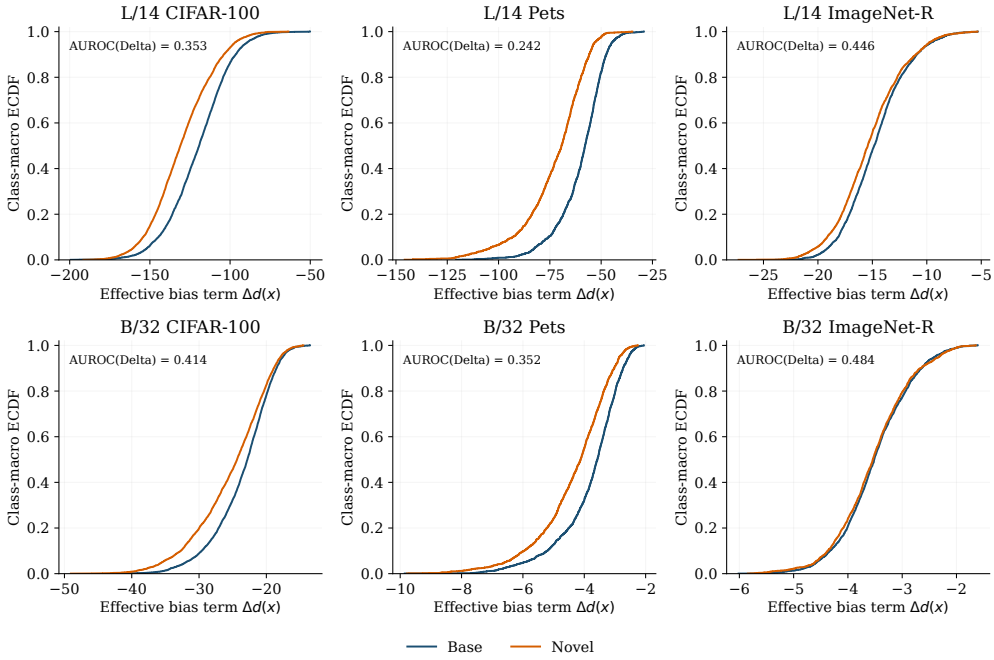


Figure S9: Class-macro ECDFs of the effective bias term for base and novel queries, pre-specified partition 0. AUROC uses fixed novel-positive orientation. Conditional separation of this term alone is neither necessary nor sufficient for improvement of the combined decision.

values are .424, .421 and .451. The term acts jointly with the uncentered margin: their combination improves group-ranking AUROC on all six conditions. ImageNet-R rises from .924 to .977 with L/14 and .856 to .904 with B/32. These ranking diagnostics do not measure correct novel-class identification. The no-centering control preserves query scales and within-group rankings, recalibrating its constant offset with the same independent images and direct accuracy test. Fusion coefficients remain fixed, so this is an offset-reselected mechanism control rather than a fully retuned architecture. Table S43 gives all six conditions; every centered and uncentered final-session test target is met. Group-margin AUROC describes old/novel discrimination, whereas the table measures correct novel-class prediction.

Appendix P.3. Error structure and paired-score variance

For a base query under the class-macro law, let $R, C \in \{0, 1\}$ denote reference and expanded correctness, $A_R = \mathbb{E}R$, $A_C = \mathbb{E}C$, $D = \Pr(R = 1, C = 0)$, and $G = \Pr(R = 0, C = 1)$. The four joint events give

$$\mathbb{E}(C - rR) = A_C - rA_R, \tag{P.6}$$

$$\mathbb{E}(C - rR)^2 = (A_R - D)(1 - r)^2 + Dr^2 + G \tag{P.7}$$

$$= A_R(1 - r)^2 + (A_R - A_C)(2r - 1) + 2rG. \tag{P.8}$$

Subtracting the squared mean proves main Eq. (15). Holding both accuracies fixed makes the variance difference between two couplings exactly $2r(G_2 - G_1)$. A zero-recovery coupling exists when $A_C \leq A_R$; it minimizes variance among couplings with those accuracies. This is a comparison of correctness distributions at fixed margins, not a claim that every retained policy has

Table S43: Centering with independently reselected offsets. Query/None denotes centered/uncentered retained fusion. Finite counts admission; all variants attain 10/10 empirical test targets. Base and Novel are fallback-inclusive test means at arrival 5, order 0; gain is Query minus None with a conditional paired 95% interval. Both use identical cap-40 calibration images, candidate construction, direct tests, query scales and fusion coefficients.

Data	Centering	Finite	Base (%)	Novel (%)	Gain (pp)
ViT-L/14					
CIFAR-100	Query	10/10	86.01	72.00	+41.73 [37.85,45.34]
	None	10/10	85.86	30.27	
Pets	Query	10/10	94.40	87.52	+17.88 [5.58,34.99]
	None	9/10	94.39	69.65	
ImageNet-R	Query	10/10	88.63	81.61	+19.41 [15.09,23.45]
	None	10/10	88.44	62.20	
ViT-B/32					
CIFAR-100	Query	10/10	76.74	52.54	+27.10 [18.60,34.32]
	None	10/10	77.12	25.44	
Pets	Query	9/10	88.59	54.58	+25.52 [11.17,38.52]
	None	10/10	88.44	29.07	
ImageNet-R	Query	10/10	72.42	52.06	+12.23 [9.01,16.09]
	None	10/10	72.44	39.83	

smaller variance than every replacement policy at their different selected accuracies.

For independent class-stratified observations, let $A_{R,c}$, $A_{C,c}$ and G_c be the conditional quantities. The identity holds within each class, and the calibration mean has variance

$$\mathrm{Var}\left(\sum_i w_i Z_i\right) = \frac{1}{B^2}\sum_c \frac{\mathrm{Var}(Z \mid c)}{n_c}. \tag{P.9}$$

If both accuracies match within every class, the weighted-mean variance difference is $2rB^{-2}\sum_c (G_{2,c} - G_{1,c})/n_c$. Matching overall macro accuracies alone does not imply this stratified comparison. The fixed-sample betting p-value also depends on the observed product, so variance order by itself is not a certification-power theorem.

The protection criterion has an additional exact relation. With $\alpha = 1 - r$,

$$\alpha - \mathbb{E}L = A_C - rA_R - G. \tag{P.10}$$

Thus preservation ($G = 0$) makes protection equivalent to the accuracy target. For replacement readouts, the direct test credits the recovery term rather than discarding it. These algebraic results explain the error-structure role separately from the empirical capacity-to-policy decomposition.

Appendix P.4. Error destinations and verification scope

For base queries, $A_B = A_{\mathrm{ref}} - D + G$ uses the same reference for every readout. D contains losses to novel labels and losses to another base label. With retained base scores, within-base ranking cannot change, so $G = 0$ and destruction to another base class is zero. A replacement readout can have both error destinations and can recover reference errors. These identities were

checked on every recorded policy; error destinations and fallback are supplied per partition.

The accompanying records provide all 120 partition/order cases, 240 exact trajectories, source hashes, calibration decisions and error destinations. Numerical reconstruction details are in the repository's reproducibility documentation. Intervals describe assignment variation conditional on fixed images. The source-benchmark mechanism analysis is complemented by the separate natural-shift evaluation in Section R.

Appendix Q. Equal-budget independent fusion retuning

Appendix Q.1. Matched search and fixed accuracy reference

The locked protocol is `experiments/kbs_full_fusion_retuning_protocol.md`. This study crosses retained linear versus dense prototype base readouts with three normalization schemes: query centering/query scaling, no centering/query scaling, and fixed base-tuning moments. Each scheme independently selects base evidence coefficient a, novel prototype coefficient u and a tuning offset. Both coefficients use $\{0, .25, .5, 1, 2, 4\}$; each pair evaluates 101 quantiles of the base-only pseudo-expansion tuning margins, giving 3,636 settings per scheme. Duplicate quantiles count toward the same budget. Maximum harmonic macro accuracy selects the setting, with deterministic first-setting ties. Adapting the offset range to each score scale avoids imposing a range suited to one normalization. Dense prototypes receive separate base and novel coefficients, extending the earlier tied-coefficient baseline.

Ridge fits and regularization, encoder, support, partitions and data roles remain fixed. Fixed moments are estimated from base-tuning queries. Selected fusion coefficients are frozen before independent calibration or test evaluation

and persist across arrivals. All schemes share the original retained table as the absolute accuracy reference and fallback. Reselecting fusion coefficients can change within-base decisions relative to that original reference; recovery G is therefore allowed in this study. Score persistence still concerns each fixed selected model as rows arrive.

Deployment candidates remain 101 base-tuning margin quantiles. The sufficient protection and direct paired tests receive identical cap-40 calibration images and their original five-session error allocation. The tuning objective is pseudo harmonic accuracy, not test-set capacity or calibration success. Test-label frontier sweeps are descriptive. Comparisons average the two orders within each of ten partitions; constrained contrasts require both orders of both schemes to be feasible within that partition. Conditional intervals resample partition means on fixed images.

Appendix Q.2. Capacity after independent optimization

Table S44 isolates normalization after fusion re-selection. On ImageNet-R, query normalization improves capacity over both alternatives for each encoder. Retaining the classifier contributes 9.69 points over the independently tuned dense readout with B/32; L/14 remains comparable at this minimum target. CIFAR L/14 benefits from query normalization and the retained readout, while Pets does not establish a common advantage across encoders. These are target-specific comparisons of independently selected fusion models.

Table S44: Fully retuned capacity contrasts (pp) at the common 95% minimum base target, fifth arrival. The first two contrasts compare retained query normalization with no centering or fixed global moments. The third compares retained and dense query-normalized readouts. Brackets give conditional paired 95% intervals; parentheses count jointly feasible partitions out of ten, requiring both orders.

Encoder	Data	Query − none	Query − fixed	Retained − dense
L/14	CIFAR-100	+4.71 (10) [2.39,6.92]	+4.32 (10) [1.26,6.95]	+3.31 (10) [0.31,6.07]
L/14	Pets	-2.61 (10) [-8.33,2.31]	-2.83 (10) [-6.73,0.65]	-2.62 (10) [-9.24,4.39]
L/14	ImageNet-R	+3.60 (10) [2.55,4.74]	+2.84 (10) [2.42,3.28]	+0.25 (10) [-0.46,0.92]
B/32	CIFAR-100	+1.02 (9) [-2.18,3.36]	+3.55 (10) [-0.30,7.28]	+4.15 (9) [1.30,7.44]
B/32	Pets	+12.49 (8) [-4.08,30.02]	+5.93 (10) [0.68,10.63]	+3.92 (8) [-8.88,15.76]
B/32	ImageNet-R	+5.29 (10) [3.74,7.25]	+4.65 (10) [3.57,5.71]	+9.69 (10) [8.08,11.21]

Appendix Q.3. Independently selected policies

Tables S45 and S46 report the direct-test policies separately from descriptive capacity. Finite admission, empirical target attainment and fallback-inclusive accuracy use the first order and ten partitions. The summary also reports protection policies, conditional accuracy and both arrival orders. Reference fallback admits no novel classes.

Table S45: Independently retuned L/14 readouts under direct accuracy calibration, final session, order 0. All methods share the original reference, support and cap-40 calibration pool. Finite and Met are admission and empirical target counts; Base/Novel are fallback-inclusive test means. Query, none and fixed specify normalization.

Data / scheme	Readout	Finite	Met	Base (%)	Novel (%)
CIFAR-100 / query	retained	10/10	10/10	85.95	71.14
CIFAR-100 / none	retained	9/10	10/10	86.11	59.61
CIFAR-100 / fixed	retained	9/10	10/10	86.15	59.75
CIFAR-100 / query	dense	6/10	10/10	86.92	34.60
CIFAR-100 / none	dense	6/10	10/10	86.92	32.75
CIFAR-100 / fixed	dense	5/10	10/10	87.17	28.66
Pets / query	retained	8/10	10/10	94.74	68.67
Pets / none	retained	9/10	10/10	94.83	75.69
Pets / fixed	retained	8/10	10/10	95.33	67.59
Pets / query	dense	8/10	10/10	95.08	65.26
Pets / none	dense	9/10	10/10	94.69	79.80
Pets / fixed	dense	8/10	10/10	94.67	67.62
ImageNet-R / query	retained	10/10	10/10	88.63	81.66
ImageNet-R / none	retained	10/10	10/10	88.59	76.42
ImageNet-R / fixed	retained	10/10	10/10	88.66	77.87
ImageNet-R / query	dense	10/10	10/10	89.24	76.35
ImageNet-R / none	dense	10/10	10/10	89.16	77.13
ImageNet-R / fixed	dense	10/10	10/10	89.35	76.47

Table S46: Independently retuned B/32 readouts. Definitions match Table S45.

Data / scheme	Readout	Finite	Met	Base (%)	Novel (%)
CIFAR-100 / query	retained	10/10	10/10	76.91	52.17
CIFAR-100 / none	retained	9/10	10/10	77.19	46.12
CIFAR-100 / fixed	retained	9/10	10/10	77.41	42.23
CIFAR-100 / query	dense	7/10	10/10	77.53	27.48
CIFAR-100 / none	dense	6/10	10/10	77.90	23.80
CIFAR-100 / fixed	dense	8/10	10/10	78.02	23.95
Pets / query	retained	7/10	10/10	88.88	44.82
Pets / none	retained	5/10	10/10	89.85	20.03
Pets / fixed	retained	9/10	10/10	89.66	46.57
Pets / query	dense	1/10	10/10	90.20	3.74
Pets / none	dense	2/10	10/10	90.25	10.06
Pets / fixed	dense	2/10	10/10	90.32	12.16
ImageNet-R / query	retained	10/10	10/10	72.35	52.44
ImageNet-R / none	retained	10/10	10/10	72.47	45.01
ImageNet-R / fixed	retained	10/10	10/10	72.45	46.25
ImageNet-R / query	dense	6/10	10/10	73.37	19.97
ImageNet-R / none	dense	6/10	9/10	73.22	20.16
ImageNet-R / fixed	dense	7/10	10/10	73.04	23.37

The 360 searches evaluate 1,308,960 settings. Reconstruction checks 28,800 metric records and 26,755 calibration tests. Boundary witnesses cover ties, extreme offsets and original-readout equivalence; source hashes and restart receipts accompany the code.

Appendix R. Frozen-source ImageNet-Sketch validation

Appendix R.1. Outcome-blind acquisition and identity separation

ImageNet-Sketch [25] supplies a natural rendering distribution for the same ImageNet class vocabulary. The execution protocol, `experiments/kb`

`s_sketch_execution_protocol_v1.md`, was fixed before target features or recognition outcomes. The author-linked Hugging Face archive is pinned to revision `0d8b8b5380efe50baae075842505918ad43c88fe`. Ranged acquisition retrieves the 200 ImageNet-R synsets, with 10,152 image entries and CRC-checked payloads, without downloading the full archive.

Identity checks compare raw bytes and resized 224-RGB pixels against all 22,280 used source fitting, tuning, support, calibration and test paths. They remove 3,624 source-byte overlaps. Target checks additionally remove eight raw duplicates, 62 pixel duplicates and 13 exact rotation/flip equivalents, leaving 6,445 images across all 200 classes (22–50 images per class). Rotation/flip grouping uses the eight square dihedral transforms. Exact fingerprints do not rule out all near-duplicates or establish absence from CLIP pre-training.

Before encoding, seed 27102026 reserves $\min(20, \max(1, \lfloor .4n_c \rfloor))$ images per class for calibration and the remainder for test, giving 2,492 calibration and 3,953 test images. Source-calibration subsets have identical per-class counts to Sketch calibration, selected without scores. Each partition uses 1,233–1,260 base calibration queries. Every method shares these identities. Local CLIP weights and source PIL224 preprocessing are reused; source-image witness encodings verify feature-space agreement before target embedding. Features are saved in resumable batches of at most 16 images.

Appendix R.2. Fixed source models and calibrated admission

The primary study transfers the original source-selected retained, dense-prototype and HyCal readouts. It is separate from Section Q's fusion re-selection. The original ten partitions, first arrival order, five sessions, source five-shot support, ridge fits, fusion coefficients and HyCal regularization

remain fixed. Candidate offsets are copied from the source-tuning archives. No Sketch labels tune these quantities. Only independent calibration selects the admission offset, using either source or Sketch queries with identical class-wise counts. Both statistical criteria retain $r = .95$, $\delta = .05$ and the five-session allocation for each fixed method/partition/calibration/policy.

With Sketch calibration, the retained readout admits all five sessions and meets every empirical test target in all ten paths for each encoder and criterion. The direct policy reaches 86.82% novel accuracy at 90.76% base accuracy with L/14, and 58.96% at 78.87% with B/32 after the fifth arrival. Source calibration also admits all retained paths and meets their targets; its final direct novel accuracies are 82.99% and 54.87%. Here target-matched calibration increases admitted novel recognition rather than repairing observed source-calibration target violations.

Tables S47 and S48 include all three readouts and both criteria. For direct target calibration, dense L/14 admits all ten partitions at 83.34% novel accuracy. Dense B/32 admits three, with fallback-inclusive/conditional novel accuracy 8.35%/27.84%. HyCal L/14 admits two, with 10.47%/52.34% novel accuracy and one final test-target miss; HyCal B/32 returns reference fallback throughout. Thus fallback, admission and empirical target attainment are reported separately.

Table S47: L/14 natural-shift policies after arrival 5, first order. Finite counts admission and Met counts empirical target attainment out of ten partitions; Base/Novel are fallback-inclusive test means. Source/Sketch calibration receive identical class-wise counts. The common closed-base reference averages 93.77% on Sketch.

Sufficient protection criterion

Readout	Calibration	Finite	Met	Base (%)	Novel (%)
Retained	Source	10/10	10/10	92.23	82.83
Retained	Sketch	10/10	10/10	90.81	86.73
Dense prototypes	Source	6/10	10/10	93.16	38.21
Dense prototypes	Sketch	10/10	10/10	91.96	78.68
HyCal	Source	0/10	10/10	93.77	0.00
HyCal	Sketch	1/10	10/10	93.41	4.50

Direct paired accuracy criterion

Readout	Calibration	Finite	Met	Base (%)	Novel (%)
Retained	Source	10/10	10/10	92.21	82.99
Retained	Sketch	10/10	10/10	90.76	86.82
Dense prototypes	Source	10/10	10/10	92.24	75.43
Dense prototypes	Sketch	10/10	10/10	91.56	83.34
HyCal	Source	0/10	10/10	93.77	0.00
HyCal	Sketch	2/10	9/10	93.04	10.47

Table S48: B/32 natural-shift policies. Definitions match Table S47; the common closed-base reference averages 81.32%.

Sufficient protection criterion

Readout	Calibration	Finite	Met	Base (%)	Novel (%)
Retained	Source	10/10	10/10	80.10	52.34
Retained	Sketch	10/10	10/10	79.10	58.23
Dense prototypes	Source	0/10	10/10	81.32	0.00
Dense prototypes	Sketch	0/10	10/10	81.32	0.00
HyCal	Source	0/10	10/10	81.32	0.00
HyCal	Sketch	0/10	10/10	81.32	0.00

Direct paired accuracy criterion

Readout	Calibration	Finite	Met	Base (%)	Novel (%)
Retained	Source	10/10	10/10	79.71	54.87
Retained	Sketch	10/10	10/10	78.87	58.96
Dense prototypes	Source	4/10	10/10	80.39	11.19
Dense prototypes	Sketch	3/10	10/10	80.82	8.35
HyCal	Source	0/10	10/10	81.32	0.00
HyCal	Sketch	0/10	10/10	81.32	0.00

Appendix R.3. Descriptive capacity and scope

Table S49 and Figure S10 distinguish attainable recognition from calibrated operating points. At the common 95% minimum target, L/14 retained and dense capacities average 88.84%, with a paired difference of 0.001 [−0.64,0.69] points over ten jointly feasible partitions. With B/32, retention adds 5.78 [4.03,7.55] points over dense prototypes on nine jointly feasible partitions. These comparisons do not identify selected-policy gaps with intrinsic classifier gains. Full trajectories for all partitions and the original five retention levels accompany the summary.

Table S49: Natural-shift descriptive capacity at the common 95% minimum base target, final session. Each mean uses that readout's feasible partitions; paired differences in the text use the joint feasible set. Infeasible readouts have no reported capacity.

ViT-L/14

Readout	Feasible	Base (%)	Novel capacity (%)
Retained	10/10	89.11	88.84
Dense prototypes	10/10	89.11	88.84
HyCal	10/10	89.11	56.50

ViT-B/32

Readout	Feasible	Base (%)	Novel capacity (%)
Retained	10/10	77.28	63.74
Dense prototypes	9/10	77.43	57.88
HyCal	0/10	–	–

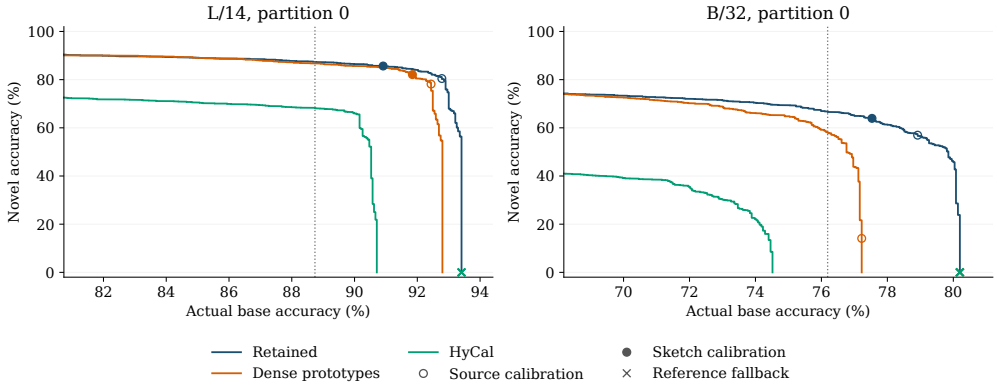


Figure S10: Sketch partition-0 trajectories and direct-calibrated policy points. Open/filled circles use source/Sketch calibration; crosses are reference fallback, which can lie outside a replacement trajectory. Dotted lines show the common requirement. Panels zoom into retention regions; connecting lines are visual guides, with no interpolation in capacity matching. All ten full trajectories are supplied separately.

The 2,700 metric records and 4,899 calibration tests are reconstructed from prediction archives, with unchanged source selection and candidate hashes.

Population confidence retains independent class-conditional sampling and calibration–target correspondence assumptions; empirical success on these fixed target images is not a guarantee for arbitrary shifts or a simultaneous confidence statement over all configurations. Source-calibrated policies carry a source-law guarantee. Intervals describe assignment variation conditional on the fixed target images.

References

[1] M. D. McDonnell, D. Gong, A. Parvaneh, E. Abbasnejad, A. van den Hengel, RanPAC: Random Projections and Pre-trained Models for Continual Learning, in: Advances in Neural Information Processing Systems, Vol. 36, 2023, pp. 12022–12053. doi:10.52202/075280-0526.
URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/2793dc35e14003dd367684d93d236847-Abstract-Conference.html

[2] K. Cao, C. Wei, A. Gaidon, N. Arechiga, T. Ma, Learning Imbalanced Datasets with Label-Distribution-Aware Margin Loss, arXiv preprint arXiv:1906.07413 (2019).
URL https://arxiv.org/abs/1906.07413

[3] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, S. Belongie, Class-Balanced Loss Based on Effective Number of Samples, arXiv preprint arXiv:1901.05555 (2019).
URL https://arxiv.org/abs/1901.05555

[4] J. Ren, C. Yu, S. Sheng, X. Ma, H. Zhao, S. Yi, H. Li, Balanced Meta-Softmax for Long-Tailed Visual Recognition, arXiv preprint arXiv:2007.10740 (2020).
URL https://arxiv.org/abs/2007.10740

[5] A. K. Menon, S. Jayasumana, A. S. Rawat, H. Jain, A. Veit, S. Kumar, Long-tail learning via logit adjustment, arXiv preprint arXiv:2007.07314 (2020).
URL https://arxiv.org/abs/2007.07314

[6] B. Kang, S. Xie, M. Rohrbach, Z. Yan, A. Gordo, J. Feng, Y. Kalantidis, Decoupling Representation and Classifier for Long-Tailed Recognition, arXiv preprint arXiv:1910.09217 (2019).
URL https://arxiv.org/abs/1910.09217

[7] J.-X. Shi, T. Wei, Z. Zhou, J.-J. Shao, X.-Y. Han, Y.-F. Li, Long-Tail Learning with Foundation Model: Heavy Fine-Tuning Hurts, in: International Conference on Machine Learning, 2024, pp. 45014–45039.
URL https://proceedings.mlr.press/v235/shi24g.html

[8] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, W. Chen, LoRA: Low-Rank Adaptation of Large Language Models, arXiv preprint arXiv:2106.09685 (2021).
URL https://arxiv.org/abs/2106.09685

[9] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, N. Houlsby, An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale, arXiv preprint arXiv:2010.11929 (2020).
URL https://arxiv.org/abs/2010.11929

[10] Z. Zhong, J. Cui, S. Liu, J. Jia, Improving Calibration for Long-Tailed Recognition, arXiv preprint arXiv:2104.00466 (2021).
URL https://arxiv.org/abs/2104.00466

[11] T. Wei, J.-X. Shi, M.-L. Zhang, Y.-F. Li, Robust long-tailed learning under label noise, Frontiers of Computer Science 20 (1) (2026) 2001321. doi:10.1007/s11704-025-40860-0.
URL https://journal.hep.com.cn/fcs/EN/10.1007/s11704-025-40860-0

[12] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, I. Sutskever, Learning Transferable Visual Models From Natural Language Supervision, in: International Conference on Machine Learning, 2021, pp. 8748–8763.
URL https://proceedings.mlr.press/v139/radford21a.html

[13] K. Zhou, J. Yang, C. C. Loy, Z. Liu, Learning to Prompt for Vision-Language Models, International Journal of Computer Vision 130 (9) (2022) 2337–2348. doi:10.1007/s11263-022-01653-1.
URL https://link.springer.com/article/10.1007/s11263-022-01653-1

[14] K. Zhou, J. Yang, C. C. Loy, Z. Liu, Conditional Prompt Learning for Vision-Language Models, arXiv preprint arXiv:2203.05557 (2022).
URL https://arxiv.org/abs/2203.05557

[15] J. Snell, K. Swersky, R. S. Zemel, Prototypical Networks for Few-shot Learning, arXiv preprint arXiv:1703.05175 (2017).
URL https://arxiv.org/abs/1703.05175

[16] Y. Wang, W.-L. Chao, K. Q. Weinberger, L. van der Maaten, SimpleShot: Revisiting Nearest-Neighbor Classification for Few-Shot Learning, arXiv preprint arXiv:1911.04623 (2019).
URL https://arxiv.org/abs/1911.04623

[17] R. Zhang, W. Zhang, R. Fang, P. Gao, K. Li, J. Dai, Y. Qiao, H. Li, Tip-Adapter: Training-free Adaption of CLIP for Few-shot Classification, arXiv preprint arXiv:2207.09519 (2022).
URL https://arxiv.org/abs/2207.09519

[18] Y. Zhu, F. Zhuang, X. Zhang, Z. Qi, Z. Shi, J. Cao, Q. He, Combat data shift in few-shot learning with knowledge graph, Frontiers of Computer Science 17 (1) (2023) 171305. doi:10.1007/s11704-022-1339-7.
URL https://journal.hep.com.cn/fcs/EN/10.1007/s11704-022-1339-7

[19] K. Książek, H. Jastrzębski, K. Pniaczek, B. Trojan, M. Karp, J. Tabor, FeNeC: Enhancing continual learning via feature clustering with neighbor- or logit-based classification, Knowledge-Based Systems 338 (2026) 115479. doi:10.1016/j.knosys.2026.115479.

[20] D. Hendrycks, S. Basart, N. Mu, S. Kadavath, F. Wang, E. Dorundo, R. Desai, T. Zhu, S. Parajuli, M. Guo, D. Song, J. Steinhardt, J. Gilmer, The Many Faces of Robustness: A Critical Analysis of Out-of-Distribution Generalization, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 8340–8349.
URL https://openaccess.thecvf.com/content/ICCV2021/html/Hendrycks_The_Many_Faces_of_Robustness_A_Critical_Analysis_of_Out-of-Distribution_ICCV_2021_paper.html

[21] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, L. Lei, Learn then test: Calibrating predictive algorithms to achieve risk control, arXiv preprint arXiv:2110.01052 (2021). doi:10.48550/arXiv.2110.01052.

[22] A. Krizhevsky, Learning multiple layers of features from tiny images, Tech. rep., University of Toronto (2009).
URL https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf

[23] O. M. Parkhi, A. Vedaldi, A. Zisserman, C. V. Jawahar, Cats and dogs, in: IEEE Conference on Computer Vision and Pattern Recognition, 2012, pp. 3498–3505. doi:10.1109/CVPR.2012.6248092.
URL https://www.robots.ox.ac.uk/~vgg/publications/2012/parkhi12a/

[24] E. Lee, M. Kim, J. Kwon, Y. Lee, J. Kim, S. Jang, Y. Kim, Hycal: A training-free prototype calibration method for cross-discipline few-shot class-incremental learning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026, pp. 29462–29471.
URL https://arxiv.org/abs/2604.15678

[25] H. Wang, S. Ge, Z. C. Lipton, E. P. Xing, Learning robust global representations by penalizing local predictive power, in: Advances in Neural Information Processing Systems, Vol. 32, 2019, pp. 10506–10518.